
Figure atlas

three experiments, no interpretation

Every graph, with what it plots — for reading together

Lior Baruch · Reichman University · 24 August 2026

Exp1 and Exp2 are PTO only. GRPO appears from the Exp3 section onward.

One figure per slide, and a strip that says what it is

- **No conclusions on any slide.** Each strip states the x axis, the y axis, what the series are, and how the numbers were produced. It does not say what the figure shows. That is what we are here to work out.
- **Sign conventions are stated wherever a difference is plotted,** because they differ between families — and three rubrics are lower-is-better (MICI, and the MI-inconsistency channels).
- **Every Exp3 figure names its grader.** The two are never averaged: one of them WAS the training reward and the other is held out, and the level offset between them is model-dependent.
- **Exp1 and Exp2 figures did not exist before this deck.** Neither experiment's EDA writes figures to disk. These were generated from the raw per-conversation files; the generator is `meetings/build/make_exp1_exp2_figs.py`.

TWO ARTIFACT DEFECTS, SO NOBODY IS MISLED

The GRPO K=5 arm's last scored iteration is 6, and every figure here plots it to 6. But 33 rendered caption files in the EDA still say it is censored at iteration 5. The figures are current; that prose is stale. Descriptions in this deck were written from the images and tables.

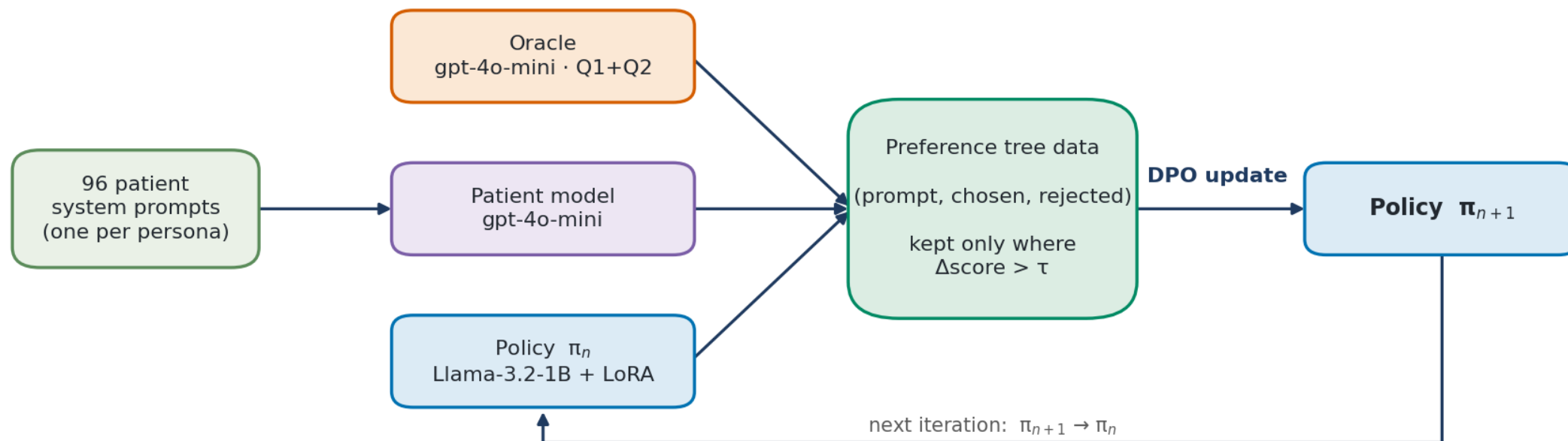
The reward-faithfulness tables pool iterations 1–6 for GRPO K=5 under a column labelled 1–5. Those two slides carry the note inline.

METHOD

How the two methods build training data

Hand-drawn schematics, not measurements. Included so the axes later in the deck have something to attach to.

PTO — preference tree + DPO



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.

WHAT IT IS

A schematic of the PTO loop, not data.

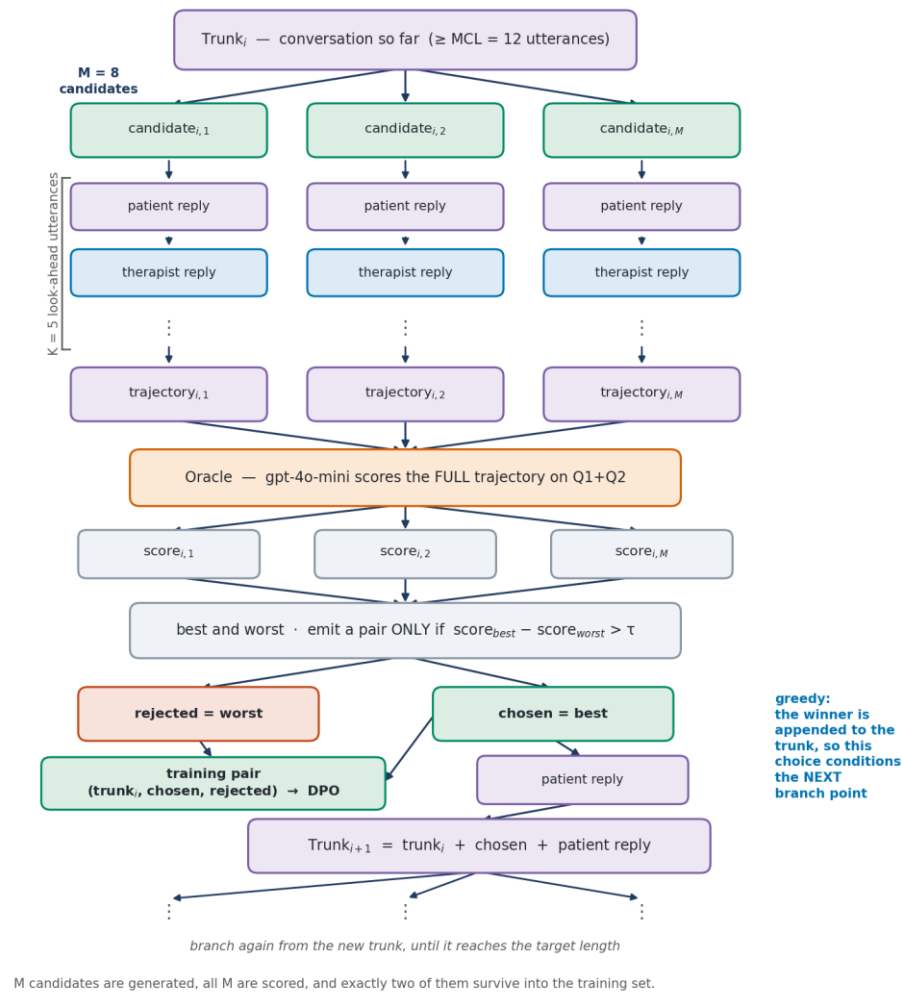
READS

Left to right: branch M candidates at a therapist turn, simulate K look-ahead turns, score each with the oracle, keep the best/worst pair.

IN EXP3

M = 8, K ∈ {0, 5}, τ = 0.1, MCL = 12.

PTO — the preference tree in detail



WHAT IT IS

Schematic of one trunk being grown.

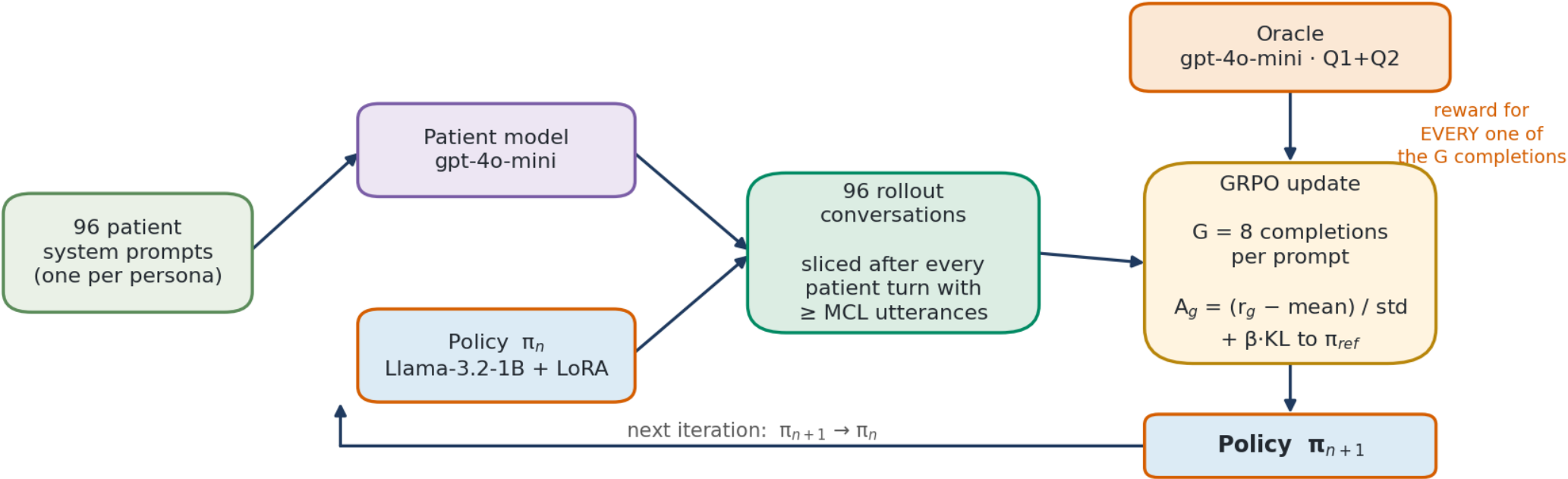
READS

At each branch point the best-of-M completion is appended to the trunk, so the chosen branch feeds the next branch point.

NOTE

This is `greedy` mode, the Exp3 default. The alternative branches a pre-recorded conversation with no feedback.

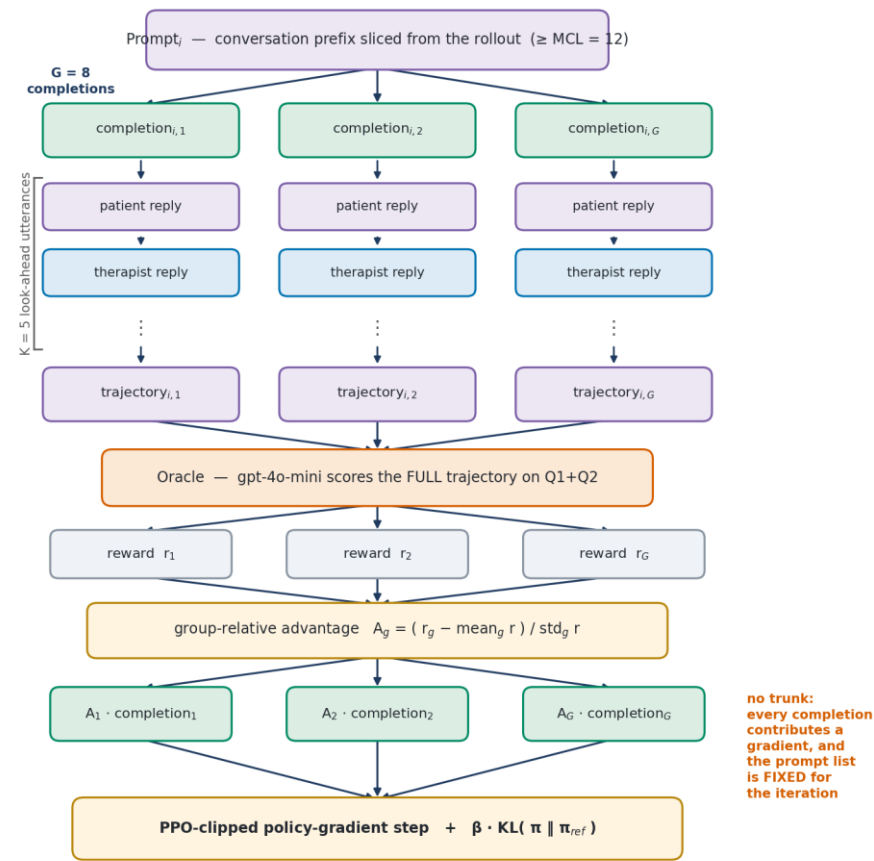
GRPO — group-relative rollout



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.
As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

WHAT IT IS A schematic of the GRPO loop, not data.	READS Slice the rollout after every patient turn; sample G completions per prompt; score all of them; use the group-relative advantage.	IN EXP3 G = 8, K ∈ {0, 5}, MCL = 12. The oracle sits inside the training loop.
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GRPO — the sampled group



The next prompt is the next slice on the list — it is not conditioned on which completion won here.

G completions are generated, all G are scored, and all G carry a gradient. Rows match the PTO figure so the two can be compared.

WHAT IT IS

Schematic of one prompt's group of G completions.

READS

All G are kept and weighted by their advantage; none is discarded.

CONTRAST

PTO keeps two of M (best and worst) and only if their margin clears τ .

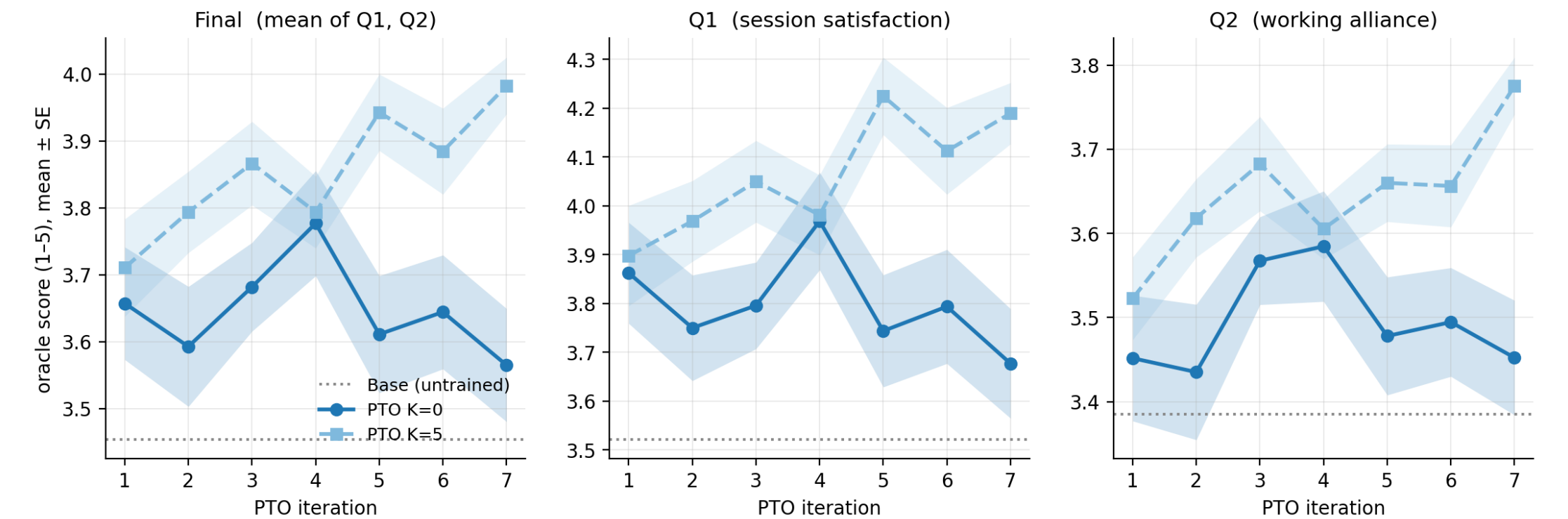
EXP1

Exp1 — ICLR 2025

Llama-2-7B therapist. GPT-3.5 as patient, oracle and training reward. PTO at $K \in \{0, 5\}$, 7 iterations, 96 personas. PTO only — there is no GRPO in this experiment.

Score by iteration, K=0 vs K=5

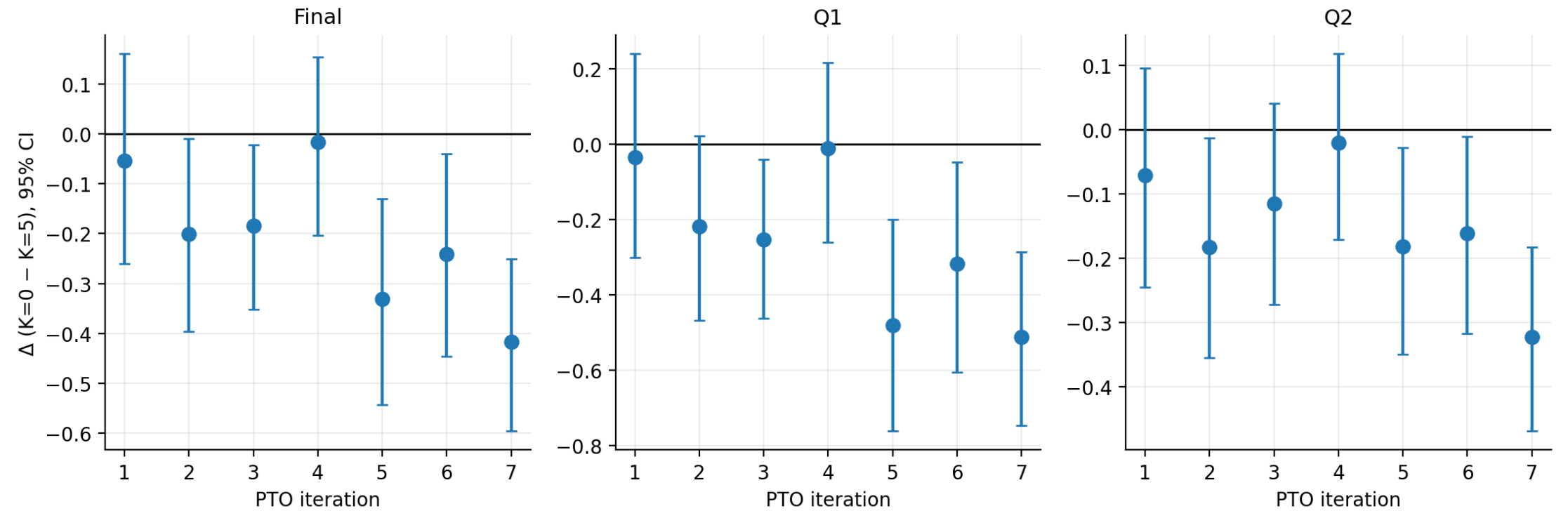
Exp1 — score by PTO iteration, K=0 vs K=5 (GPT-3.5 oracle, n = 96 per point)



X AXIS PTO iteration, 1–7.	Y AXIS Oracle score, 1–5. Band = ±1 SE of the mean over 96 conversations.	SERIES K=0 solid circle, K=5 dashed square. Dotted grey = the untrained baseline. Panels: Final = mean(Q1, Q2), then Q1 and Q2 separately.	DATA The 15 models of the paper's Table 1, recomputed from the shipped per-conversation score files. Grader: GPT-3.5.
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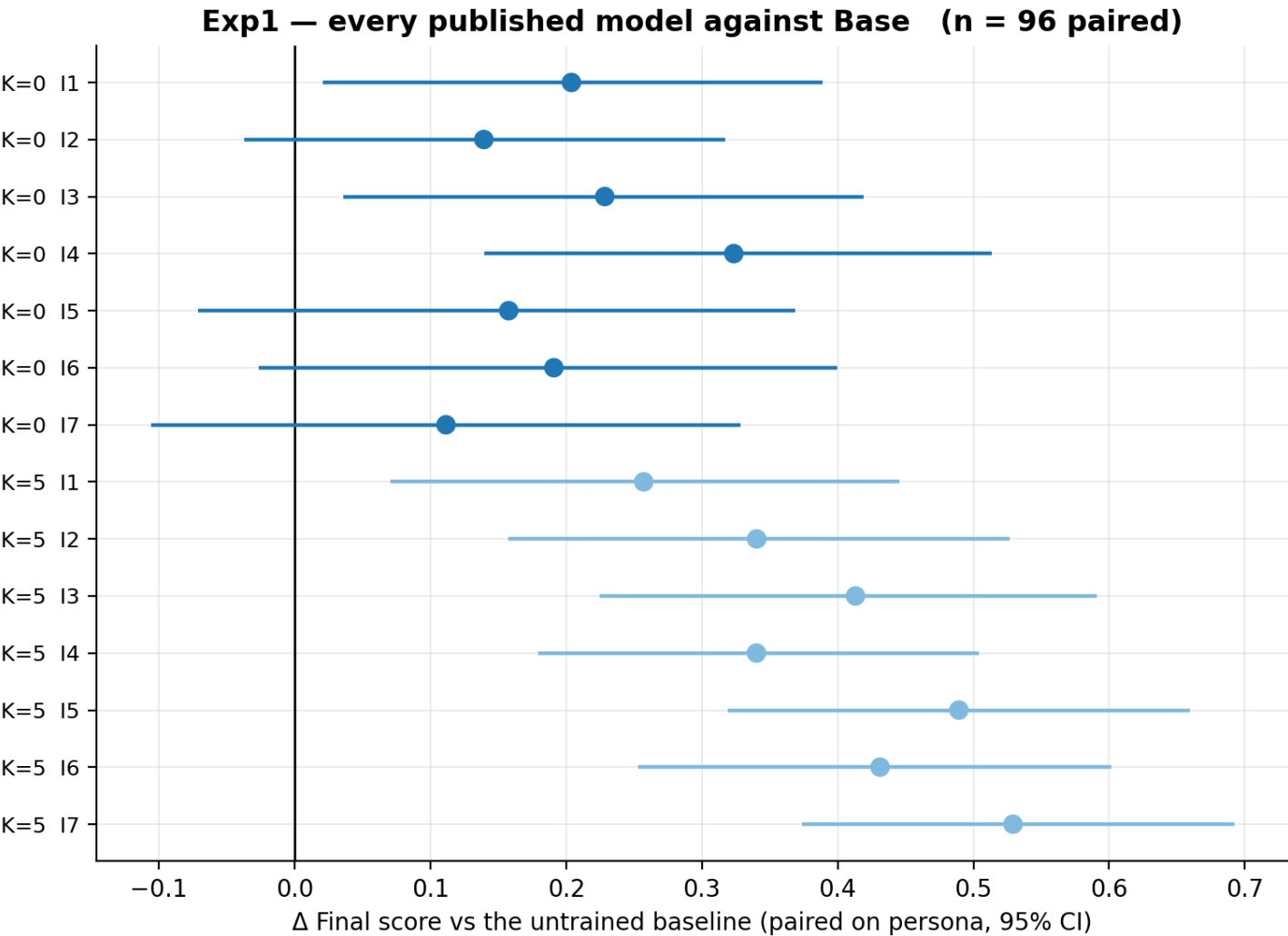
Paired difference between the two look-ahead arms

Exp1 — paired K=0 minus K=5 by iteration (same 96 personas; below 0 = K=5 scored higher)



X AXIS PTO iteration, 1–7.	Y AXIS Mean of (K=0 score – K=5 score) over the 96 shared personas. Positive = K=0 scored higher. Whiskers = 95% percentile bootstrap CI, 5,000 resamples.	SERIES One point per iteration; the three panels are Final, Q1, Q2.	DATA Same conversation index = same patient permutation in both arms, so the pairing is exact. n = 96.
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Every published model against the untrained baseline



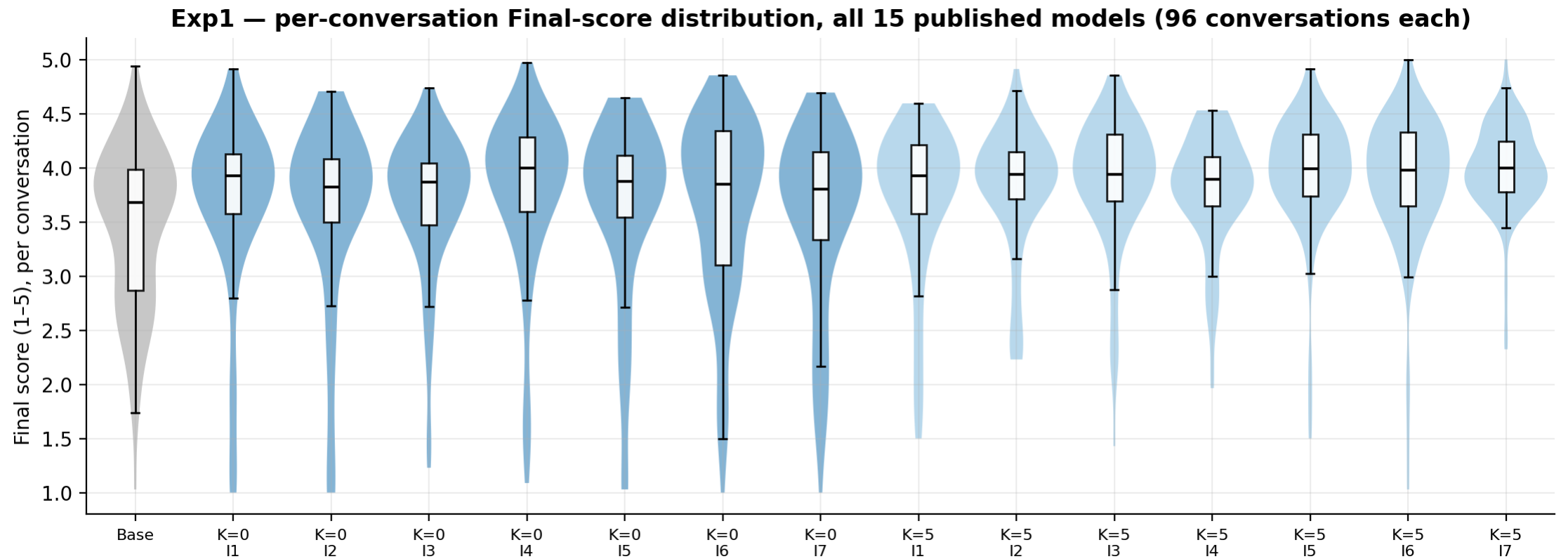
X AXIS
Δ Final score vs Base, paired on persona. 95% bootstrap CI.

Y AXIS
The 14 trained models: the K=0 block above the K=5 block, iteration 1 at the top of each block and iteration 7 at the bottom.

SERIES
Colour = look-ahead arm.

DATA
n = 96 paired per row. Uncorrected — no multiplicity adjustment is applied in this figure.

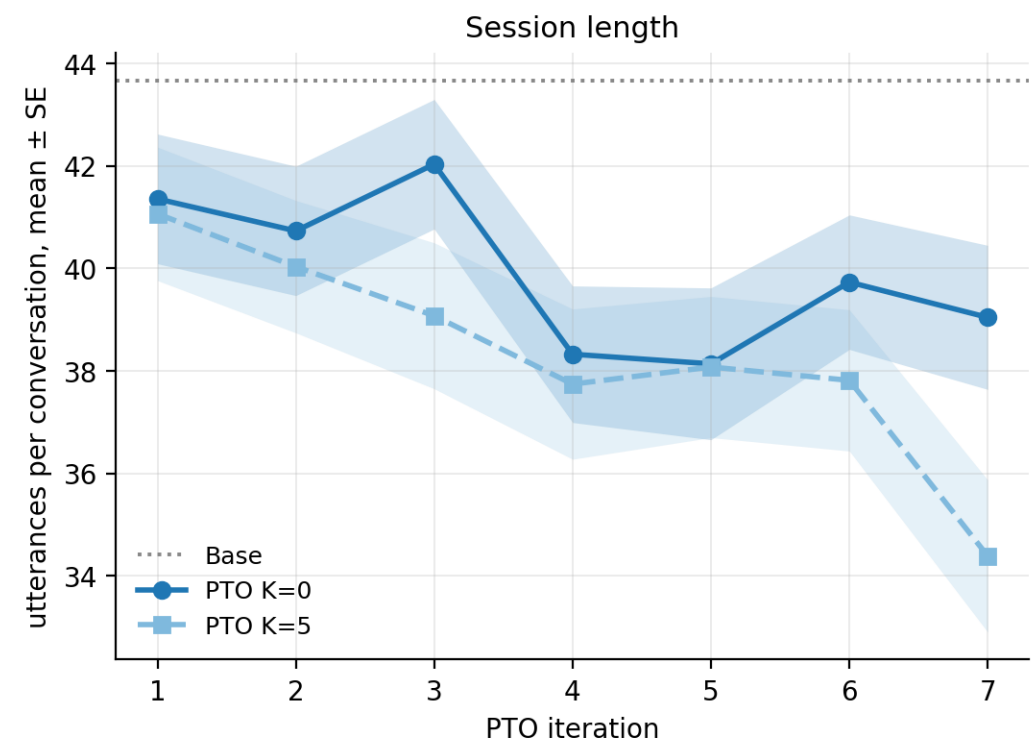
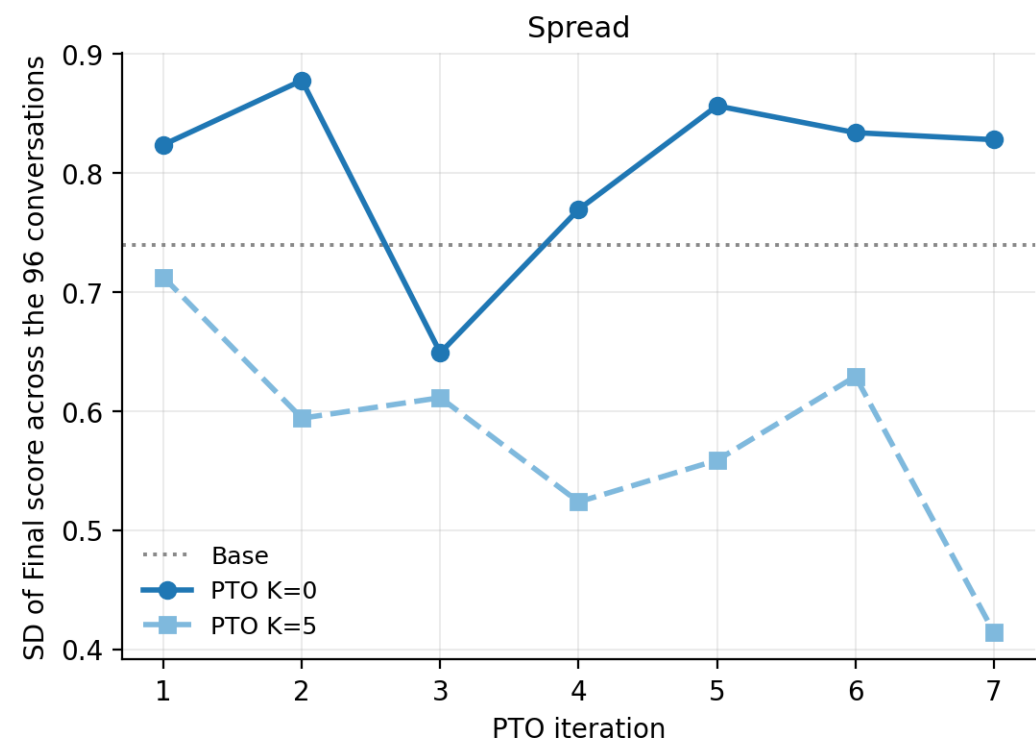
The spread behind each mean



X AXIS The 15 models: Base, then K=0 iterations 1–7, then K=5 iterations 1–7.	Y AXIS Final score of a single conversation, 1–5.	SERIES Violin = full distribution, box = median and quartiles, outliers hidden.	DATA 96 conversations per model, 1,440 in total.
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Score spread and session length by iteration

Exp1 — score spread and session length by iteration (n = 96 per point)



LEFT PANEL

Y = standard deviation of the Final score across the 96 conversations (ddof = 1). One point per model.

RIGHT PANEL

Y = utterances per conversation, therapist and patient combined, mean ± SE. Counted as rows of the stored transcript.

SERIES

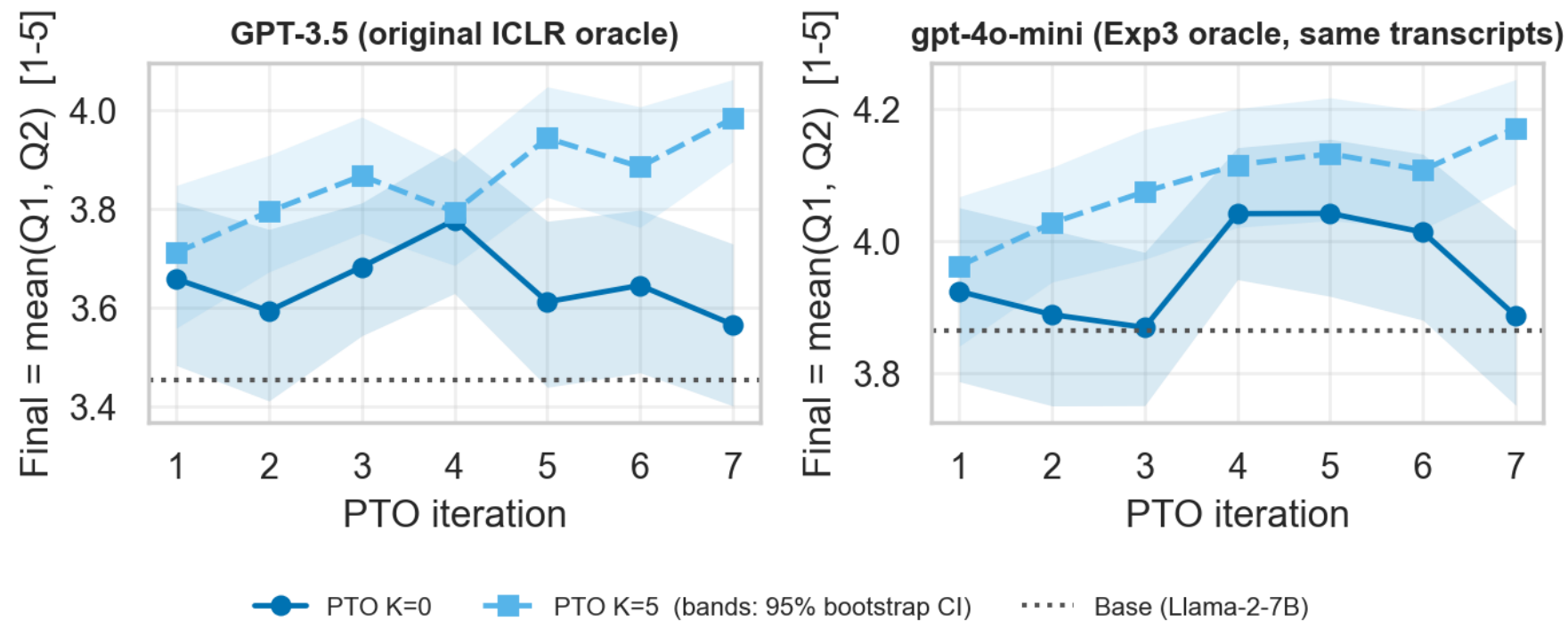
K=0 solid circle, K=5 dashed square, dotted grey = Base.

NOTE

The paper underlines the lowest SD per column in its Table 1 but runs no test on variance.

The same transcripts under two different graders

Exp1 (ICLR 2025) conversations under two graders — K=0 solid, K=5 dashed



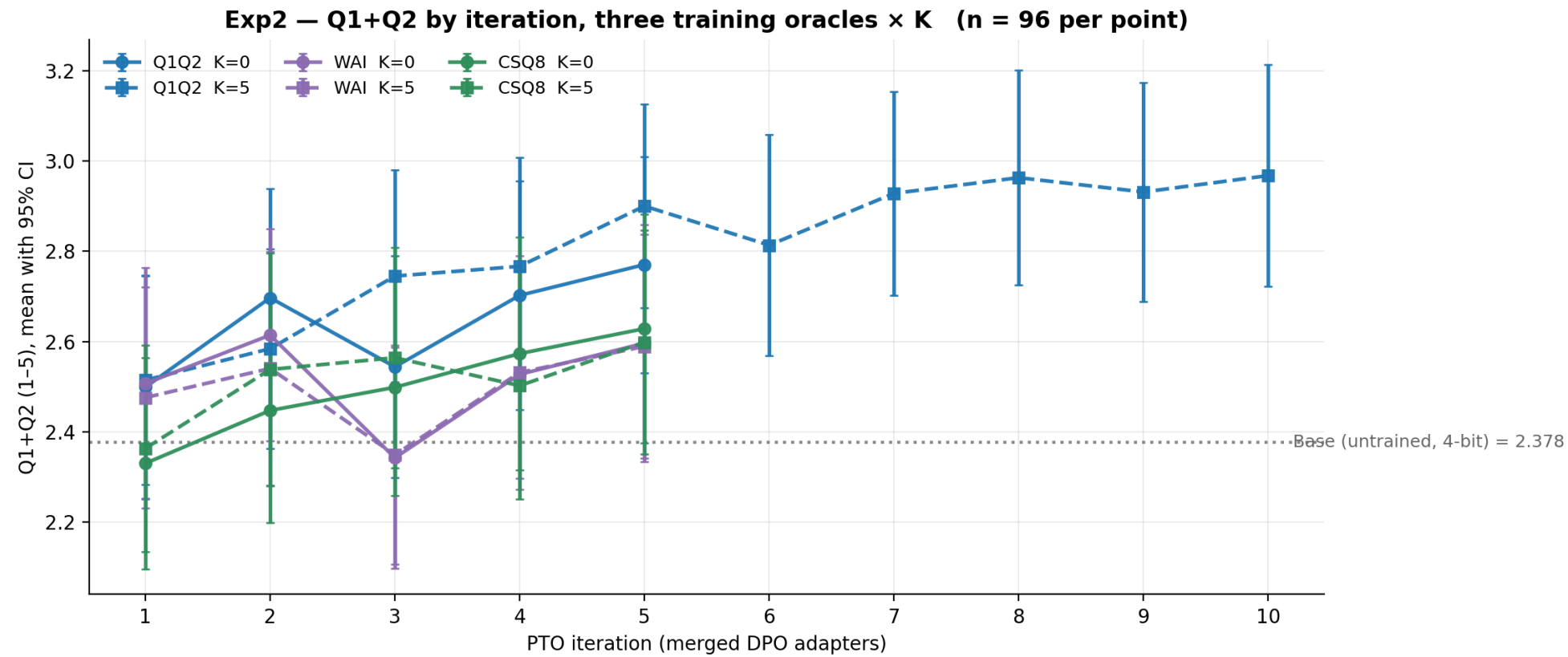
X AXIS PTO iteration, 1–7. Both arms complete 7 — no censoring here.	Y AXIS Final = mean(Q1, Q2) over the 96 conversations. Band = 95% percentile bootstrap CI. Note the two panels have SEPARATE y scales.	SERIES Left: the original GPT-3.5 oracle, i.e. the paper's Table 1. Right: the same stored transcripts re-scored by gpt-4o-mini on the Exp3 rubric.	DATA 1,440 conversations re-scored. The scales differ by roughly 0.19–0.43 points and are never averaged.
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EXP2

Exp2 — the regime change

Llama-3.2-1B therapist in 4-bit NF4. gpt-4o-mini as patient and oracle, on a JSON schema. Less cooperative patients. PTO across three training instruments $\times K \in \{0, 5\}$. PTO only.

Q1+Q2 by iteration, three training oracles × K



X AXIS	Y AXIS	SERIES	DATA
PTO iteration — the number of merged DPO adapters, 1–10.	Q1+Q2 (mean of Q1 and Q2 per conversation), 1–5, with 95% CI.	Colour = which instrument the arm was TRAINED on (Q1+Q2, WAI-SR, CSQ-8). Solid circle = K=0, dashed square = K=5. Dotted = untrained base.	n = 96 per point. Only the Q1+Q2 K=5 arm ran past iteration 5.

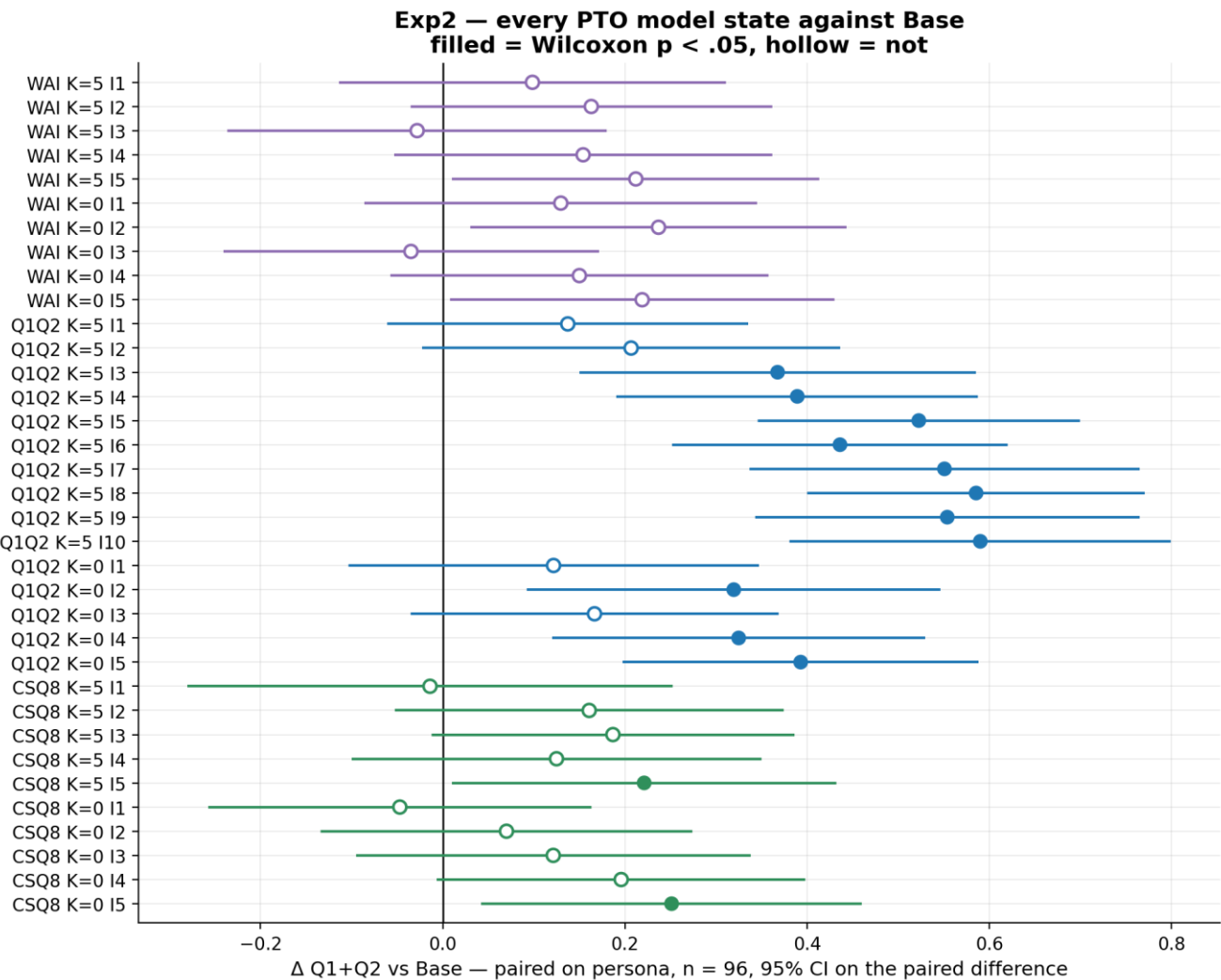
Each arm scored on the instrument it was trained on

Exp2 — each arm measured on its OWN training instrument (n = 96 per point)



X AXIS	Y AXIS	SERIES	DATA
PTO iteration.	Score on that arm's OWN training instrument — note each panel has its own scale. Mean with 95% CI.	K=0 solid circle, K=5 dashed square. Dotted = the base model on that same instrument.	n = 96 per point. This is the like-for-like test of whether the optimisation moved the thing it optimised.

Every model state against the base model



X AXIS

Δ Q1+Q2 vs Base, paired on persona. Bar = 95% CI on the PAIRED difference, from the paired SD.

Y AXIS

The 35 trained model states, grouped by training oracle then K.

SERIES

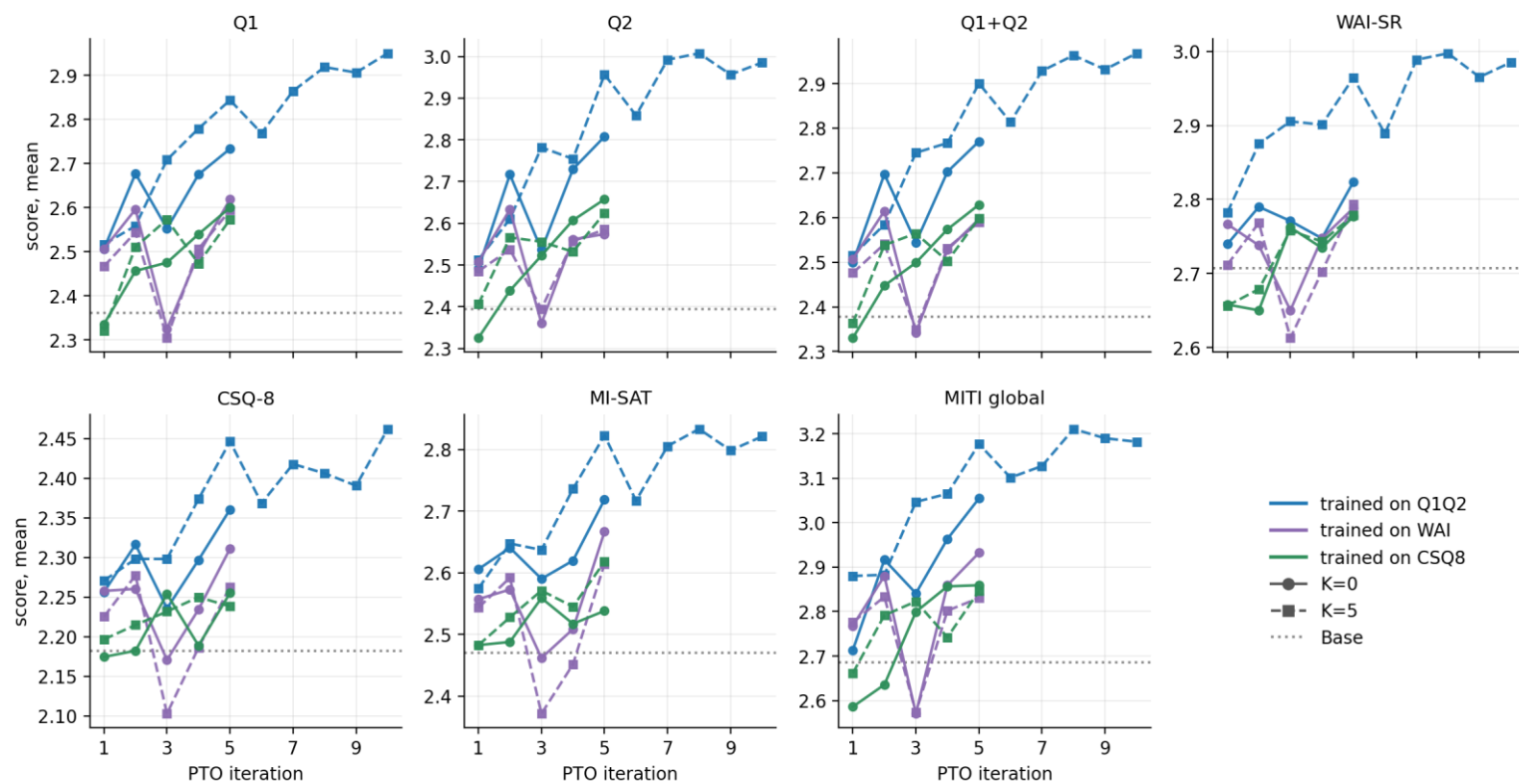
Filled marker = Wilcoxon signed-rank $p < .05$; hollow = not. Colour = training oracle.

DATA

n = 96 paired per row, uncorrected. The marker is a rank test and the bar a normal-approximation CI, so near $p \approx .05-.10$ they can disagree — three WAI rows are hollow with a bar clear of zero.

All six instruments read on every arm

Exp2 — all six instruments read on every arm (training reward varies by colour; n = 96 per point)



X AXIS

PTO iteration, in every panel.

Y AXIS

Score on the panel's instrument — each panel has its own scale. Dotted = base on that instrument.

SERIES

Colour = training oracle, line style = K. Every arm is read on every instrument, including ones it never optimised.

DATA

n = 96 per point. SEVEN panels — the six instruments plus the Q1+Q2 composite, which is the mean of two panels already shown. MITI global is the 1–5 clinician-style global rating.

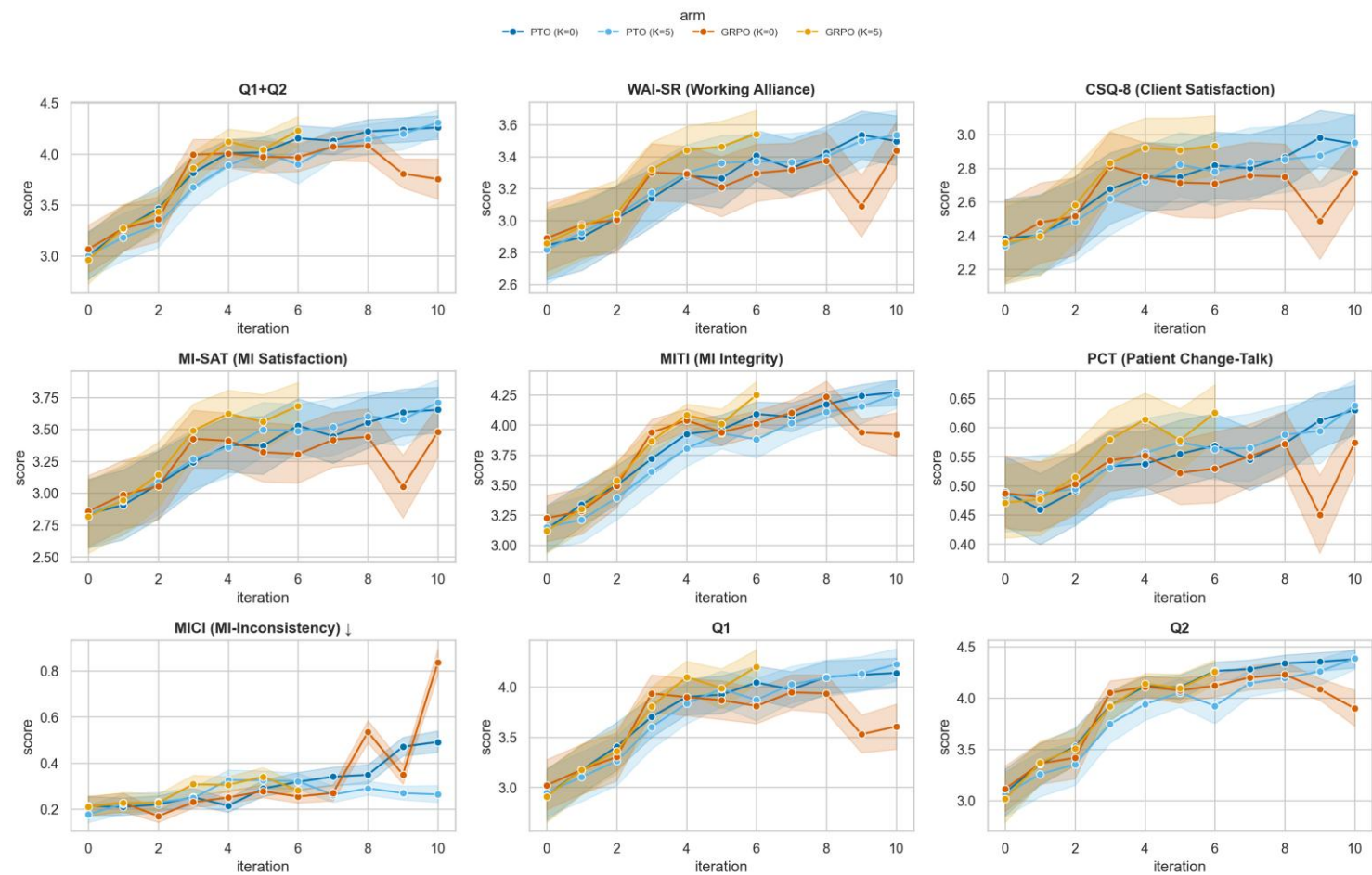
EXP3 · OUTCOMES

Exp3 — four arms, two graders

Llama-3.2-1B in bf16. gpt-4o-mini patient + training oracle, Claude Haiku 4.5 held out. PTO and GRPO at $K \in \{0, 5\}$, matched MCL, $M = G = 8$, temperature and oracle. 40 model states $\times$ 8 instruments $\times$ 96 personas, per grader.

All nine rubrics by iteration — training oracle

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor, PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

source: [arms/outcomes/figures/gpt-4o-mini/trajectories_all_metrics.png](#)

X AXIS

Training iteration, 0–10, in every panel.

Y AXIS

The panel's rubric, on its own scale. MICI and PCT are 0–1 rates; the rest are 1–5.

SERIES

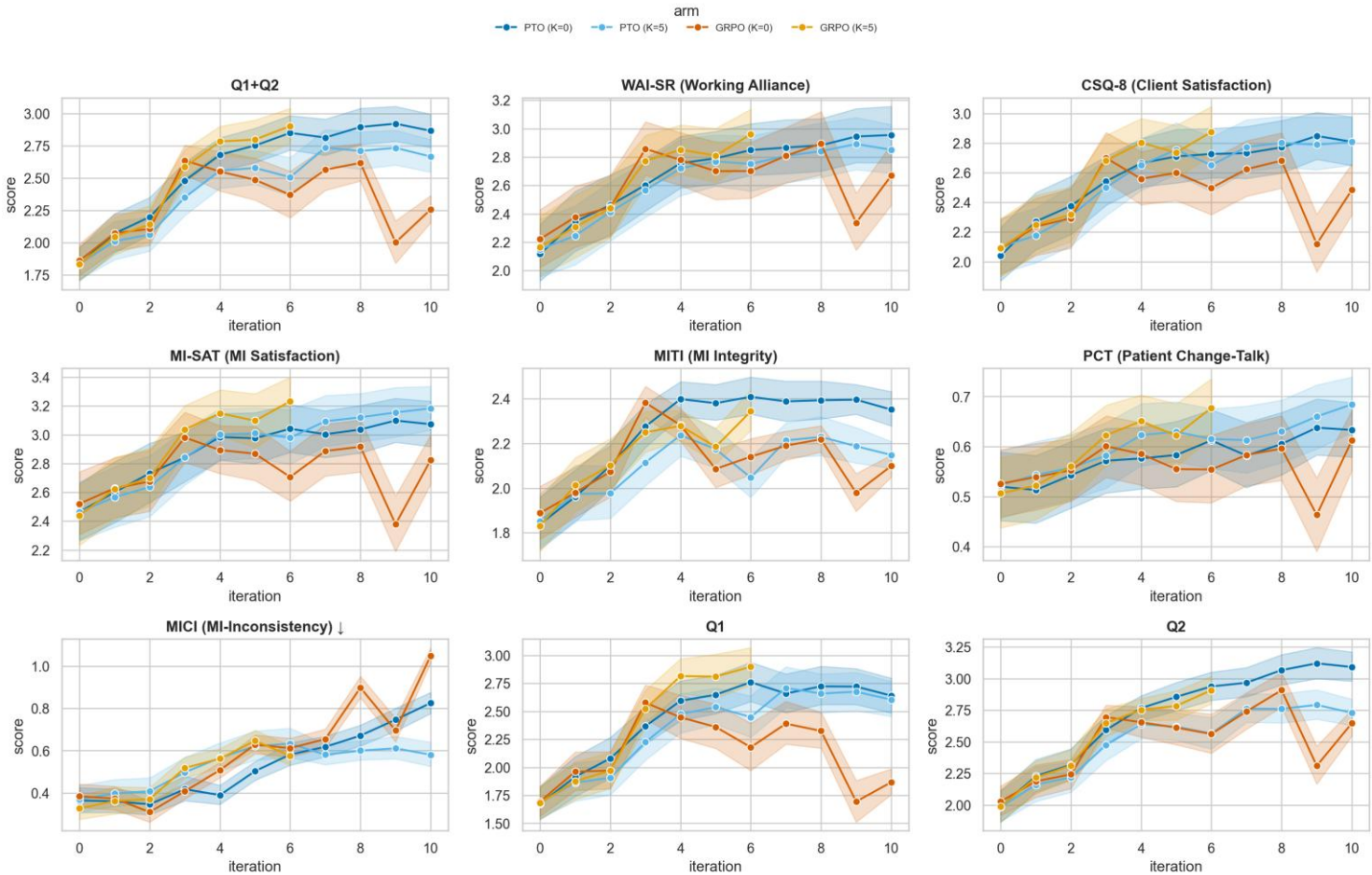
Four arms. Colour alone carries the arm here — dark blue / light blue = PTO K=0 / K=5, orange / gold = GRPO K=0 / K=5. All four lines are solid with circle markers on this figure. GRPO K=5's line ends at iteration 6, its last scored state.

DATA

Grader: gpt-4o-mini, which was also the training reward. 96 personas, one conversation each. Contrasts are paired on persona identity.

All nine rubrics by iteration — held-out judge

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor, PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

X AXIS

Training iteration, 0–10.

Y AXIS

Same rubrics, same arms as the previous slide, but the y scales differ. The held-out judge's offset is RUBRIC-DEPENDENT: Q1, Q2 and MITI about 1.3–1.7 points lower; WAI-SR 0.6, MI-SAT 0.45, CSQ-8 0.16; on MICI and PCT it scores slightly higher.

SERIES

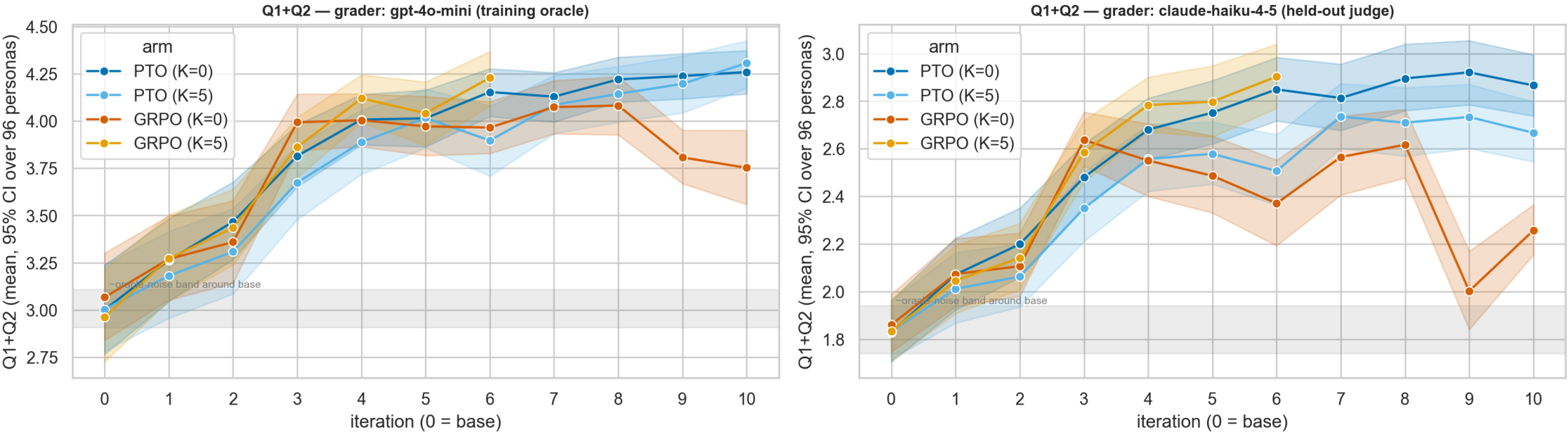
Four arms. Colour alone carries the arm here — dark blue / light blue = PTO K=0 / K=5, orange / gold = GRPO K=0 / K=5. All four lines are solid with circle markers on this figure. GRPO K=5's line ends at iteration 6, its last scored state.

DATA

Grader: Claude Haiku 4.5, held out. Same 96 conversations per state as the previous slide — only the grader changes.

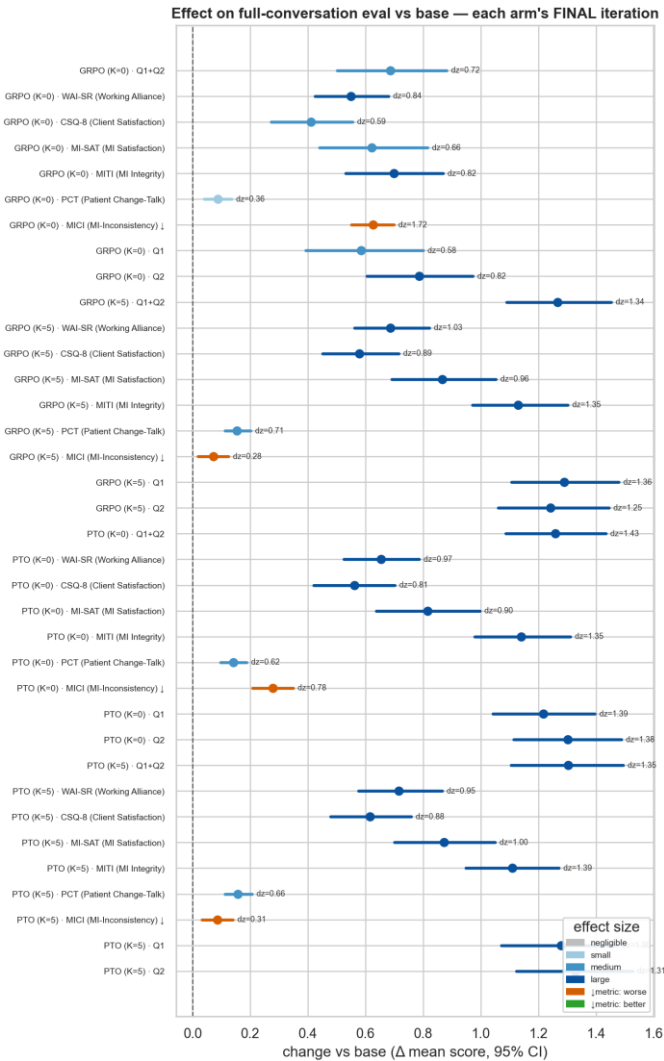
The training-reward rubric alone, both graders

RQ-i in one frame: both look-ahead arms of both methods (K=5 lines stop at their last scored iteration)



X AXIS Training iteration, 0–10.	Y AXIS Q1+Q2, mean with 95% CI. Left = gpt-4o-mini, right = Claude Haiku 4.5, on SEPARATE scales. Grey band = ±0.10 around the pooled base, the measured oracle repeatability.	SERIES Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state.	DATA 96 personas, one conversation each. Contrasts are paired on persona identity. Where a K=5 line stops, everything further right is that arm's K=0 sibling only.
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Each arm's endpoint against its own base — training oracle



X AXIS

Change vs the arm's own untrained base draw (Δ mean score) with 95% bootstrap CI, at the arm's FINAL iteration. Cohen's dz is annotated per row as text and has no interval drawn.

Y AXIS

Rubric $\times$ arm.

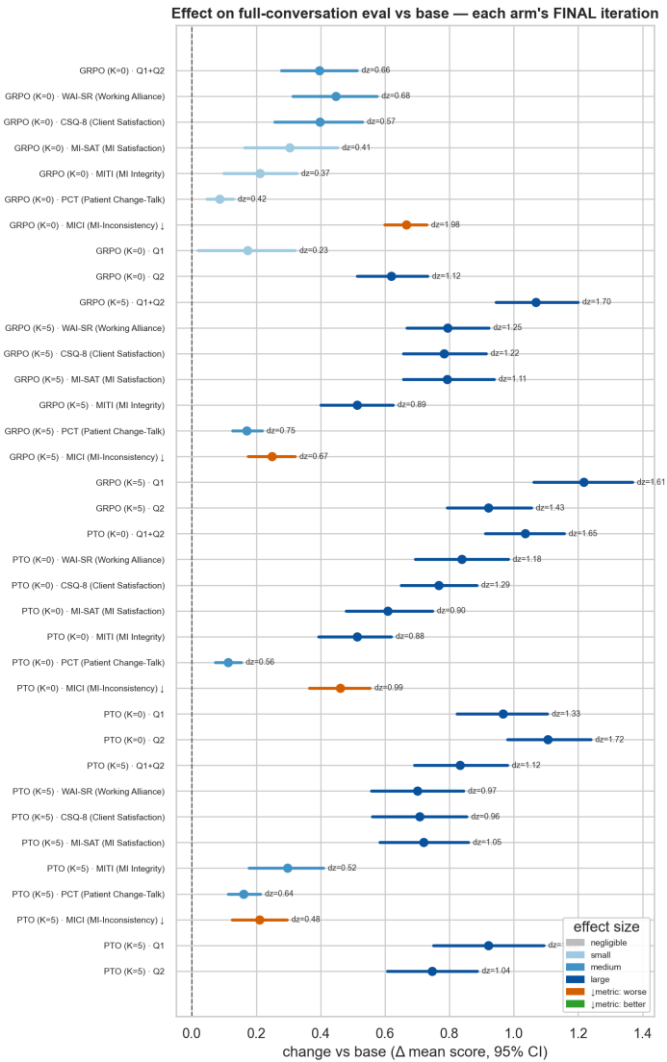
SERIES

One row per (arm, rubric). Lower-is-better rubrics are flagged in the row label.

DATA

Grader: gpt-4o-mini. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

Each arm's endpoint against its own base — held-out judge



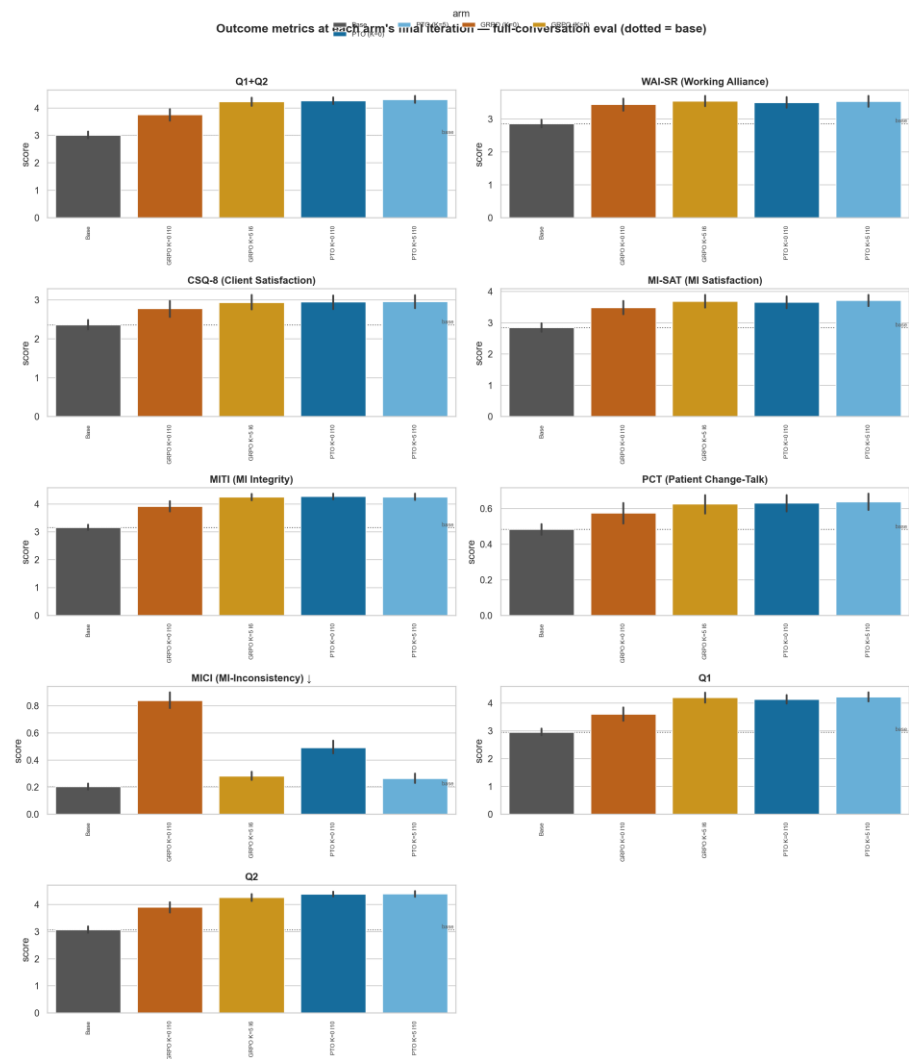
X AXIS
Change vs own base (Δ mean score) with 95% bootstrap CI, at the final iteration; dz annotated per row as text.

Y AXIS
Rubric $\times$ arm, same layout as the previous slide.

SERIES
One row per (arm, rubric).

DATA
Grader: Claude Haiku 4.5. Same conversations as the previous slide. Persona-paired, $n = 96$. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

Arm by arm across every rubric, at the final iteration



source: arms/outcomes/figures/gpt-4o-mini/outcomes_by_model_final.png

X AXIS
Model state.

Y AXIS
Score on each rubric, panel by panel.

SERIES
One bar or point per arm; base included.

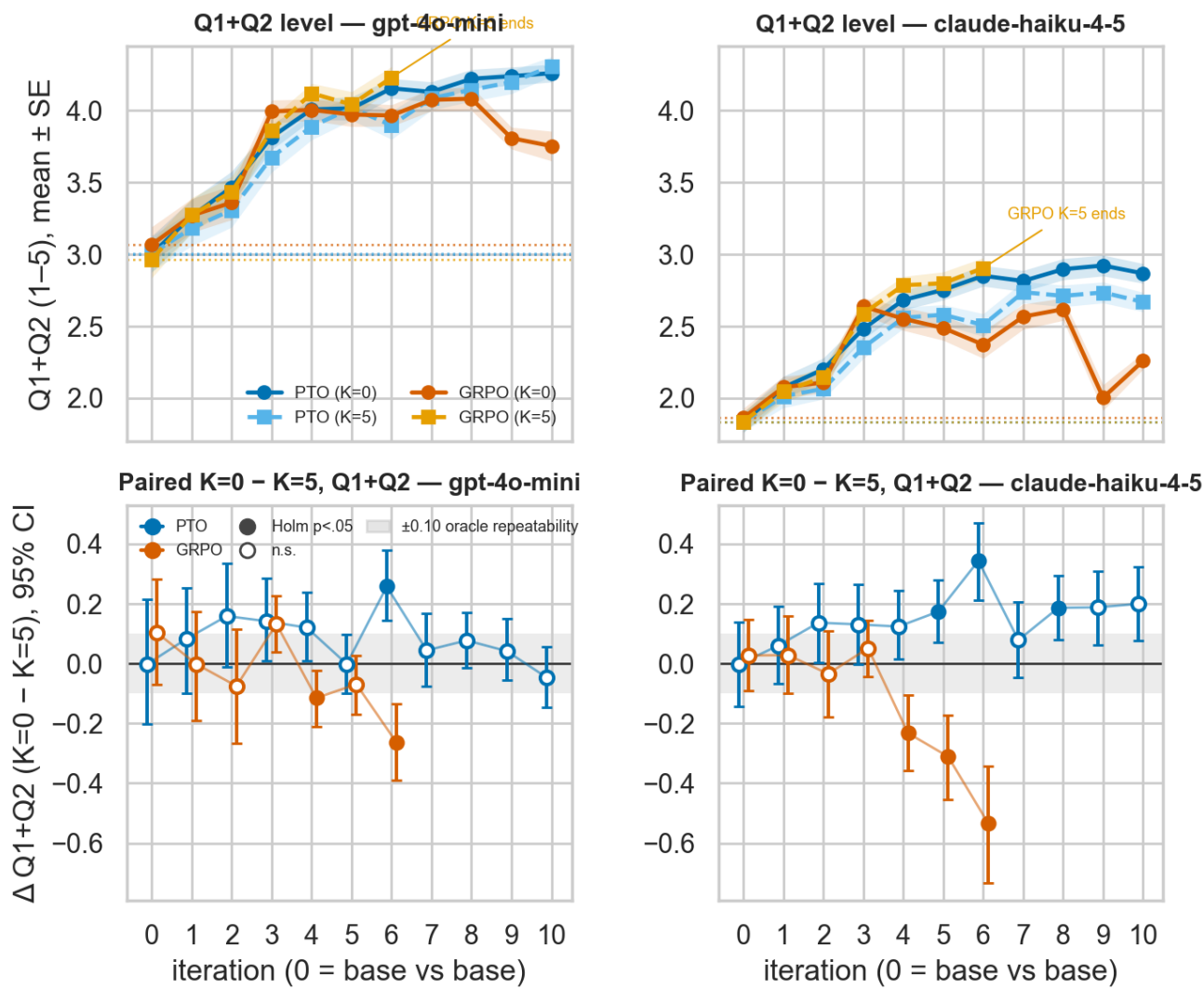
DATA
Grader: gpt-4o-mini, final iteration of each arm.
96 personas, one conversation each. Contrasts are paired on persona identity.

EXP3 · LOOK-AHEAD

Exp3 — the look-ahead contrast

$K = 0$ vs $K = 5$ within each optimizer, on the same 96 personas, under both graders. Sign convention throughout: positive = $K=0$ higher.

Level and paired difference in one frame



source: lookahead/reward/figures/k_headline_q1q2.png

TOP ROW

Y = Q1+Q2 level, mean ± SE. Each arm's own base is the dotted line. Left = training oracle, right = held-out judge, shared row scale.

BOTTOM ROW

Y = paired (K=0 - K=5) on Q1+Q2, 95% CI. Positive = K=0 higher. Grey band = ±0.10 oracle repeatability. Filled = Holm p < .05 across iterations; hollow = not.

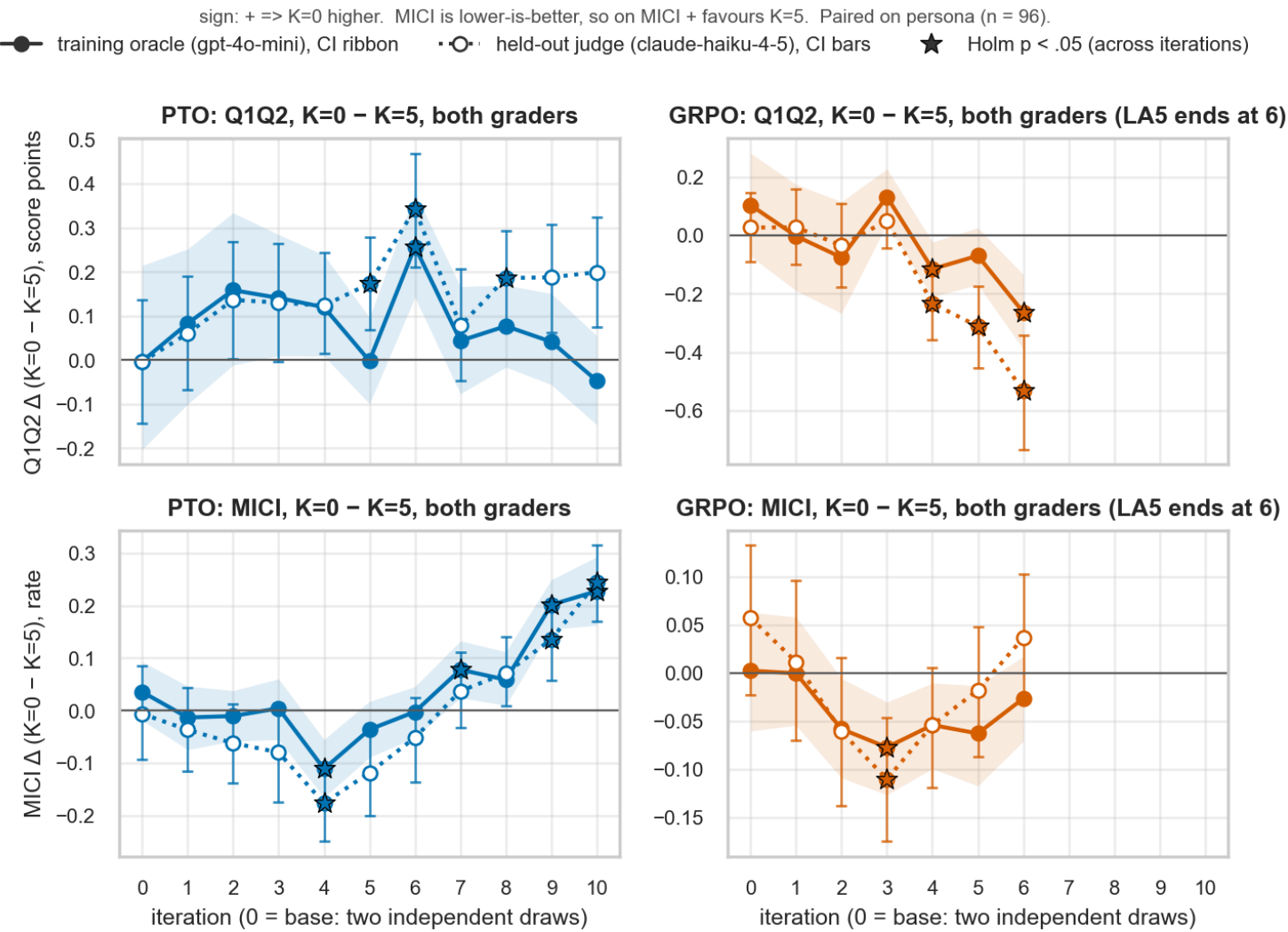
X AXIS

Training iteration. 0 = two independent draws of the same untrained base — the noise floor of this design.

DATA

Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

The contrast on the reward rubric and the harm rubric



X AXIS

Training iteration.

Y AXIS

Paired K=0 – K=5. Rows are Q1+Q2 and MICI; columns are PTO and GRPO.

SERIES

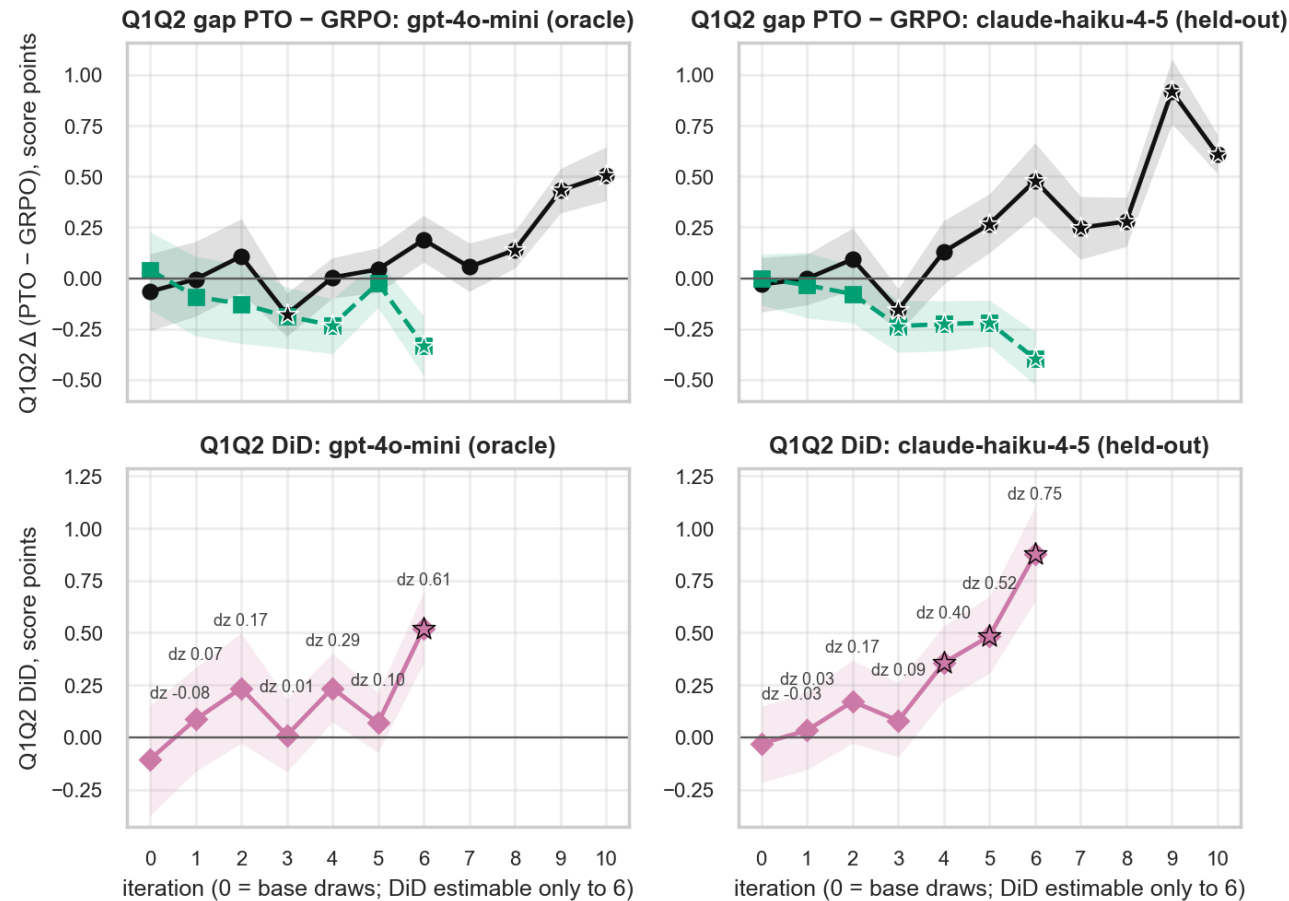
Solid line + filled circle + ribbon = training oracle; dotted line + open circle + bars = held-out judge. A star replaces the marker on WHICHEVER grader's series is significant, so a star on the solid line is the training oracle and on the dotted line the held-out judge.

SIGN

Positive = K=0 higher. MICI is lower-is-better, so on the MICI row a positive value means K=0 was more MI-inconsistent. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

The method gap at each K, and the difference between them

signs: gap + => PTO higher; DiD + => PTO's lead over GRPO is larger at K=0 than at K=5. Ribbons = persona-bootstrap 95% CI, n = 96 personas.
● K=0: PTO_LA0 – GRPO_LA0 ■ K=5: PTO_LA5 – GRPO_LA5 (to iter 6) ◆ DiD = gap(K=0) – gap(K=5) ★ Holm p < .05 (across iterations)



TOP ROW

Y = PTO minus GRPO on Q1+Q2 at a fixed K. Black solid = K=0, green dashed = K=5. Ribbons = persona-bootstrap 95% CI. Positive = PTO higher.

BOTTOM ROW

Y = difference-in-differences: the K=0 gap minus the K=5 gap, computed per persona. dz annotated at each iteration.

COLUMNS

Left = training oracle, right = held-out judge. Row y-limits shared.

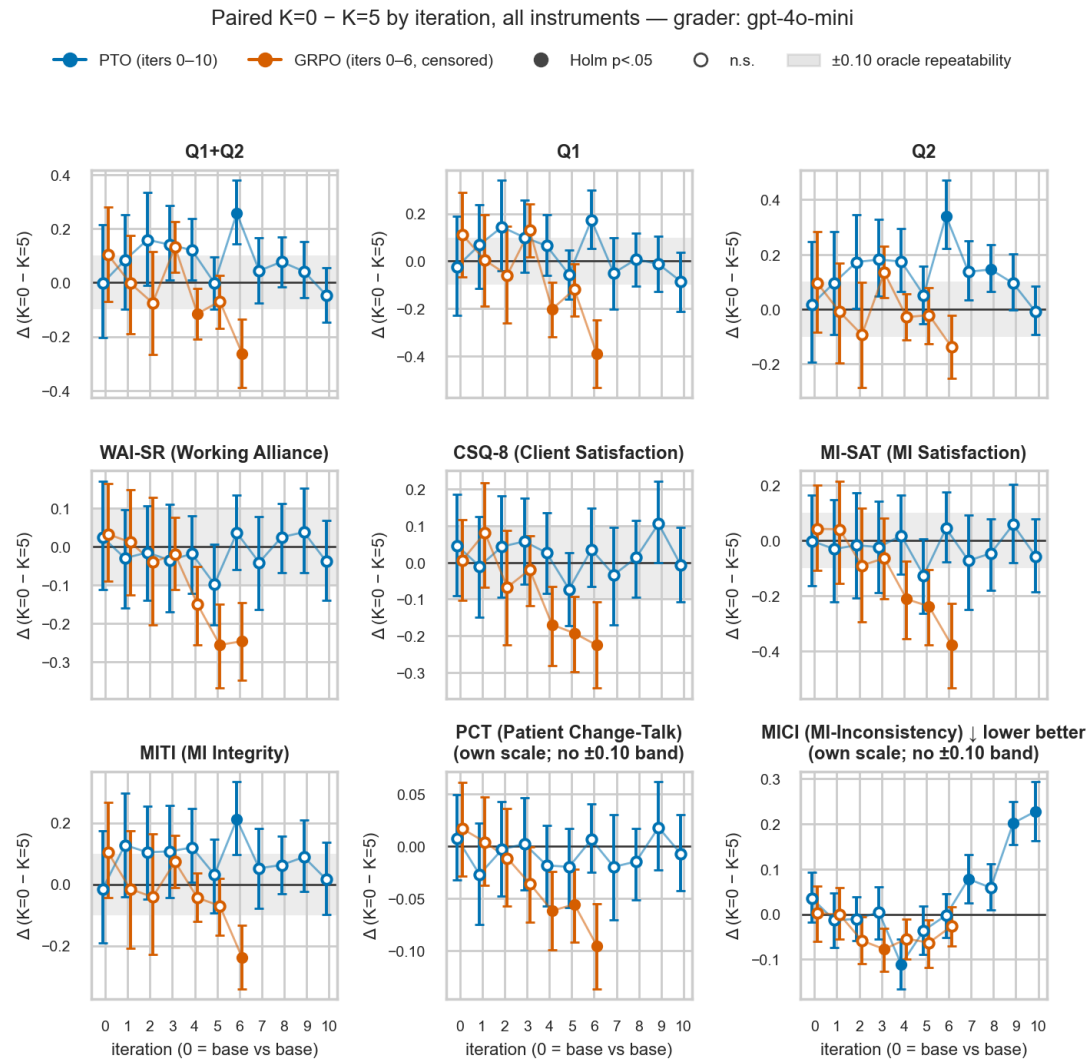
STARS

Holm p < .05, family = iterations within (grader, K). That is a DIFFERENT family from the method-gap slide later on, where Holm runs across the rubrics inside one (grader, K, iteration) — so the same series can carry stars at different iterations on the two slides.

DATA

The DiD needs all four arms, so it is defined while GRPO K=5 runs. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

The contrast on every rubric — training oracle



X AXIS

Training iteration, in each of the nine panels.

Y AXIS

Paired K=0 – K=5 on that panel's rubric, 95% CI whiskers.

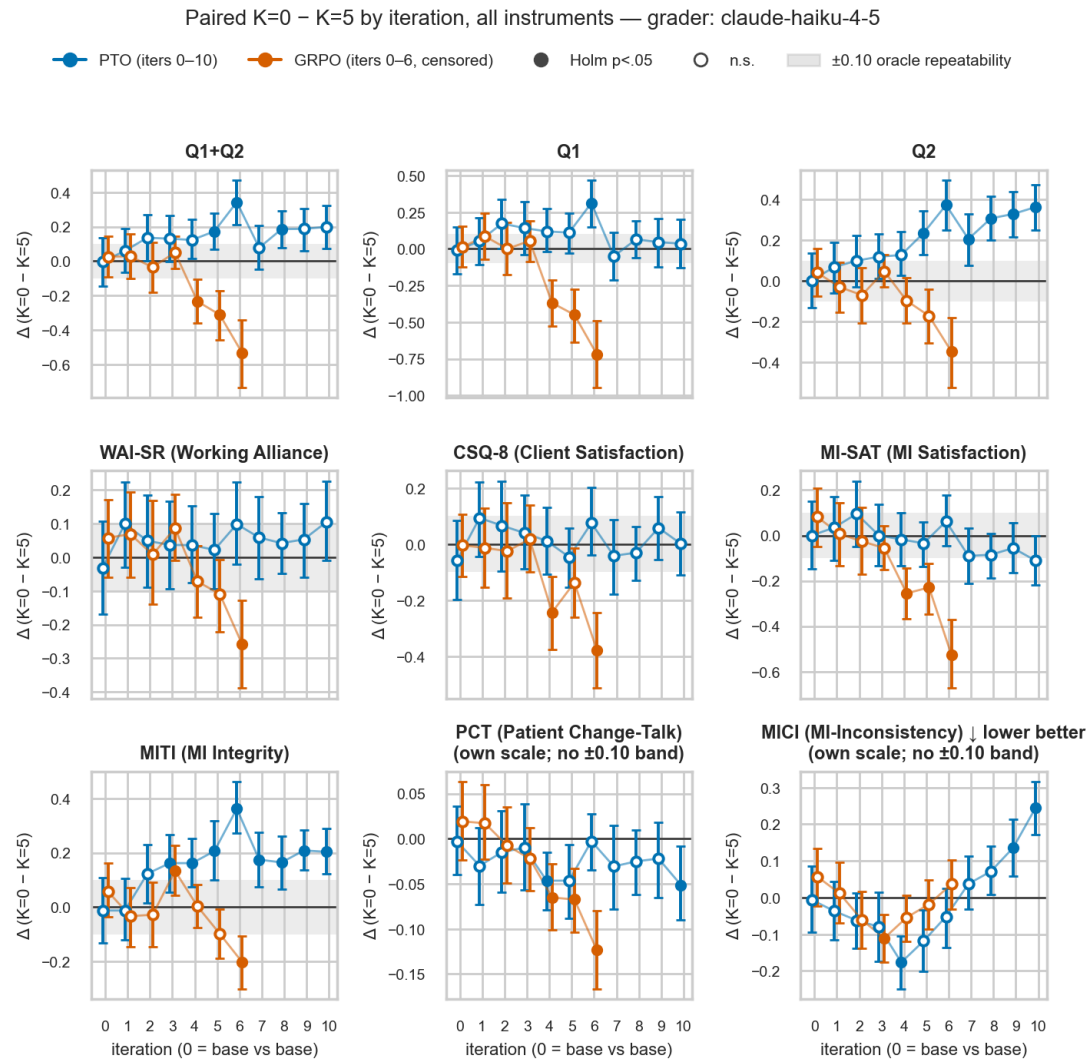
SERIES

PTO points shifted left, GRPO shifted right. Filled = Holm p < .05 within (grader, method, rubric); hollow = not.

SCALES

Point rubrics carry the ±0.10 repeatability band; PCT and MICI are 0–1 rates and get a bare zero line. Lower-is-better rubrics are flagged.

The contrast on every rubric — held-out judge



X AXIS

Training iteration.

Y AXIS

Paired K=0 – K=5, same nine rubrics and same layout as the previous slide.

SERIES

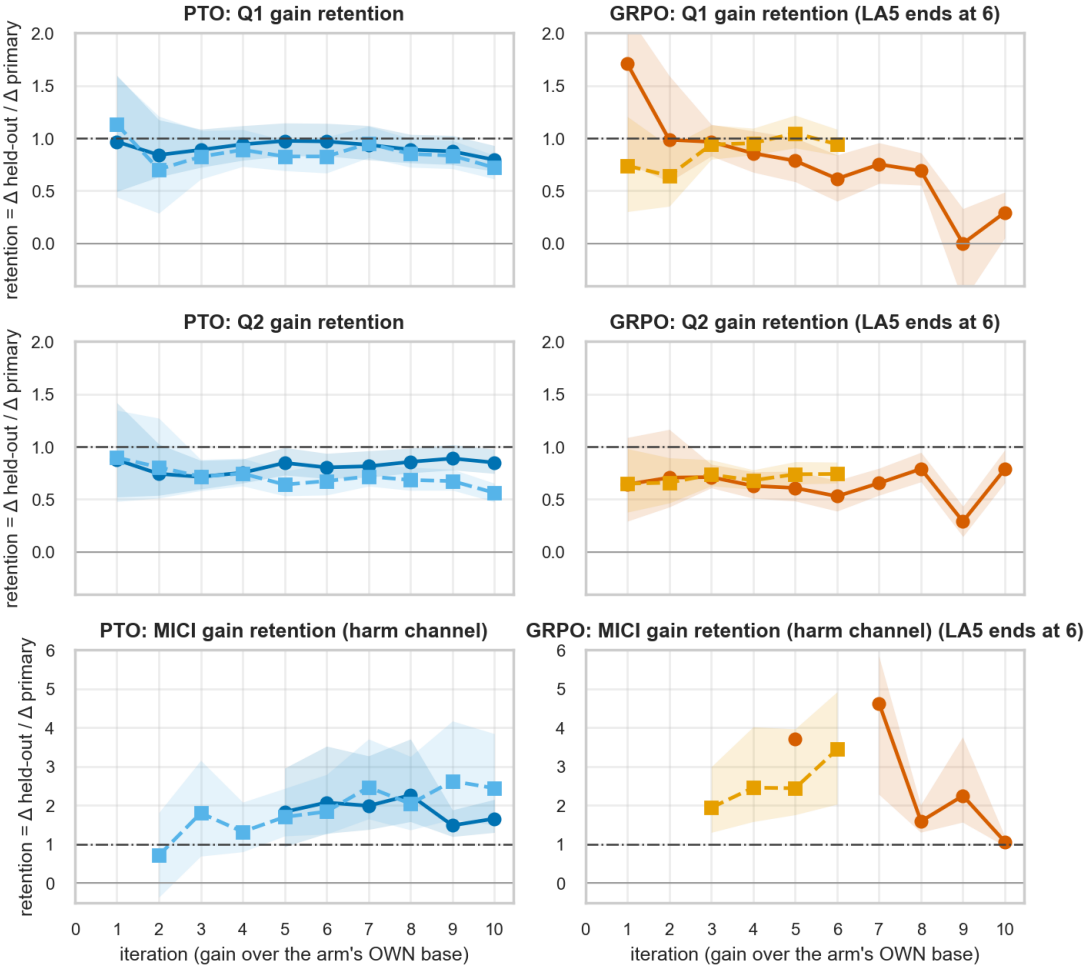
PTO shifted left, GRPO shifted right. Filled = Holm p < .05.

DATA

Grader: Claude Haiku 4.5, on the same conversations. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

How much of each gain the held-out judge also sees

Δ held-out = Claude Haiku 4.5 gain over base; Δ primary = gpt-4o-mini (training oracle) gain over base; ribbons = persona-bootstrap 95% CI; blank where $|\Delta$ primary| < 0.15 (point-scale rubrics) or < 0.05 (PCT / MICI)
● PTO K=0 ■ PTO K=5 ● GRPO K=0 ■ GRPO K=5 -.- retention = 1 (gain fully real to the held-out judge)



X AXIS

Training iteration.

Y AXIS

Retention = (gain measured by the held-out judge) ÷ (gain measured by the training oracle), each against that arm's own base. Dash-dot line at 1.0.

SERIES

Rows = Q1, Q2, MICI. Columns = PTO, GRPO. K=0 solid circle, K=5 dashed square, ribbons = persona-bootstrap 95% CI.

BLANKS

The curve is blanked where the training-oracle gain is under 0.15 points (0.05 on MICI), because the ratio is unstable there. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

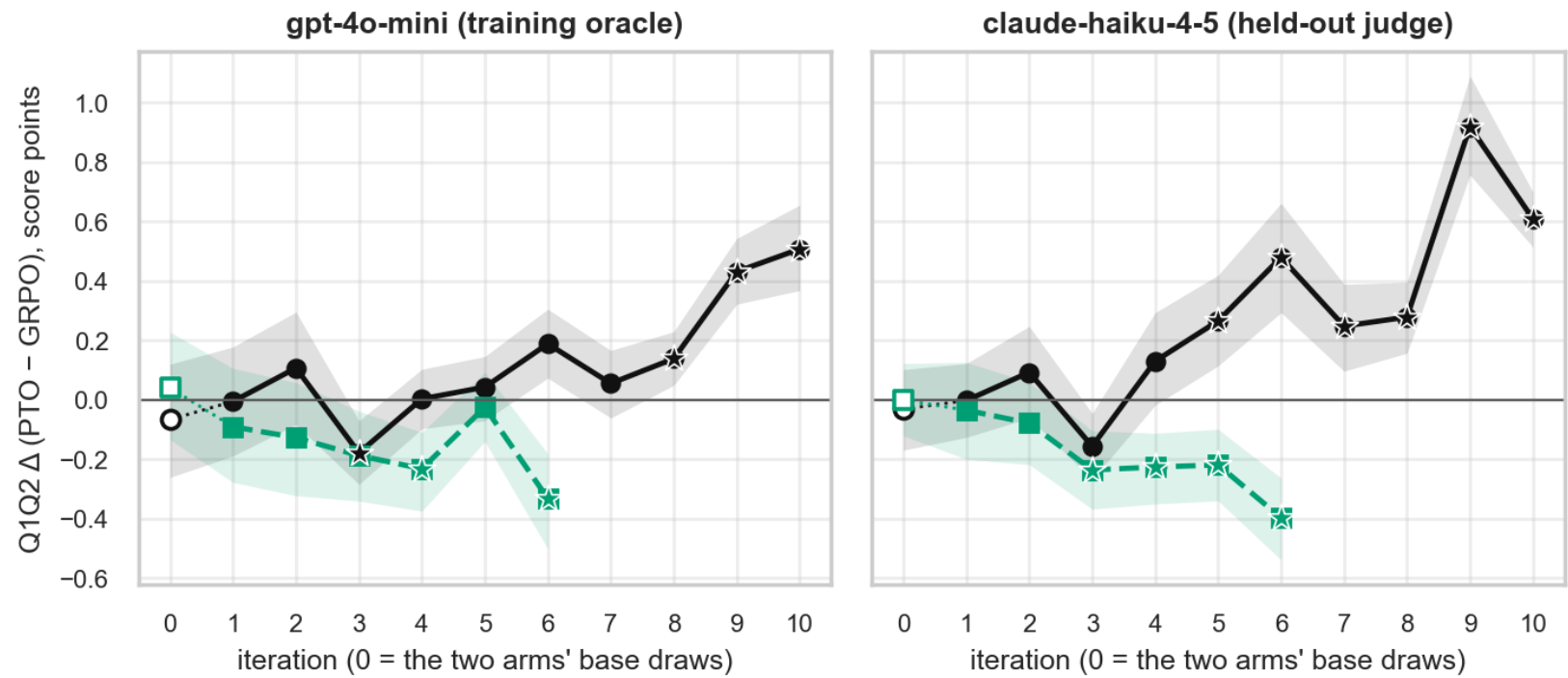
EXP3 · METHOD

Exp3 — PTO against GRPO

The two optimizers at matched K and matched iteration, and then on the compute axis.

The method gap per iteration, at each K

Q1Q2 gap PTO – GRPO by iteration, one panel per grader. + => PTO higher. Ribbons = persona-bootstrap 95% CI, n = 96 personas.
● K=0: PTO_LA0 – GRPO_LA0 ■ K=5: PTO_LA5 – GRPO_LA5 (to iter 6, GRPO_LA5 censored) ★ Holm p < .05 (across rubrics within the (judge, K, iteration) contrast)



X AXIS Training iteration.	Y AXIS PTO minus GRPO on Q1+Q2. Positive = PTO higher. Ribbons = persona-bootstrap 95% CI over the 96 shared personas.	SERIES Black solid circle = K=0, green dashed square = K=5. Stars = cleared Holm across rubrics within that grader / K / iteration.	NOTE Iteration 0 (hollow) is the two arms' independent base draws of the same untrained model — the noise floor. Matched on iteration, not on budget.
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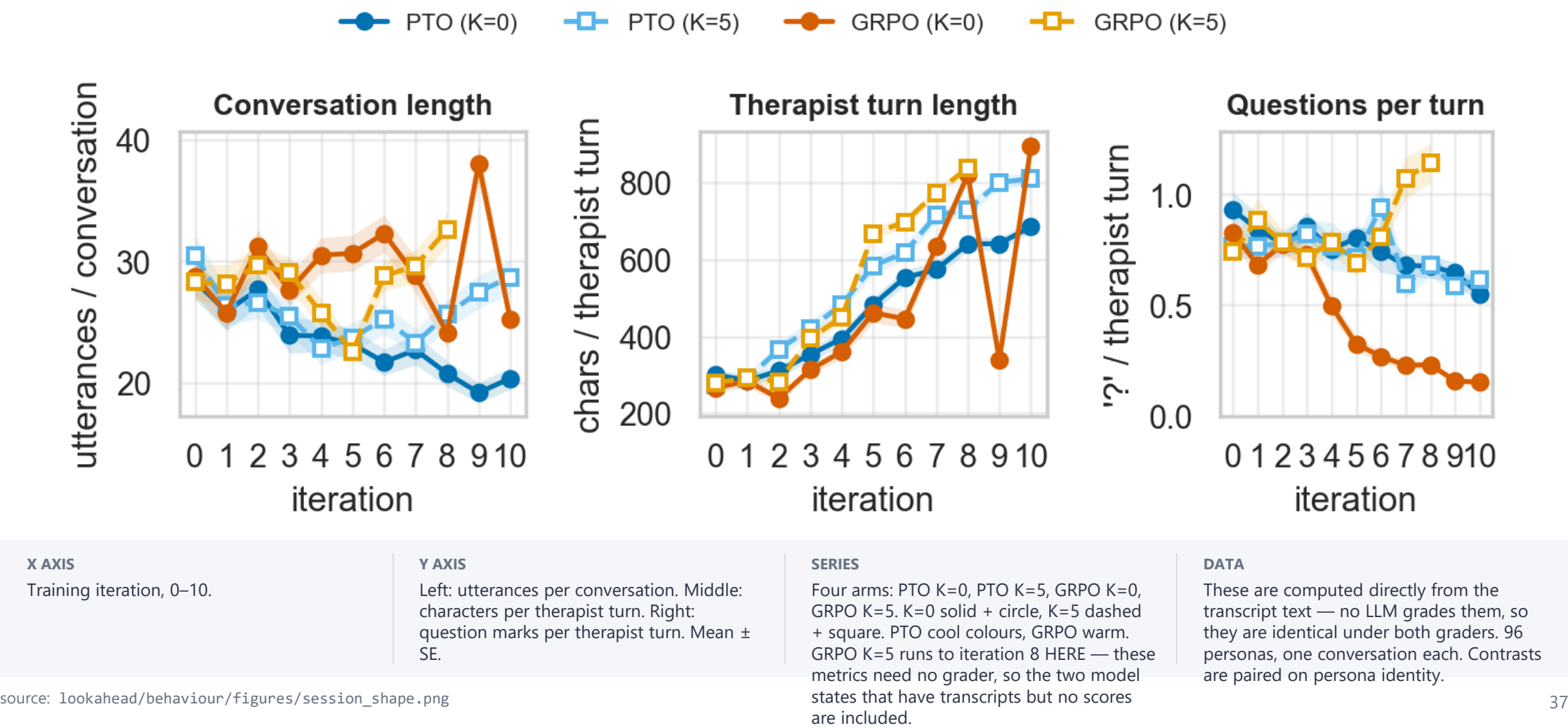
EXP3 · BEHAVIOUR

Exp3 — what the therapist actually does

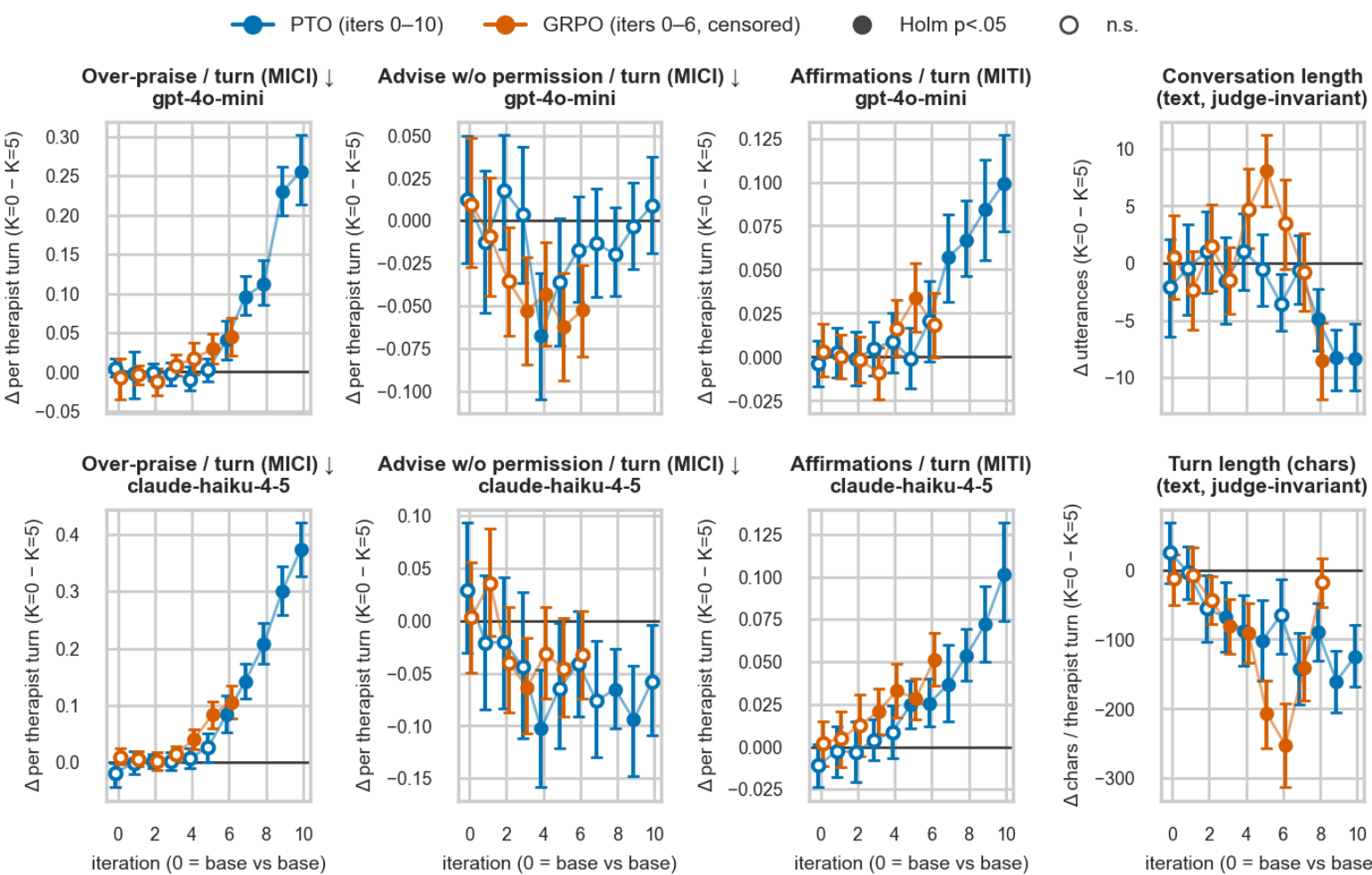
Coded behaviour channels and deterministic text metrics, rather than rubric totals.

Session shape — no grader involved

Session shape by iteration — deterministic text metrics (mean ± SE over 96 personas; grader-free)

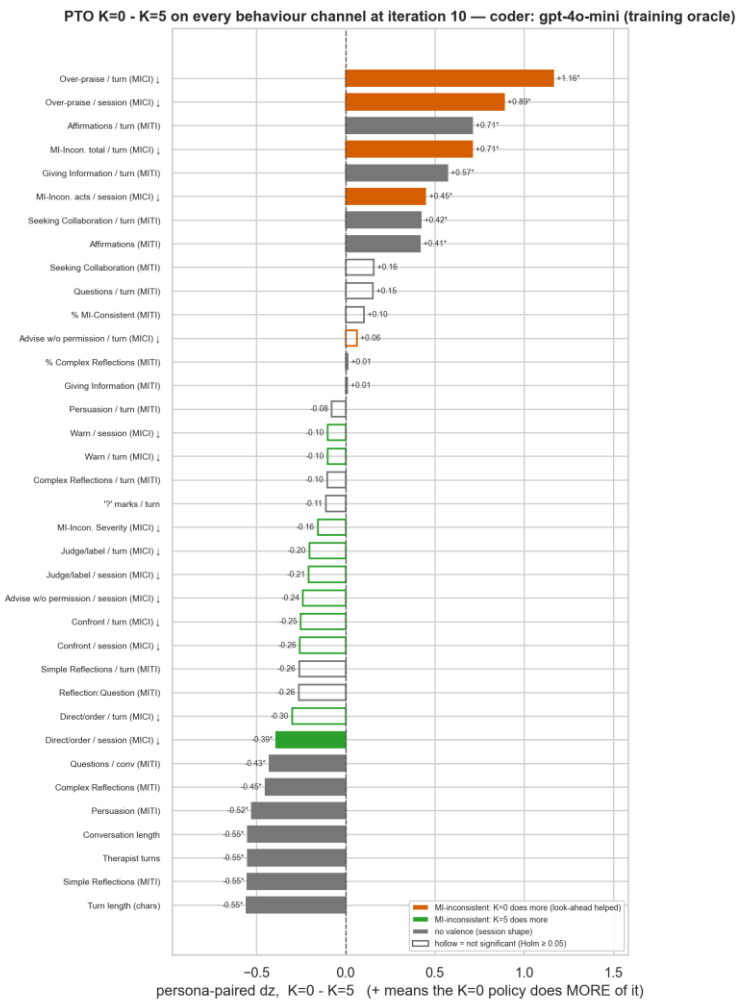


The K contrast on coded behaviour channels



source: lookahead/behaviour/figures/k_channels_grid.png

Every channel at once, at PTO's last iteration — training oracle



Sign convention + = the K=0 policy does MORE of the channel. Colour is VALENCE, not direction: red = a MI-inconsistent behaviour K=0 does more of, green = one K=5 does more of, grey = unvalenced session shape. Red AND green bars both being large would mean the violations were swapped, not removed. Hollow = did not clear Holm within its (iteration, family) set.

X AXIS

Paired dz of K=0 – K=5 at iteration 10. Positive = the K=0 policy does more of it.

Y AXIS

One row per behaviour channel.

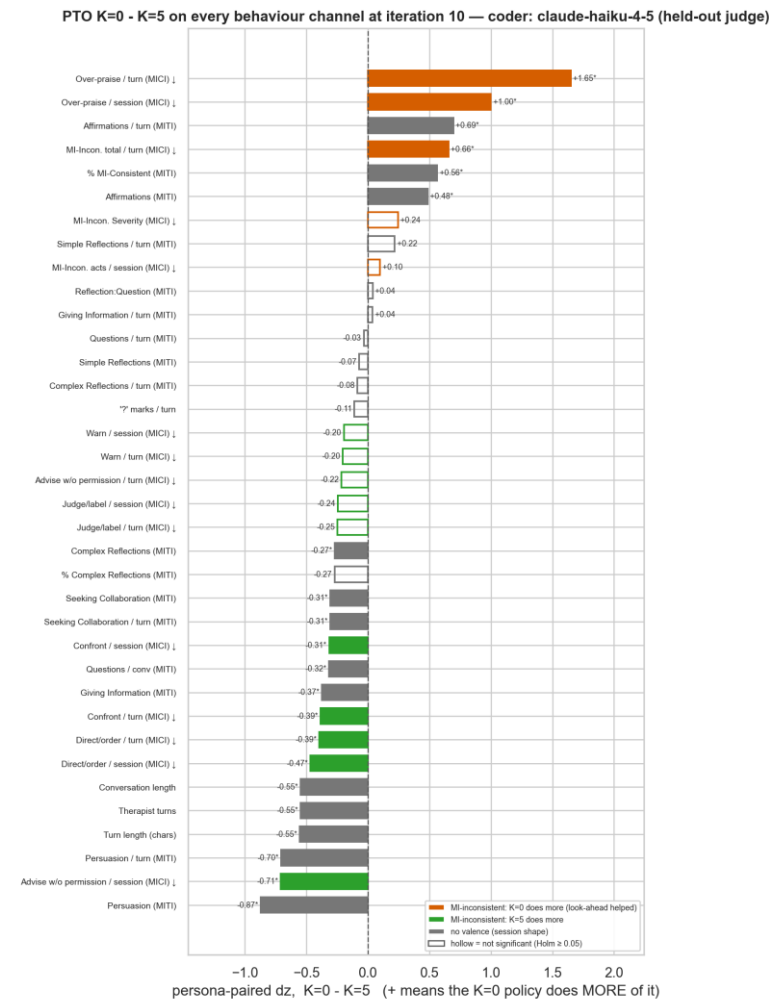
COLOUR

Red = MI-inconsistent and higher under K=0;
green = MI-inconsistent and higher under K=5;
grey = unvalenced session shape.

MARKERS

Hollow = did not clear Holm within its (iteration, family) set. Coder: gpt-4o-mini. n = 96.

Every channel at once, at PTO's last iteration — held-out judge



X AXIS

Paired dz of K=0 – K=5 at iteration 10.

Y AXIS

Same channels, same layout as the previous slide.

COLOUR

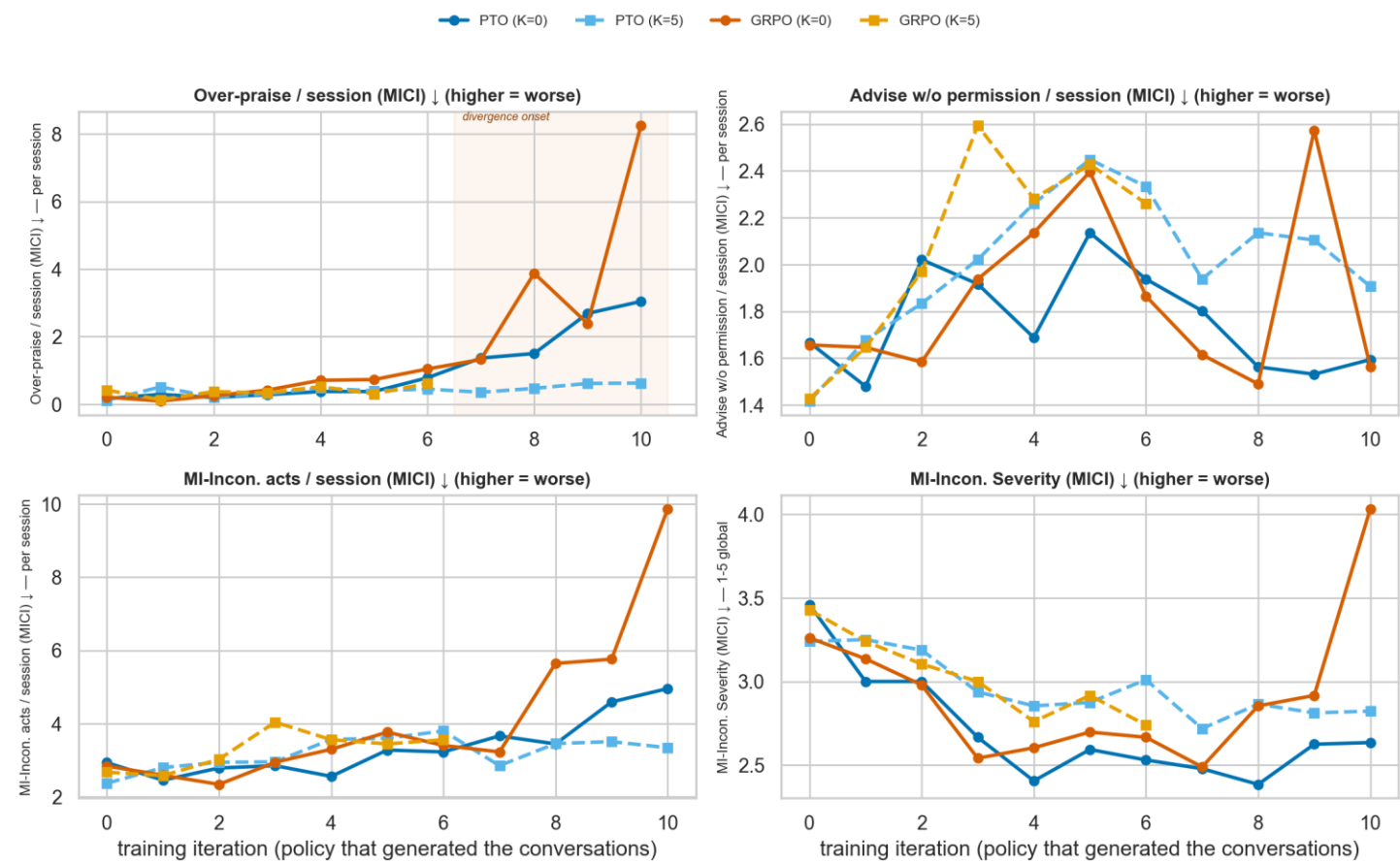
Same valence encoding. Hollow = did not clear Holm.

DATA

Coder: Claude Haiku 4.5, on the same 96 conversations.

MI-inconsistent acts, counted per session

Does look-ahead reduce MI-inconsistency, or recompose it? (K=0 solid · K=5 dashed) — coder: gpt-4o-mini (training oracle)



X AXIS

Training iteration.

Y AXIS

Counts PER SESSION, so no denominator is involved. Panels: over-praise, advice without permission, the TOTAL number of MI-inconsistent acts, and the severity global.

SERIES

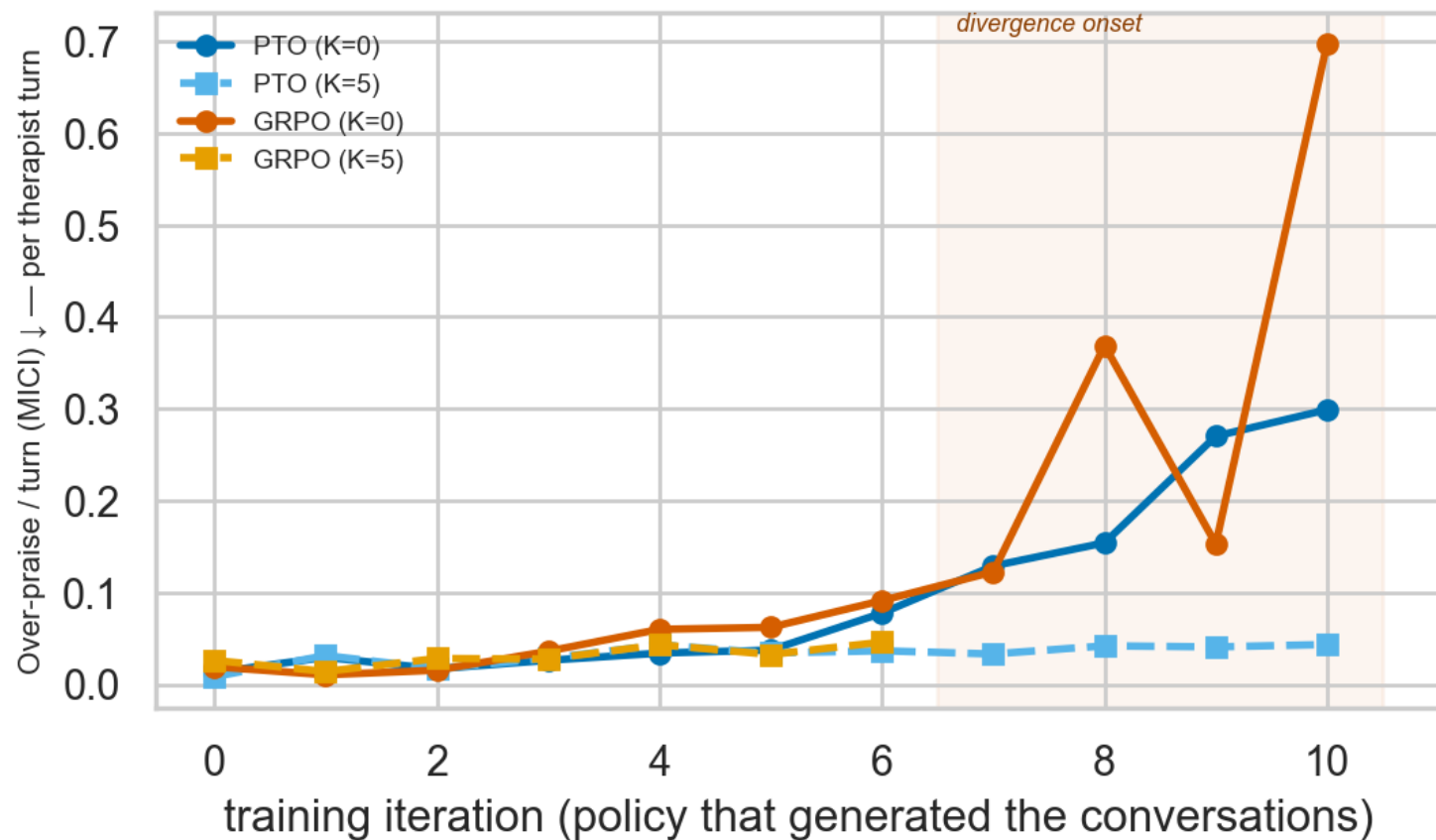
Arm means over 96 personas. Higher is worse on every panel.

WHY PER SESSION

The per-turn rate has a moving denominator — the arms differ in turn count. The total panel is the aggregate control. Coder: gpt-4o-mini.

The over-praise channel per therapist turn

The over-praise channel, per policy iteration — coder: gpt-4o-mini (training oracle)



X AXIS

Training iteration.

Y AXIS

Over-praise instances per therapist turn. Higher is worse.

SERIES

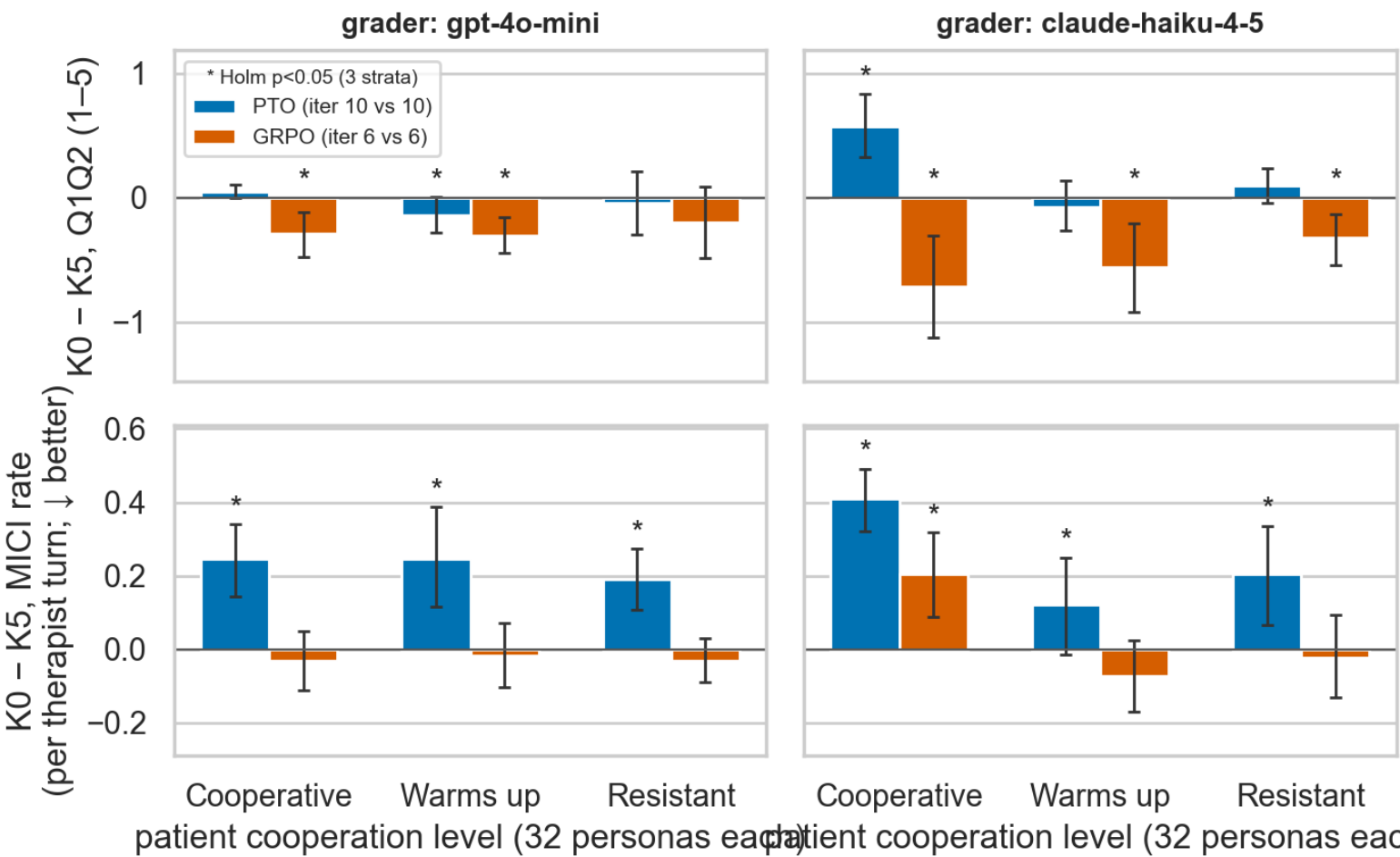
All four arms; K=0 solid, K=5 dashed. Shading marks the first iteration at which the PTO K contrast cleared Holm within its family.

CAVEAT

Per-turn rate — the denominator moves between arms. Read against the per-session panel two slides back. Coder: gpt-4o-mini.

The contrast split by how cooperative the patient is

Look-ahead contrast by patient cooperation (persona-paired; + = K=0 higher)



X AXIS
Patient stratum: Cooperative / Warms up / Resistant, 32 personas each.

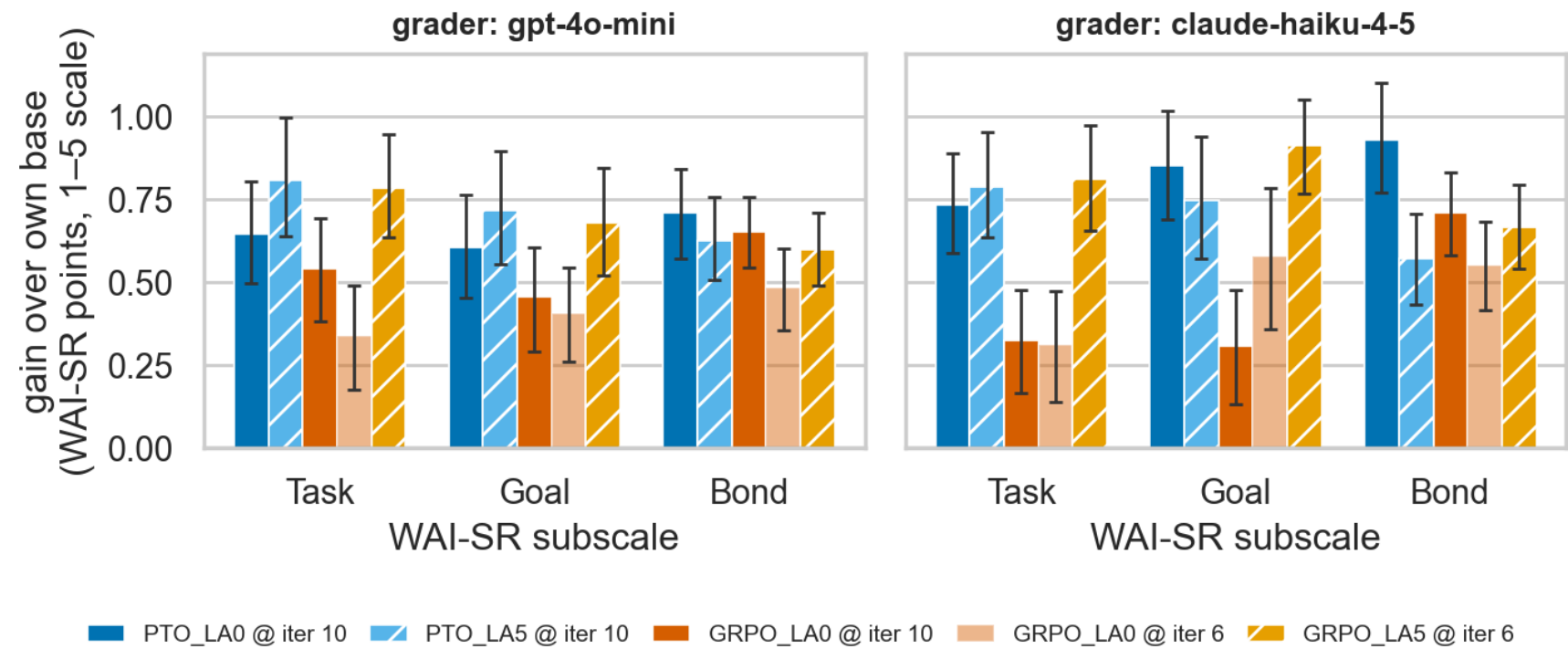
Y AXIS
Paired K=0 – K=5 at each method's matched endpoint. Rows: Q1+Q2 (higher better) and MICI (lower better, so positive = K=0 worse).

SERIES
PTO and GRPO bars, coloured by the method. Whiskers = 95% bootstrap CI. Star = Holm p < .05 across the three strata.

COLUMNS
Left = training oracle, right = held-out judge.

WAI-SR subscale gains at each arm's endpoint

WAI-SR subscale gain at the endpoint (persona-paired vs own base, 95% bootstrap CI)



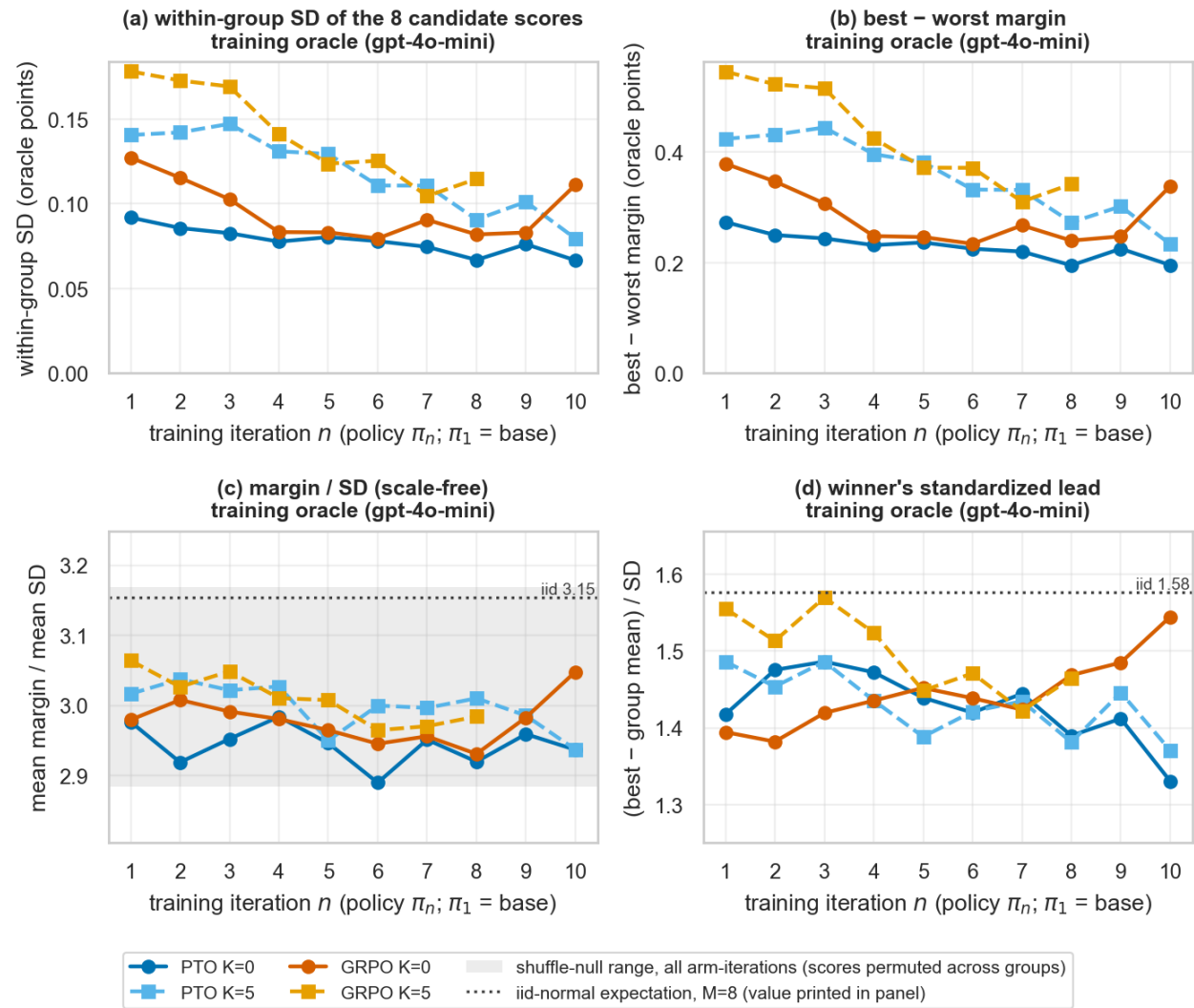
X AXIS WAI-SR subscale: Bond, Goal, Task.	Y AXIS Gain over that arm's OWN base, paired on persona, 95% bootstrap CI.	SERIES One bar per arm; K=5 bars hatched. Left panel = training oracle, right = held-out judge.	NOTE This is an arm-vs-own-base figure, NOT a K contrast. n = 96 paired.
---	--	---	--

EXP3 · MECHANISM

Exp3 — inside the training signal

These read the trainer's own per-candidate records: what the oracle scored, what it selected, and how faithful a short-prefix score is to the finished conversation.

Dispersion of the reward within each sampled group



X AXIS

Training iteration.

Y AXIS

(a) within-group SD of the 8 candidate scores;
(b) best-minus-worst margin; (c) margin ÷ SD; (d)
the winner's standardized lead.

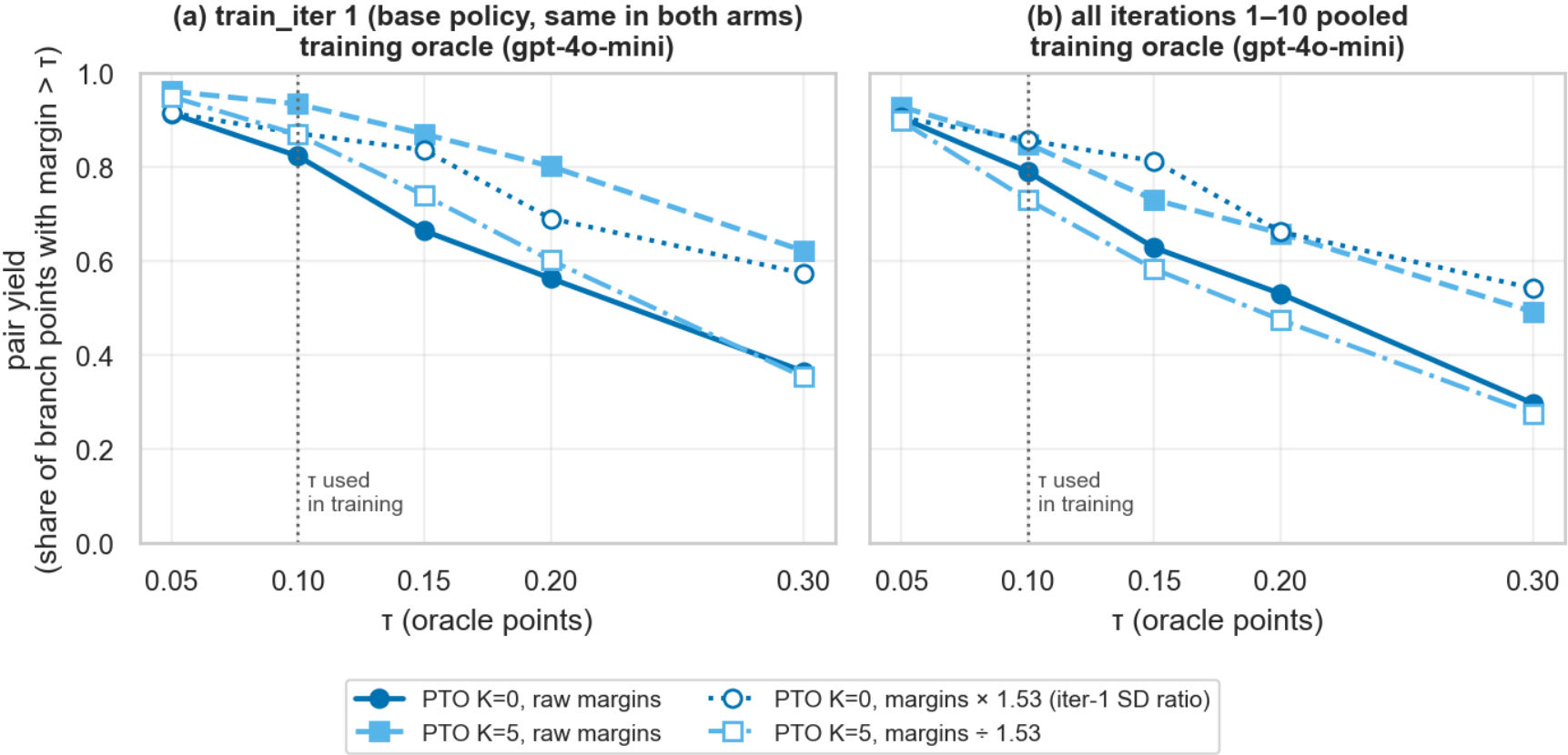
REFERENCES

Panel (c) has the iid-normal expectation 3.15
dotted and a shuffle null shaded; panel (d) has
1.58 dotted.

UNIT

One GROUP = the 8 candidates sampled at one
branch point, keyed (arm, iteration,
conversation_id, branch_id). Grader: gpt-4o-mini.

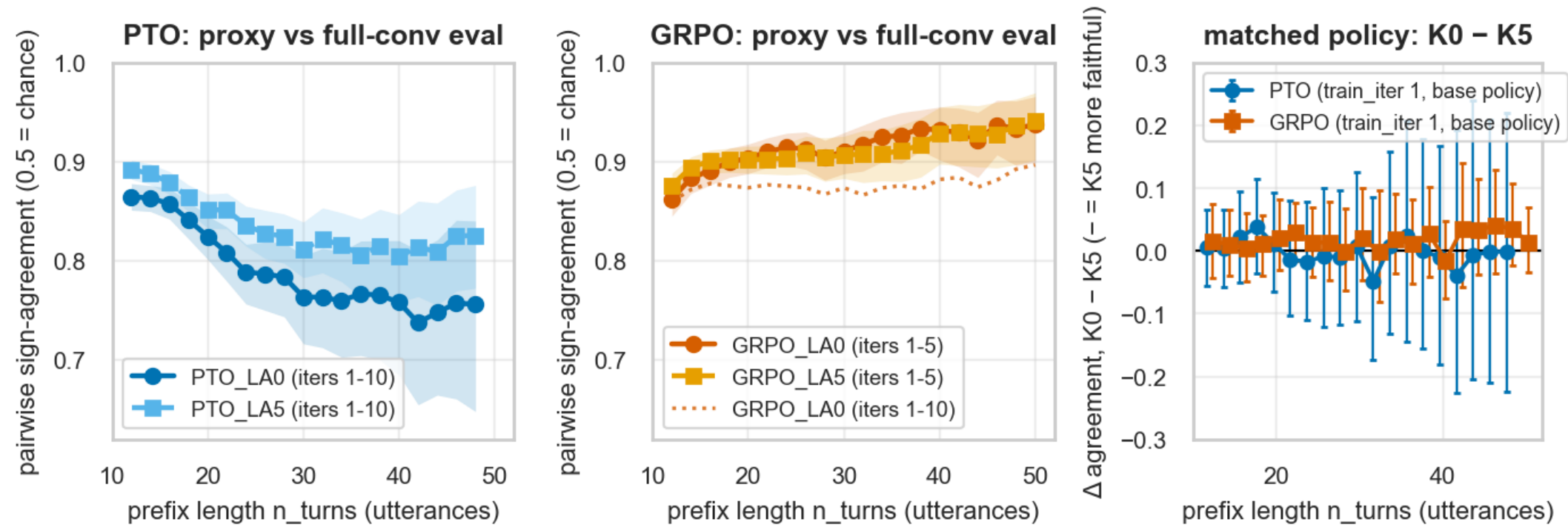
How many preference pairs survive the τ filter



X AXIS	Y AXIS	SERIES	OPEN MARKERS
τ , the minimum best-worst margin required to emit a preference pair.	Share of branch points that yield a pair at that τ .	(a) iteration 1 only, where both arms share the same base policy; (b) iterations 1–10 pooled. Solid circle = K=0, dashed square = K=5.	The same curves after rescaling by the iteration-1 within-group SD ratio (1.53). Vertical dotted line = $\tau = 0.1$, the value used.

Does a short-prefix score predict the finished conversation?

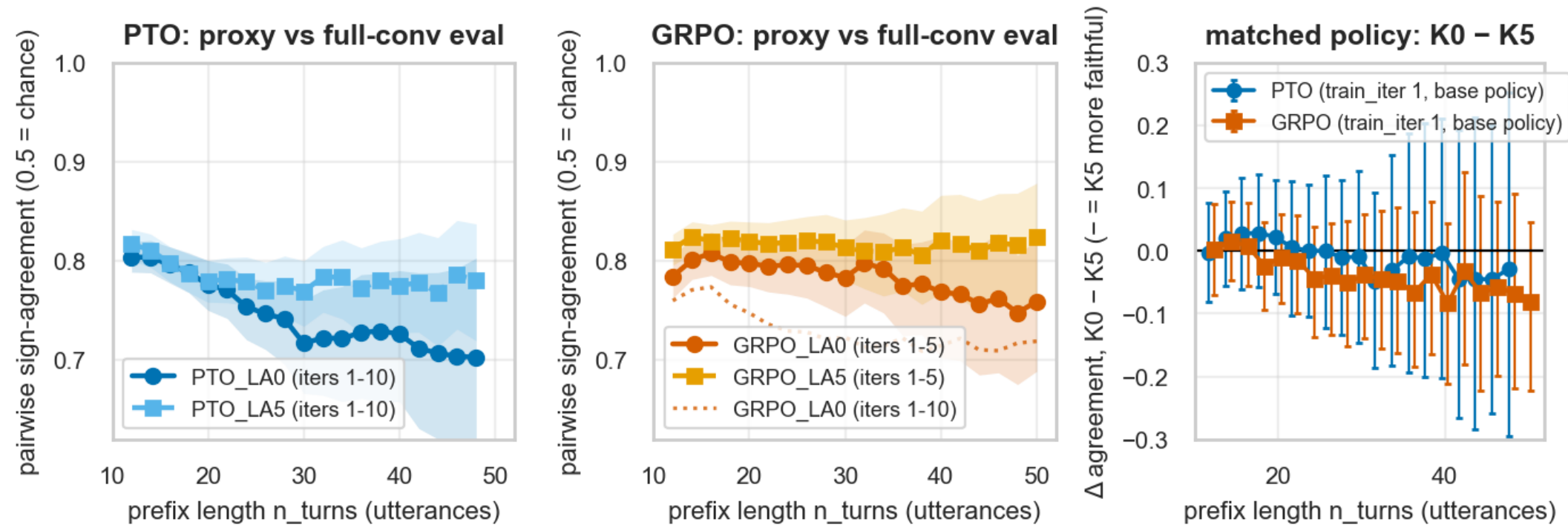
Training-reward faithfulness — eval graded by the training oracle (gpt-4o-mini); proxy = training oracle



X AXIS	Y AXIS	SERIES	DEFECT
Prefix length in turns — how much of the conversation the training reward saw.	Pairwise rank agreement between that short-prefix score and the full-conversation Q1+Q2. 0.5 = chance.	(a) PTO, (b) GRPO; K=0 solid circle, K=5 dashed square; ribbons = 95% cluster bootstrap over conversations. (c) = the K difference at a matched policy (iteration 1), where negative means K=5 agreed more.	The GRPO K=5 series pools iterations 1–6 under a column labelled 1–5 (faithfulness.py:110). The GRPO panel is not like-for-like.

The same question, scored by the held-out judge

Training-reward faithfulness — eval graded by the held-out judge (Claude Haiku 4.5); proxy = training oracle



X AXIS

Prefix length in turns.

Y AXIS

Rank agreement between the training-oracle prefix score and the full-conversation score AS GRADED BY Claude Haiku 4.5. 0.5 = chance.

SERIES

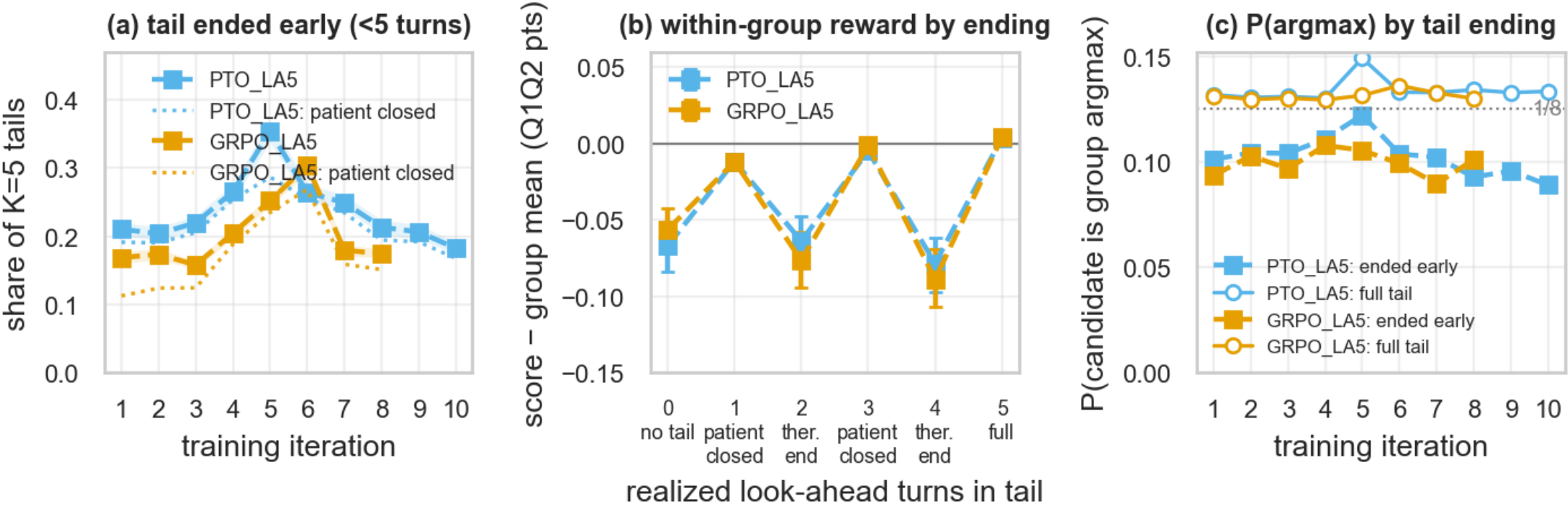
Same layout as the previous slide.

DEFECT

Same GRPO K=5 pooling defect as the previous slide.

What happens in the 5 simulated look-ahead turns

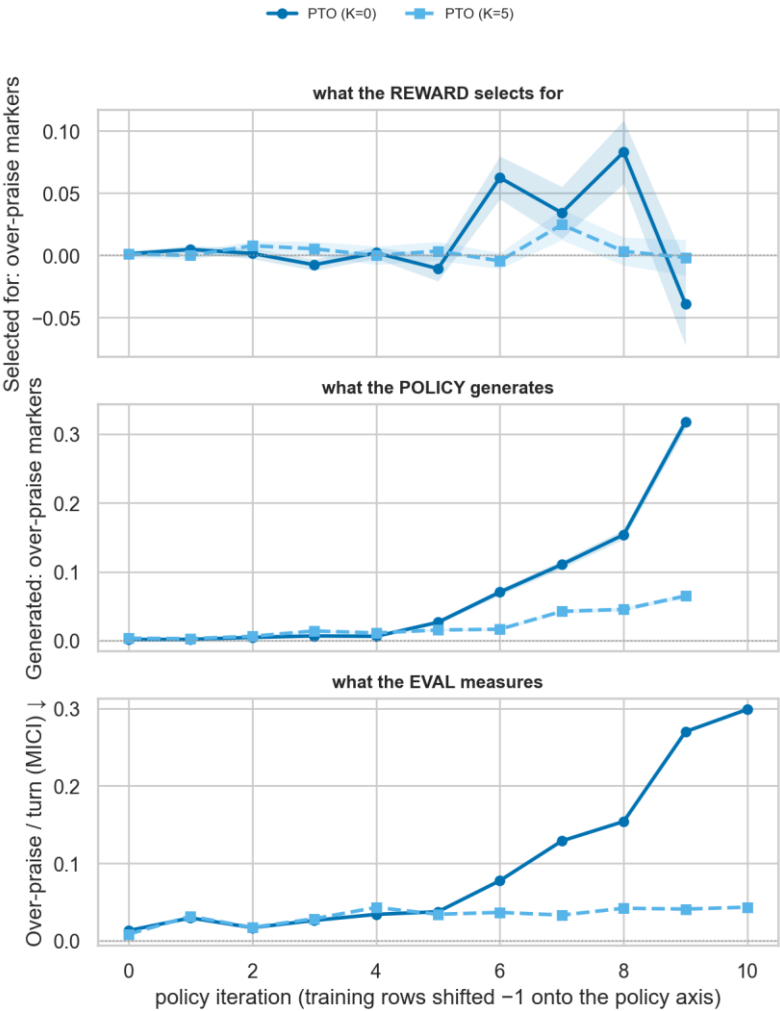
Look-ahead tails (K=5 arms) — grader: training oracle (gpt-4o-mini); GRPO_LA5 censored at iteration 8



PANEL A Share of candidates whose 5-turn tail ended early, by iteration, with the patient-closed share dotted underneath. 95% bootstrap CI.	PANEL B Within-group reward (score minus group mean, in Q1+Q2 points) against the realized tail length. Odd = the patient closed; even = ended on a therapist turn.	PANEL C Probability that a candidate is its group's argmax, split by whether the tail ended early. 1/8 = chance.	DATA K=5 arms only — K=0 has no tail. Grader: gpt-4o-mini.
---	---	--	--

One channel traced through selection, generation and evaluation

Over-praise: where look-ahead intervenes (PTO, K=0 solid · K=5 dashed)



X AXIS

Policy iteration. Training rows are shifted by -1 so all three describe the same policy.

TOP ROW

What the training reward SELECTS for within a group (± 1 SE over groups).

MIDDLE ROW

What the policy GENERATES, over every sampled candidate.

BOTTOM ROW

What the full-conversation eval MEASURES per therapist turn.

SERIES

Both PTO look-ahead arms; K=0 solid, K=5 dashed. Top two rows record the training oracle's own choices and are grader-invariant.

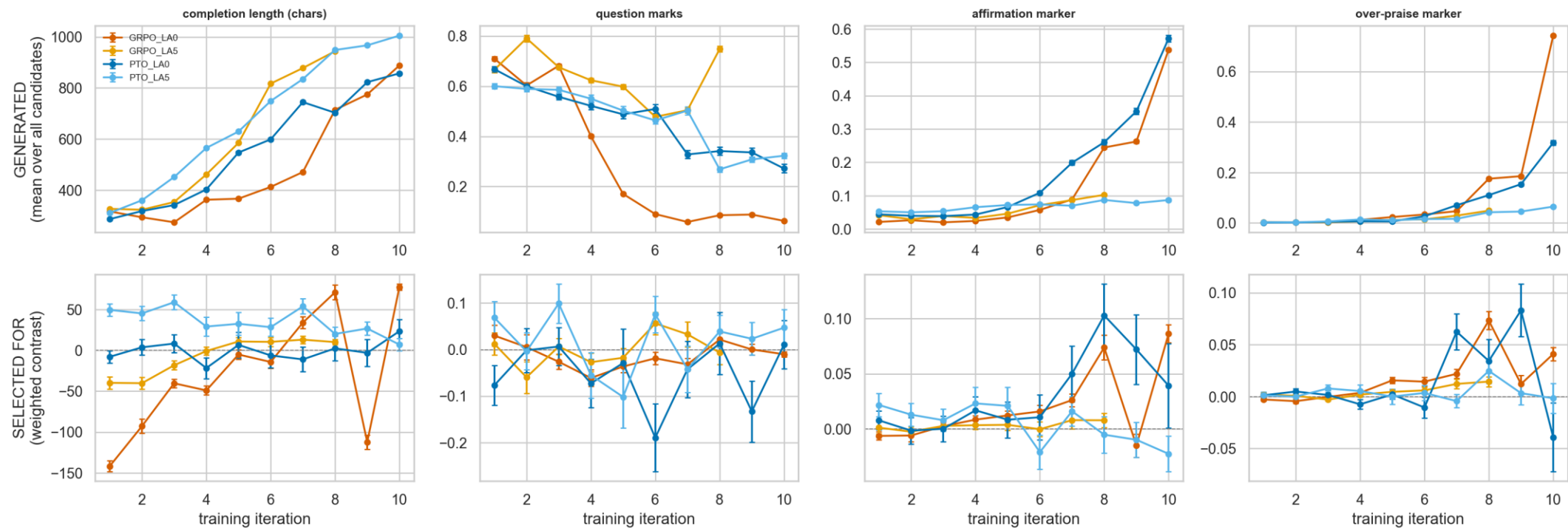
EXP3 · TRAINING SIGNAL

Exp3 — the preference data itself

Built from the per-generation records each trainer writes: every candidate, its score, and which one was chosen.

What the policy generates vs what the oracle selects

Generation vs selection — does the update pull the drift, or follow it?

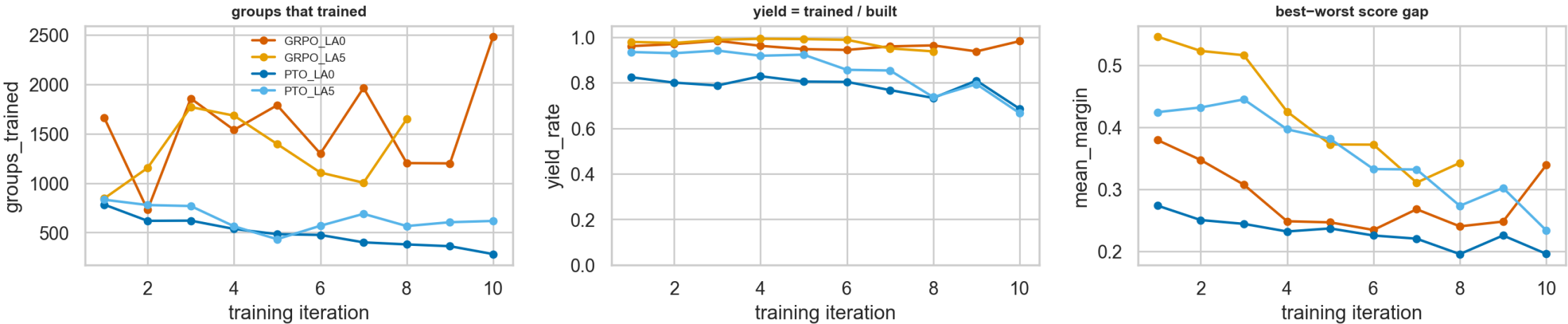


Top: what the policy produces (unweighted mean over every candidate). Bottom: what the update pushes toward within a group (0 = indifferent).

X AXIS	Y AXIS	SERIES	DATA
Training iteration.	Two quantities on the same axis: the rate at which a behaviour appears in generated candidates, and the weight the oracle's selection puts on it.	One line per arm.	Read from iteration_N/eda/generations.jsonl — the trainer's own record of every candidate it sampled. Grader: gpt-4o-mini.

How many usable training rows each iteration produced

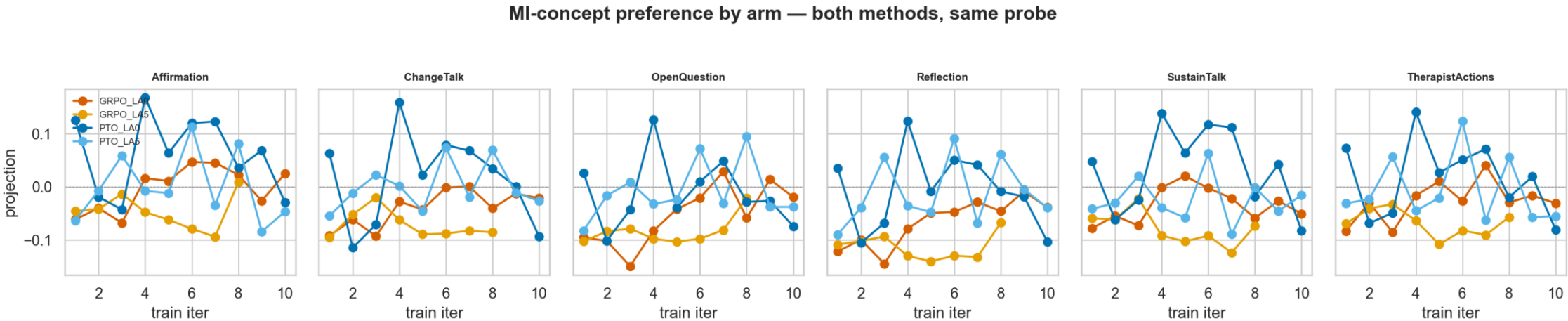
How much usable training signal each iteration actually produced



GRPO trains on every prompt it builds; PTO emits a pair only where the best and worst branch differ by more than tau, so its yield can fall as branches converge.

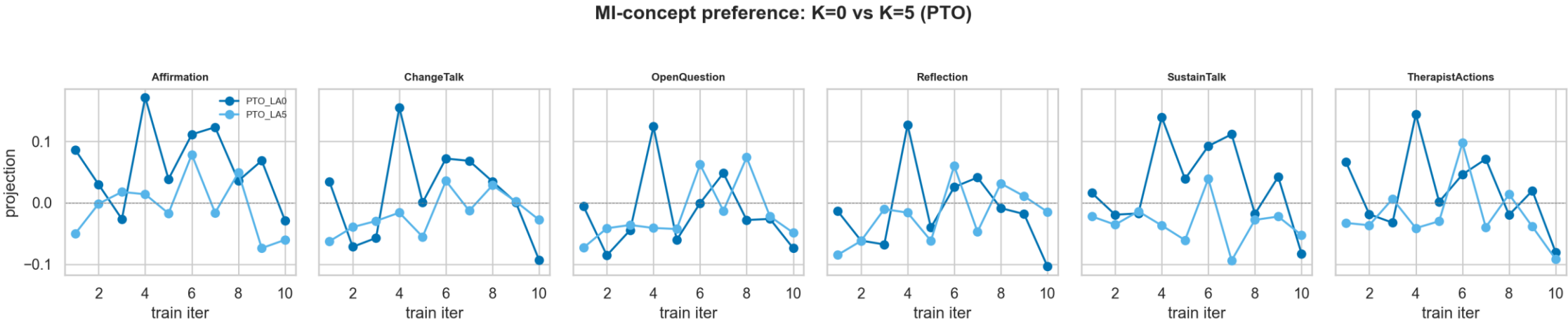
X AXIS	PANELS	SERIES	DATA
Training iteration.	Three, each on its own scale: groups that trained (a count), yield = trained ÷ built (a rate), and the best-worst training-reward gap.	One line per arm — all four, both methods, in every panel.	PTO emits a pair only where the best-worst margin clears $\tau = 0.1$, so its yield can fall below 1. GRPO trains on every prompt it builds. Grader: gpt-4o-mini.

What the chosen-minus-rejected direction pushes toward



X AXIS Training iteration, in each of the six panels.	Y AXIS Projection of that panel's MI-concept word group onto the iteration's chosen-minus-rejected embedding direction. The dotted line is 0.	SERIES One line per arm, all four. Panels = the six MI concepts (Affirmation, ChangeTalk, OpenQuestion, Reflection, SustainTalk, TherapistActions).	DATA Computed over every emitted preference pair. Grader: gpt-4o-mini.
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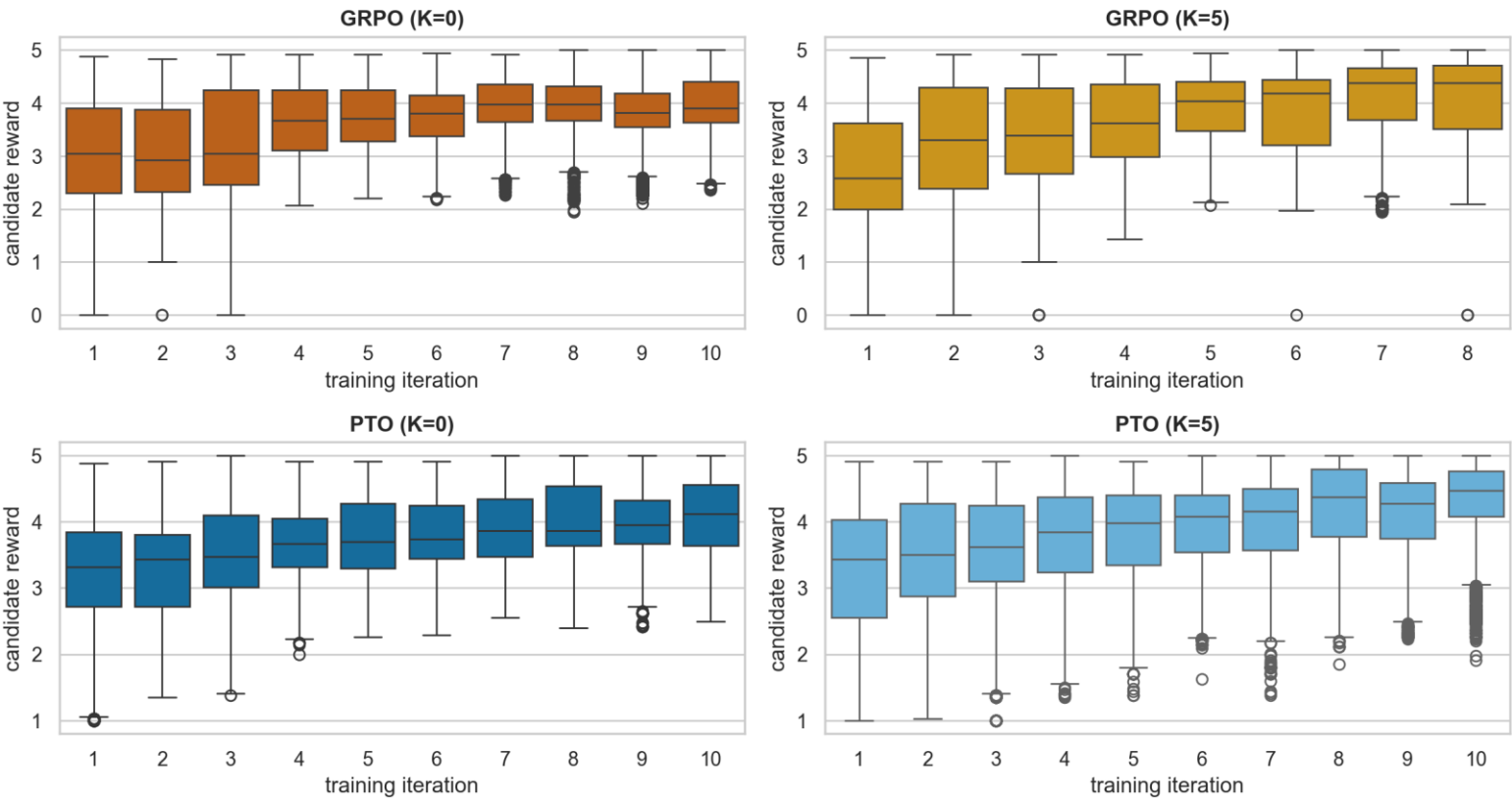
The same categories, K=0 against K=5



X AXIS Training iteration, in each of the six panels.	Y AXIS Projection onto the chosen-minus-rejected direction; 0 = indifferent. Same construction as the previous slide.	SERIES PTO K=0 and PTO K=5 only. Panels = the six MI concepts.	DATA Every emitted preference pair, both PTO arms. Grader: gpt-4o-mini.
---	---	--	---

Distribution of the training reward, per arm

TRAINING signal — candidate reward distribution per iteration (oracle on partial-conv branches, NOT the full-conv eval)



X AXIS

Training iteration, one panel per arm.

Y AXIS

Training reward of a single scored candidate, 0–5. Box = median and quartiles, whiskers 1.5 IQR, dots = outliers.

SERIES

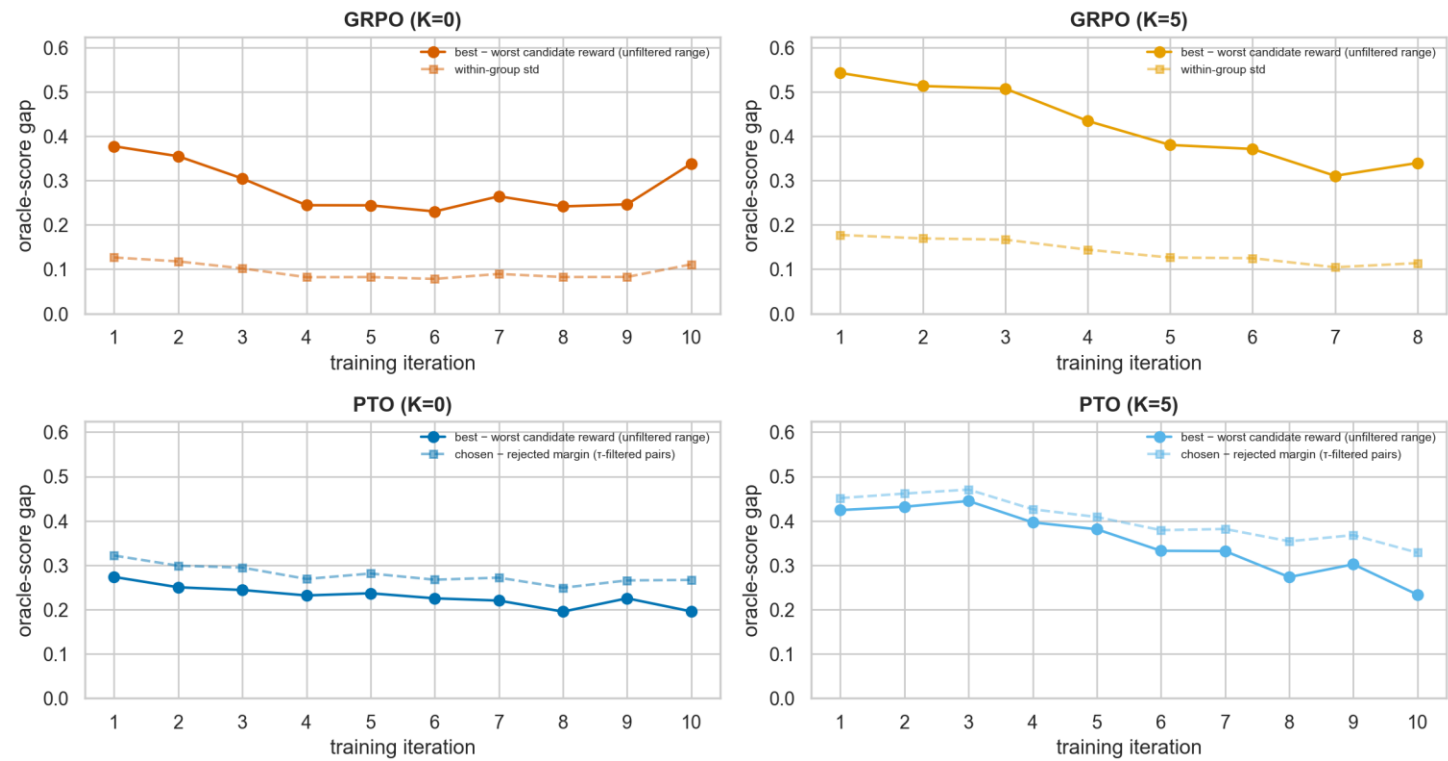
Four panels — GRPO K=0, GRPO K=5, PTO K=0, PTO K=5 — on a shared y scale.

DATA

Every scored candidate across all iterations, from generations.jsonl. Grader: gpt-4o-mini.

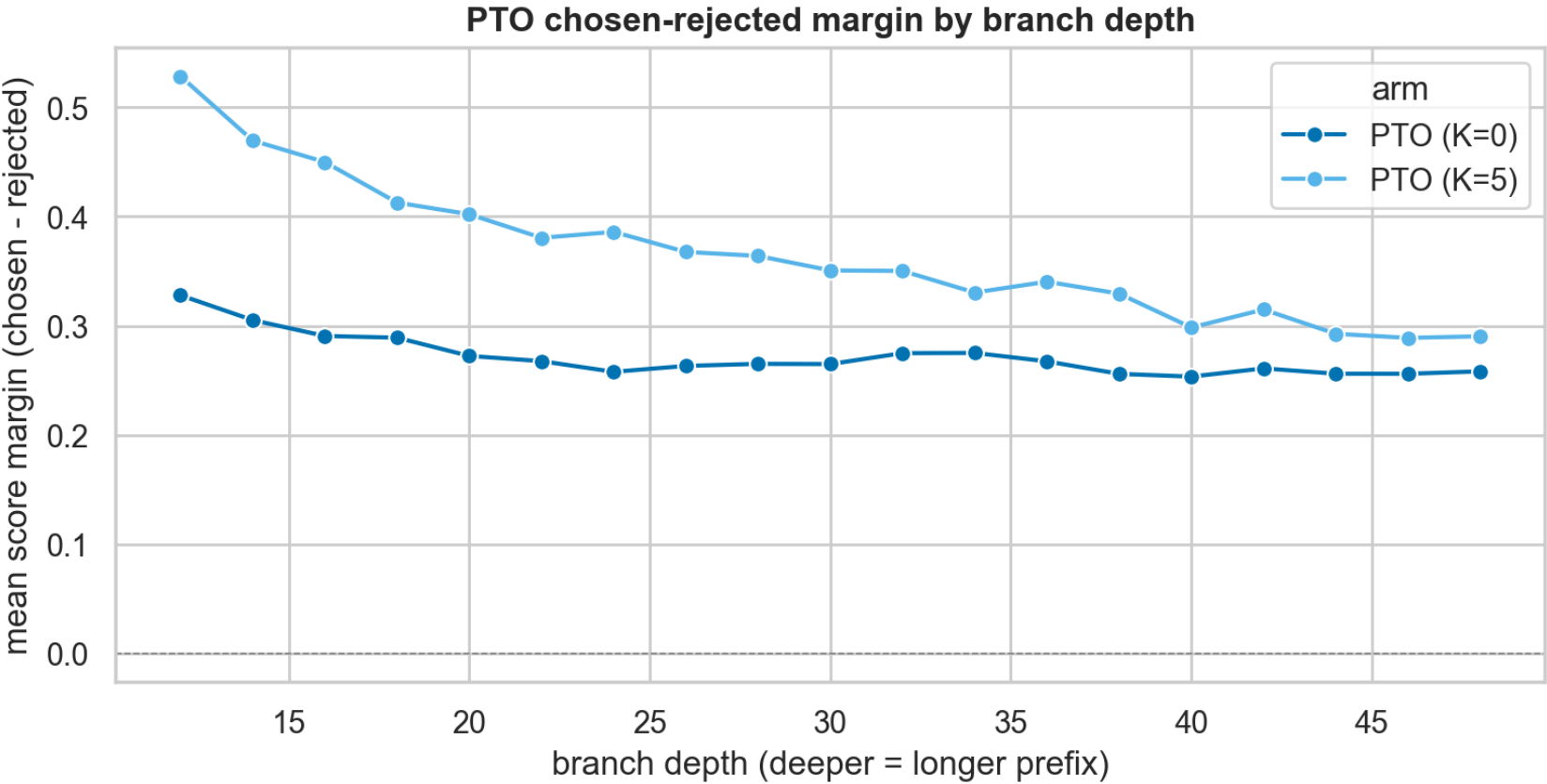
The advantage signal the two methods see

Training decisiveness (same oracle-score-gap scale): GRPO vs PTO best-worst candidate range — unfiltered, like-for-like; PTO τ -filtered margin faint



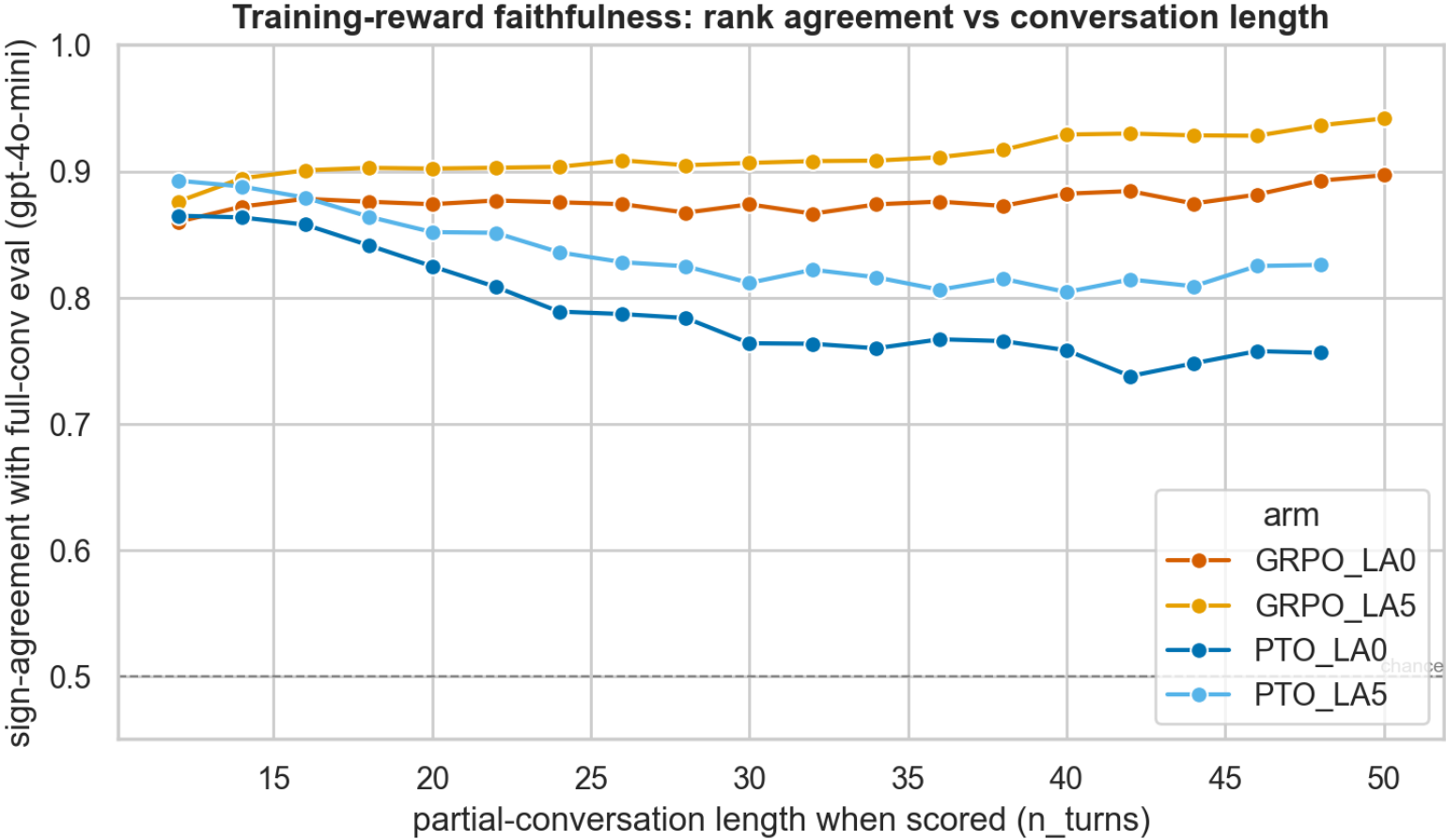
X AXIS	Y AXIS	SERIES	NOTE
Training iteration, in all four panels.	Oracle-score gap, on a shared scale across panels.	Four panels, one per arm. The solid line in every panel is best minus worst candidate reward, unfiltered — the same quantity for both methods. Faint secondary series: within-group SD for GRPO, the τ -filtered chosen-rejected margin for PTO.	GRPO's computed advantages are NOT in this figure; what is plotted is the raw spread of the oracle scores within each sampled group.

PTO's best-worst margin by depth in the trunk



X AXIS Depth in the preference trunk — how many therapist turns into the conversation the branch point sits.	Y AXIS Best-minus-worst score margin among the 8 candidates.	SERIES One line per PTO arm.	NOTE PTO's branch_id IS the trunk depth and repeats across conversations, so aggregation is keyed on (conversation_id, branch_id).
--	--	--	--

Reward reliability against prefix length



X AXIS

Conversation-so-far length in turns.

Y AXIS

Agreement between the score at that prefix and the full-conversation score. 0.5 = chance.

SERIES

Per arm.

DATA

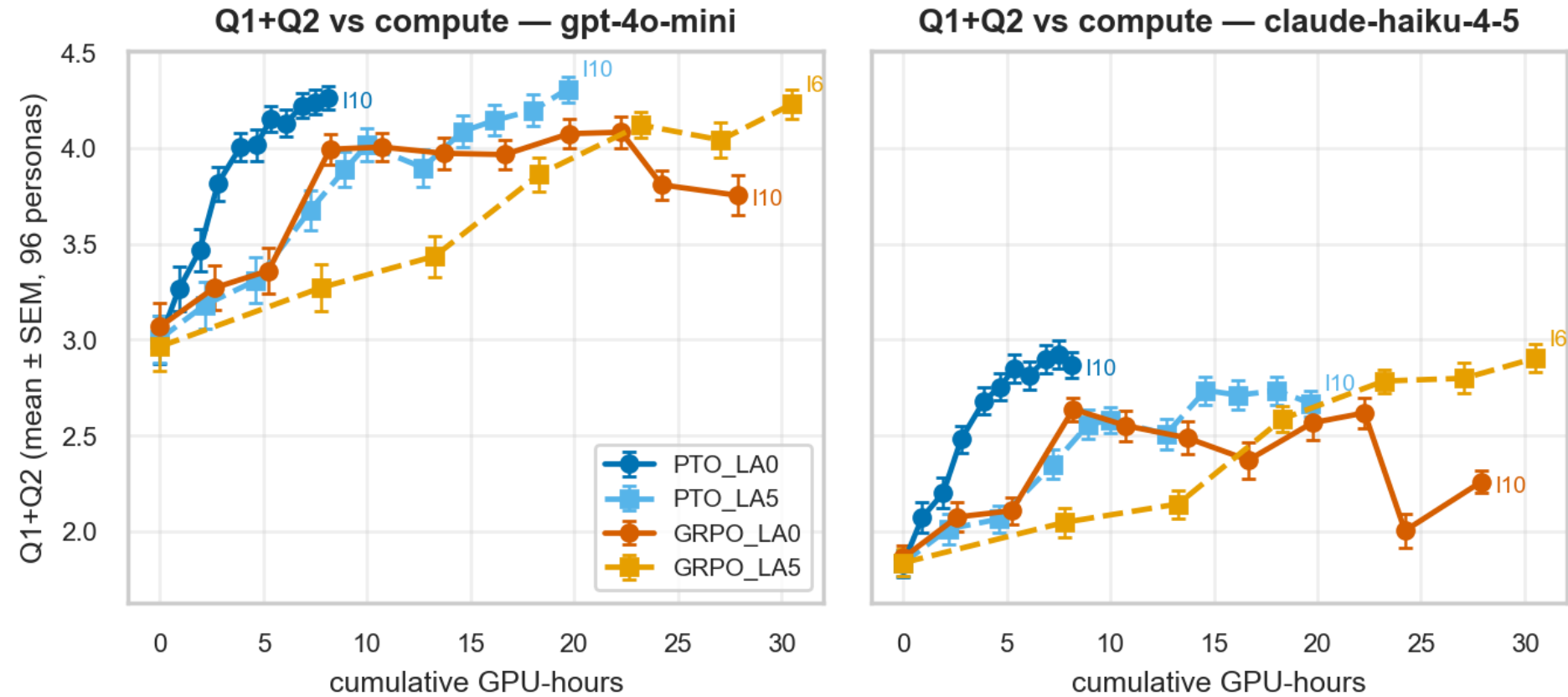
The curve STARTS at 12 turns because MIN_CONV_LENGTH = 12 already removed everything shorter from the training data, so it cannot speak to prefixes below that. The MCL knob was motivated by the earlier Exp2 partial-conversation diagnostic, not by this curve.

EXP3 · COMPUTE

Exp3 — what each arm cost

GPU-hours reconstructed from artifact modification times, and API call counts. An iteration is not a fixed unit of spend.

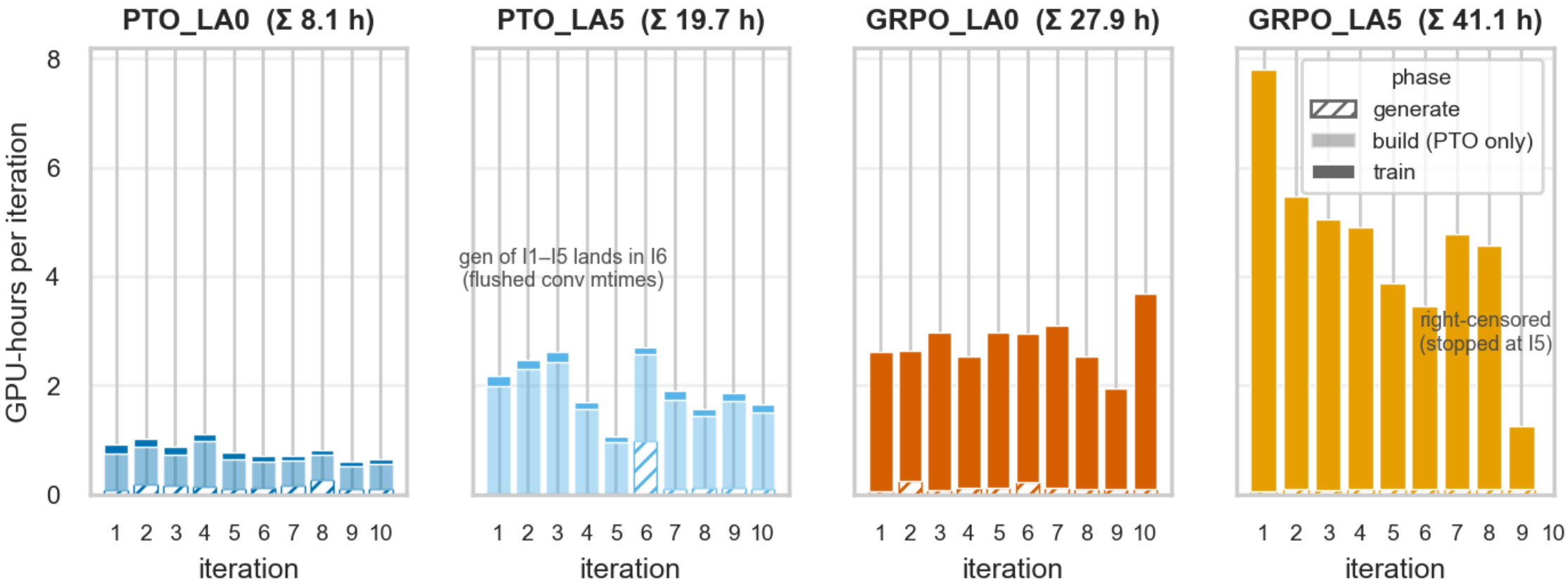
Score against cumulative GPU-hours instead of iteration



X AXIS Cumulative GPU-hours, reconstructed from artifact mtimes. Iteration 0 sits at 0 h.	Y AXIS Q1+Q2, mean ± SEM over the 96 personas.	SERIES Four arms: PTO K=0, PTO K=5, GRPO K=0, GRPO K=5. K=0 solid + circle, K=5 dashed + square. GRPO K=5's line ends at iteration 6, its last scored state. The last point of each arm is labelled with its iteration.	READ THE SPACING Marker spacing along x is uneven by construction: an arm with cheap iterations gets many markers close together.
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source: compute/cost/figures/compute_trajectory.png

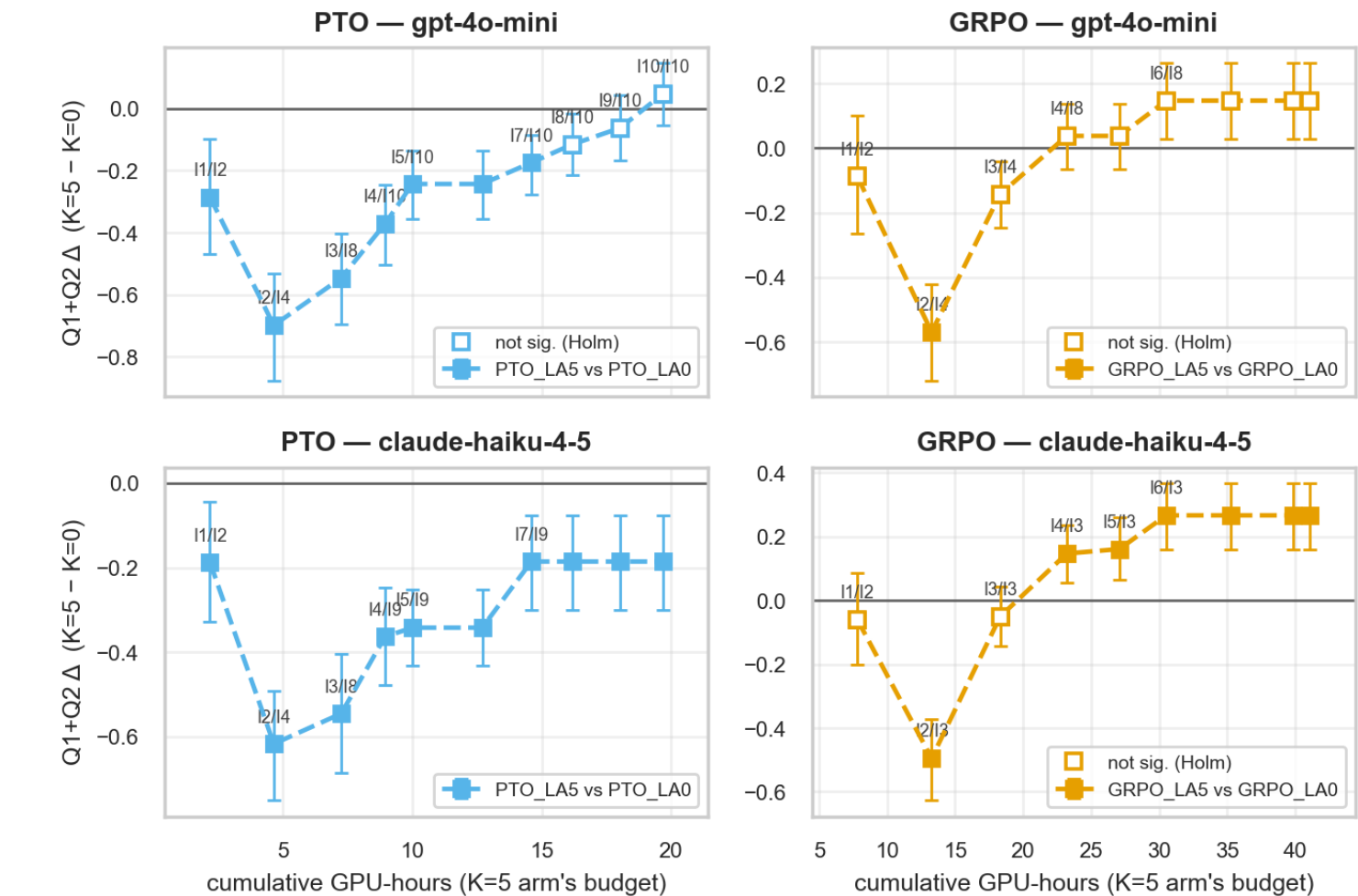
Where each arm's GPU-hours went



X AXIS Training iteration, one panel per arm.	Y AXIS GPU-hours, stacked by phase: generate (hatched), build (light, PTO only), train (solid).	SERIES Panel title = that arm's total.	KNOWN ARTEFACT PTO K=5's generation for iterations 1–5 lands in iteration 6 because conversation mtimes were batch-flushed; the cumulative total is right, the per-iteration split for that arm is not.
---	---	--	---

The K contrast read at matched budget rather than matched iteration

Look-ahead vs budget: paired Q1+Q2 delta between best-within-budget checkpoints (labels I_K5/I_K0 = selected iterations)



source: compute/cost/figures/budget_sweep.png

X AXIS

The K=5 arm's cumulative GPU-hours — the budget.

Y AXIS

Paired Q1+Q2 difference between the best checkpoint each arm reached WITHIN that budget. Above zero = the K=5 arm is ahead.

SERIES

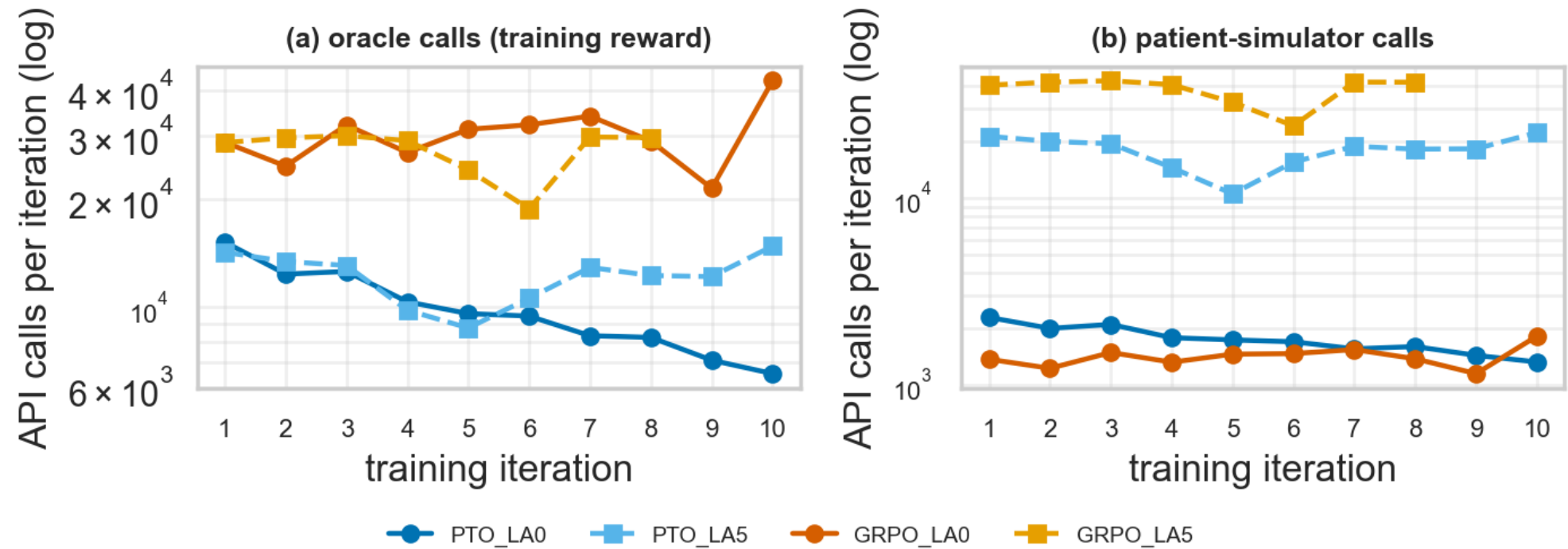
Rows = grader, columns = method. Bars = bootstrap 95% CI; hollow = Holm $p \geq .05$. Labels show which iterations were selected.

NOTE

Sign here is the opposite of the k_table1 convention — this one is K=5 minus K=0. Persona-paired, n = 96. The trainer reshuffles the 96 personas every iteration, so pairing is on persona identity, not file index.

API calls per training iteration

API calls per training iteration = eval convs of $\text{pi}_{\{n-1\}}$ + training-time calls (GRPO rescaled to its true step count); GRPO_LA5 censored at iteration 8



X AXIS Training iteration.	Y AXIS Number of API calls, LOG scale.	SERIES (a) oracle calls for the training reward — Q1 and Q2 per scored candidate; (b) patient-simulator calls, including the K=5 tail turns. One line per arm.	DATA Counts, not scores, so this figure is identical under both graders.
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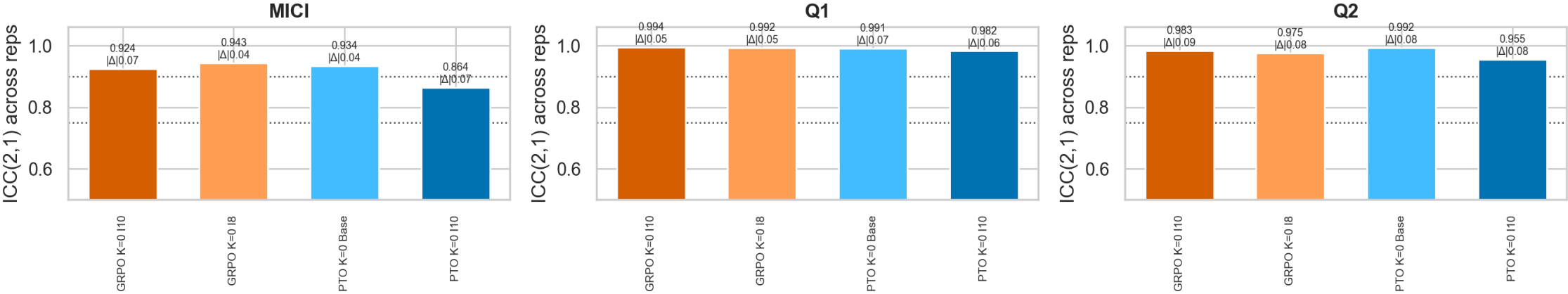
EXP3 · MEASUREMENT

Exp3 — how much the grader matters

Repeatability of the primary oracle, and everything the held-out judge does and does not reproduce.

Repeatability of the primary oracle

[EVAL] Oracle repeatability — same conversations re-scored 4× (seeds differ, nothing else); dotted = Koo & Li good (0.75) / excellent (0.90)



X AXIS

Model state. Only four states were re-scored, all of them K=0 arms.

Y AXIS

ICC(2,1) across FOUR independent re-scoring of the same 96 conversations by gpt-4o-mini (seeds differ, nothing else).

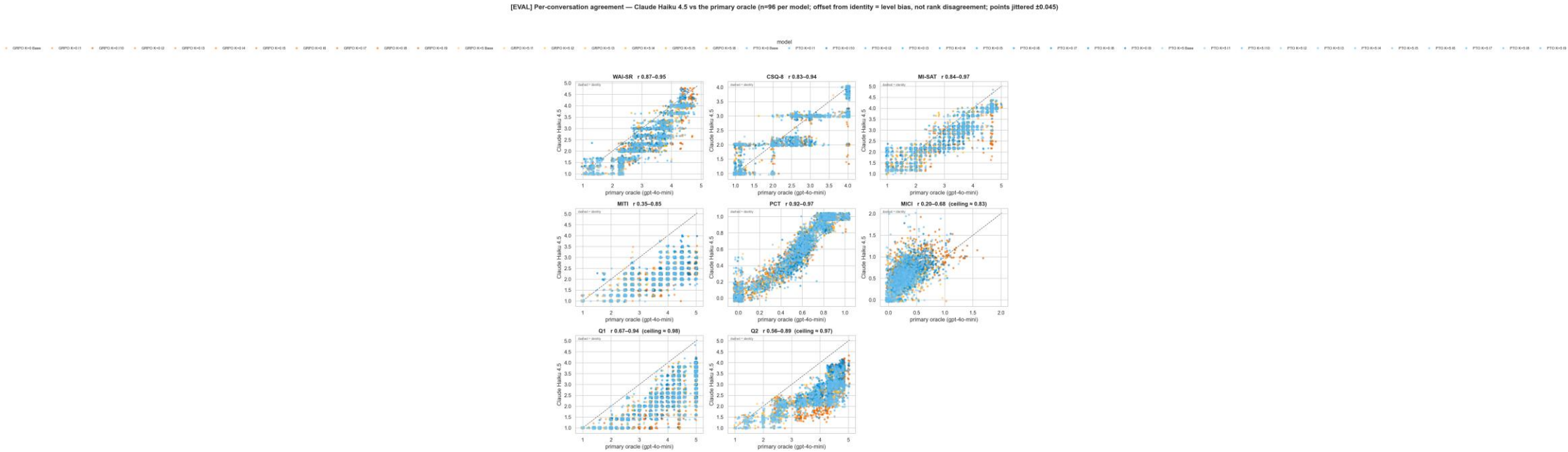
PANELS

Three: MICI, Q1, Q2 — the metrics with repeat draws.

REFERENCE LINES

Dotted at 0.75 and 0.90 — the Koo & Li 'good' and 'excellent' thresholds.

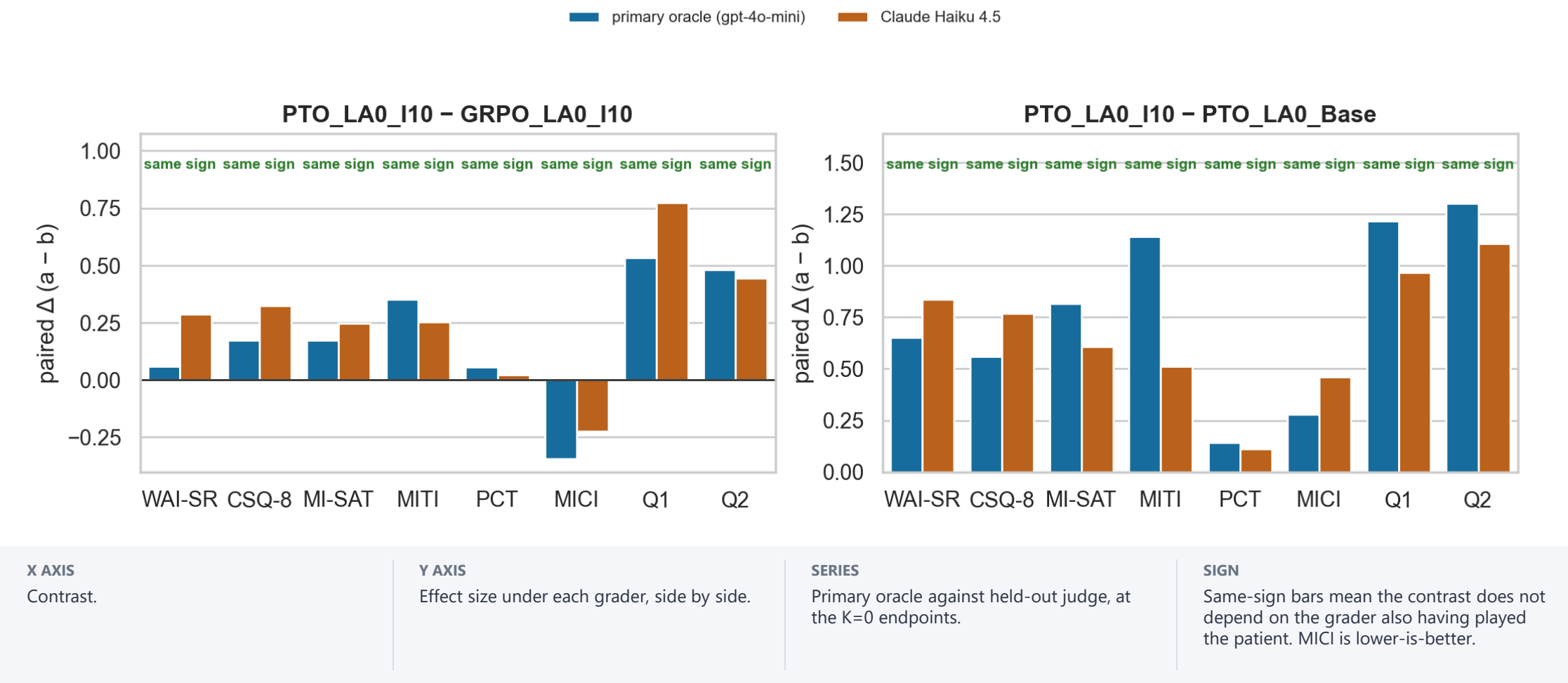
The two graders on the same conversations



X AXIS Score given by the primary oracle, gpt-4o-mini.	Y AXIS Score given by Claude Haiku 4.5 on the same conversation.	SERIES One panel per metric; all 40 model states pooled; one dot per conversation.	REFERENCE Dashed identity line. Vertical distance from it is level offset; scatter around the trend is rank disagreement.
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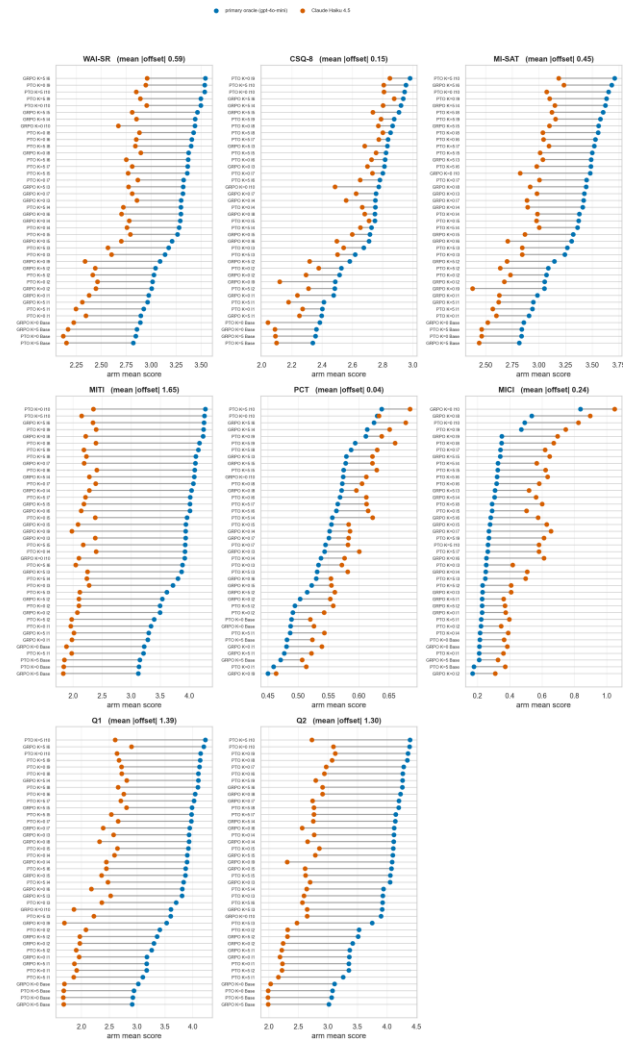
Do the two graders agree on the DIRECTION of each contrast?

[EVAL] Contrast preservation — does the result survive a judge that never played the patient? (MICI: lower is better)



Arm means under each grader

[EVAL] Arm means under both judges — bar LENGTH is level bias (cancels in every contrast); bar ORDER is what the thesis claims. Never averaged across judges.



X AXIS
Score.

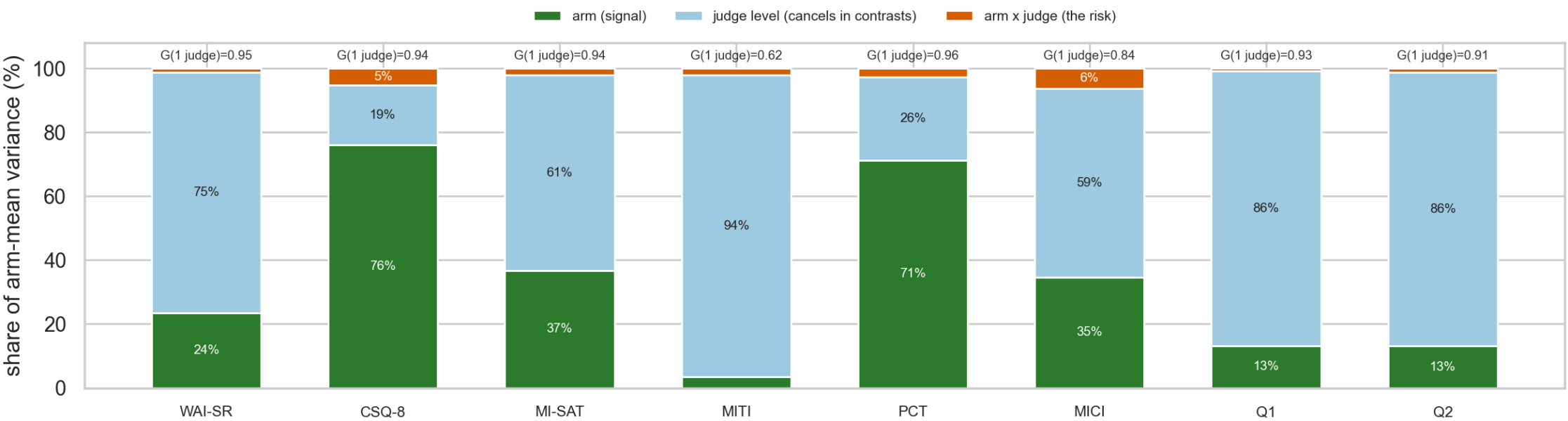
Y AXIS
Model state, all four arms.

SERIES
Each bar joins that state's mean under the primary oracle to its mean under Claude Haiku 4.5.

READ
Bar LENGTH is the level offset; bar ORDER is the ranking. The two are deliberately not averaged.

Where the variance in arm means comes from

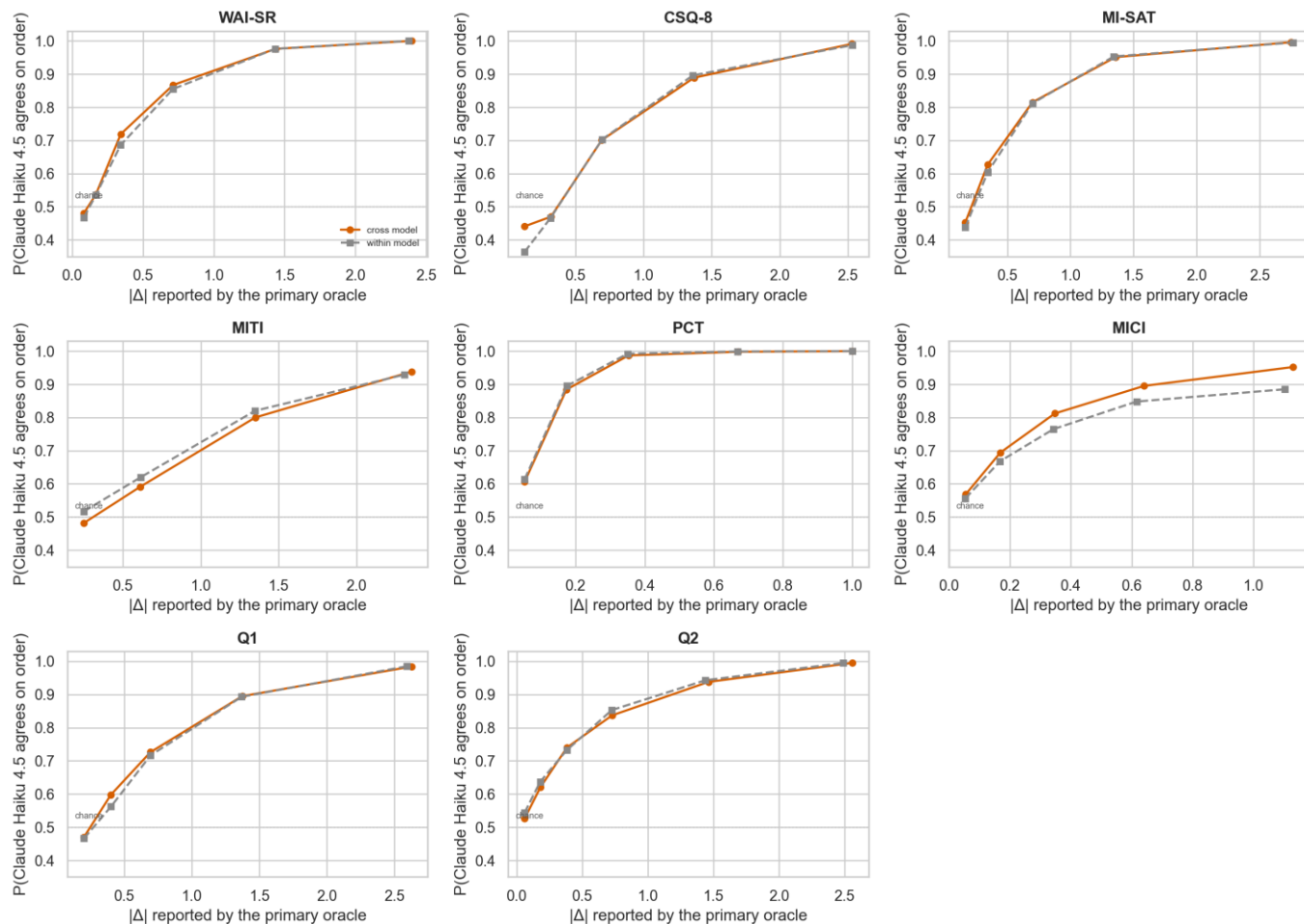
[EVAL] Sources of variance in the numbers the thesis reports (two-way random effects over arms x judges)



X AXIS Metric.	Y AXIS Share of arm-mean variance attributed to each source: arm, judge level, and arm × judge.	SERIES Stacked components across all judged model states.	NOTE A judge-level component cancels in any contrast; the arm × judge component is the one that does not.
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Cross-grader agreement as a function of effect size

[EVAL] Ordering agreement vs effect size — per CONVERSATION PAIR, not per arm (exact primary-judge ties excluded: they state no order to reproduce)



X AXIS

$|\Delta|$ between two conversations under the primary oracle. Each panel has its own x range.

Y AXIS

Probability that Claude Haiku 4.5 orders that pair the same way. Chance is 0.5.

SERIES

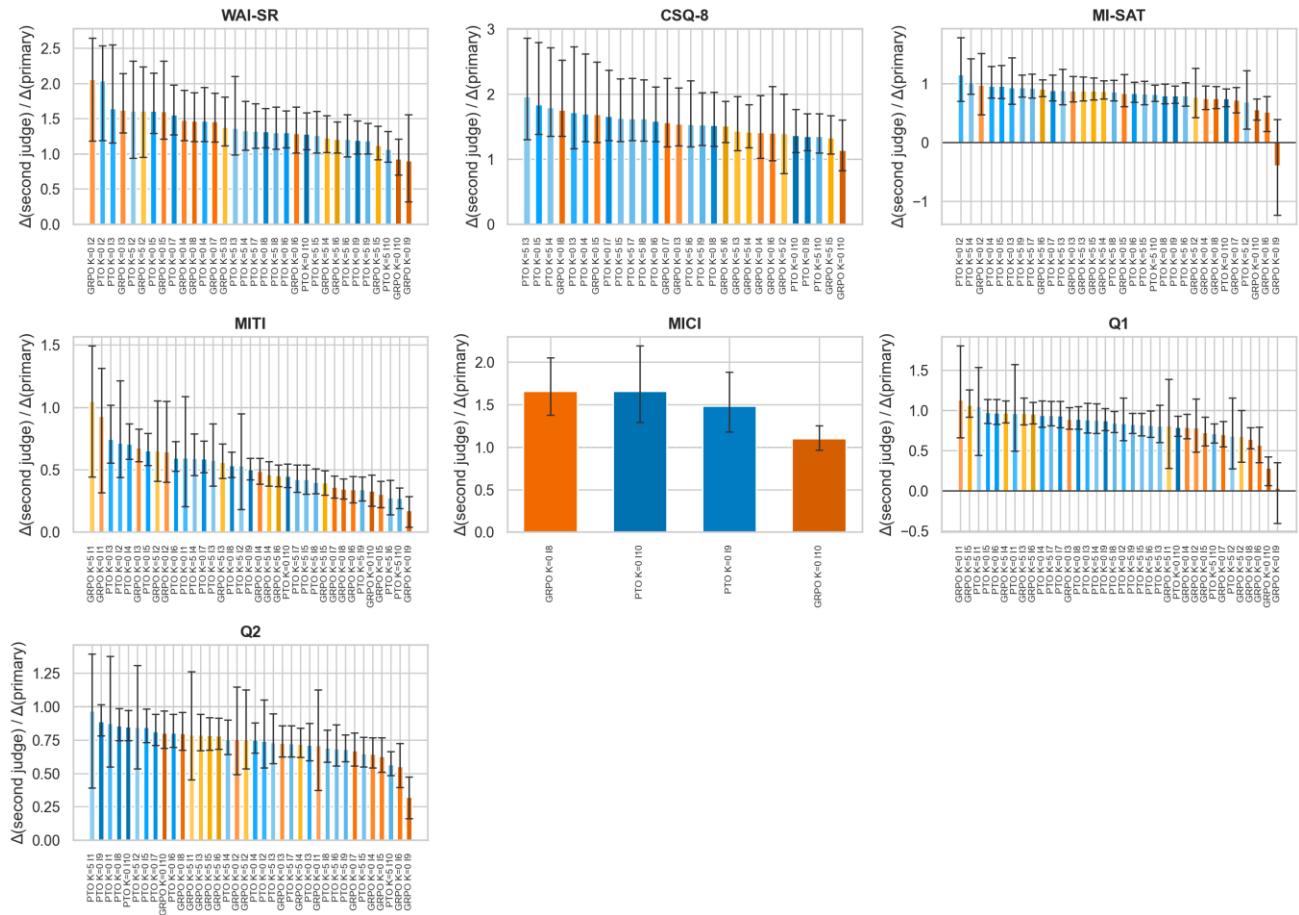
Eight panels, one per metric, TWO curves in each: cross-model (solid) = pairs drawn from different model states, within-model (dashed) = pairs from the same state.

DATA

Computed over conversation pairs; exact ties under the primary oracle are excluded from the denominator.

Gain retention, every model state

[EVAL] Does the improvement transfer? Gain over PTO K=0 Base as seen by Claude Haiku 4.5, relative to the primary oracle it was trained against (dashed = full transfer)



X AXIS

Model state, all four arms.

Y AXIS

(gain under the held-out judge) ÷ (gain under the primary oracle), with persona-bootstrap CI.

SERIES

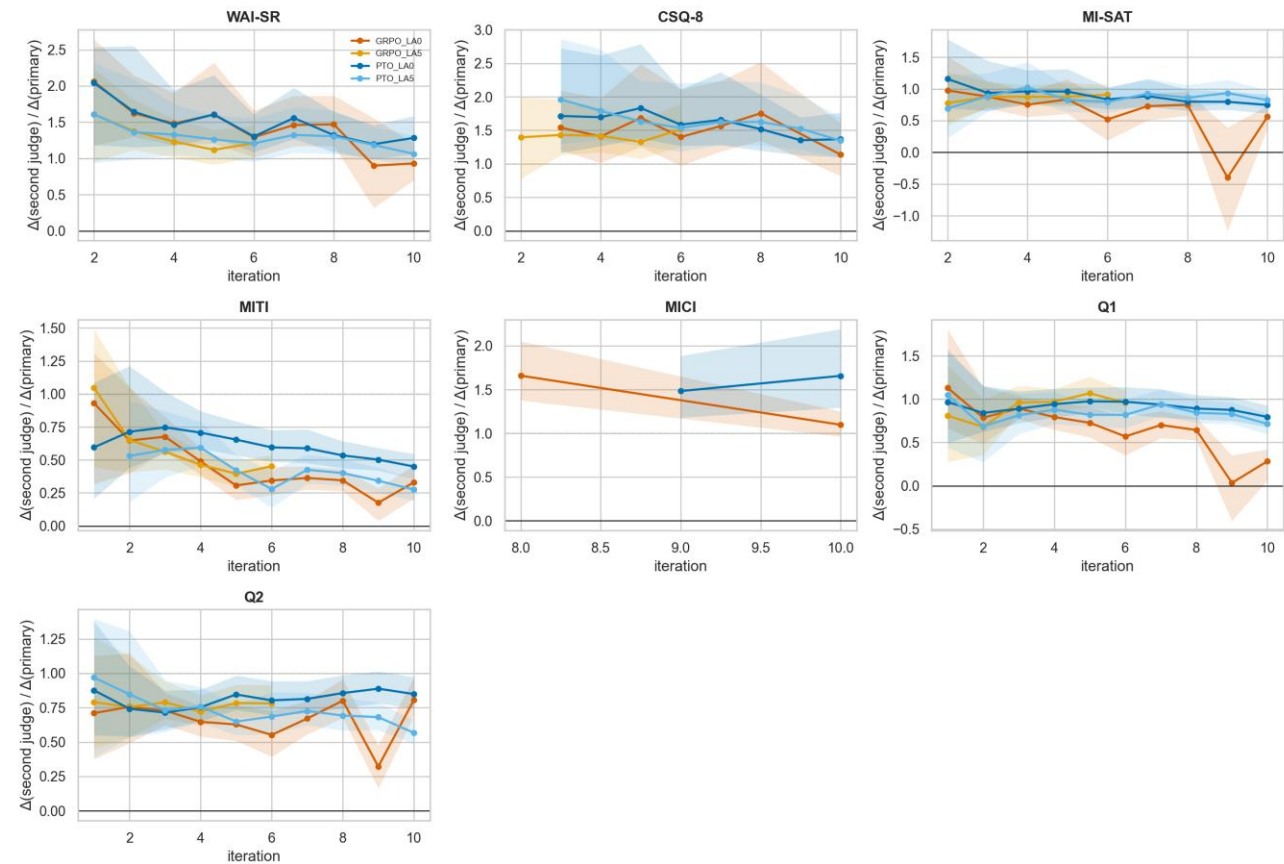
One bar per state.

NOTE

Uniform bars across arms and one arm dropping alone are different situations; the figure shows which.

Gain retention against iteration

[EVAL] When do the gains stop transferring? Retention vs iteration under Claude Haiku 4.5 (dashed = full transfer; a declining line = progressively fitting the trained-against grader)



X AXIS

Training iteration.

Y AXIS

Retention under Claude Haiku 4.5, referenced to the PTO K=0 base.

SERIES

One line per arm.

REFERENCE

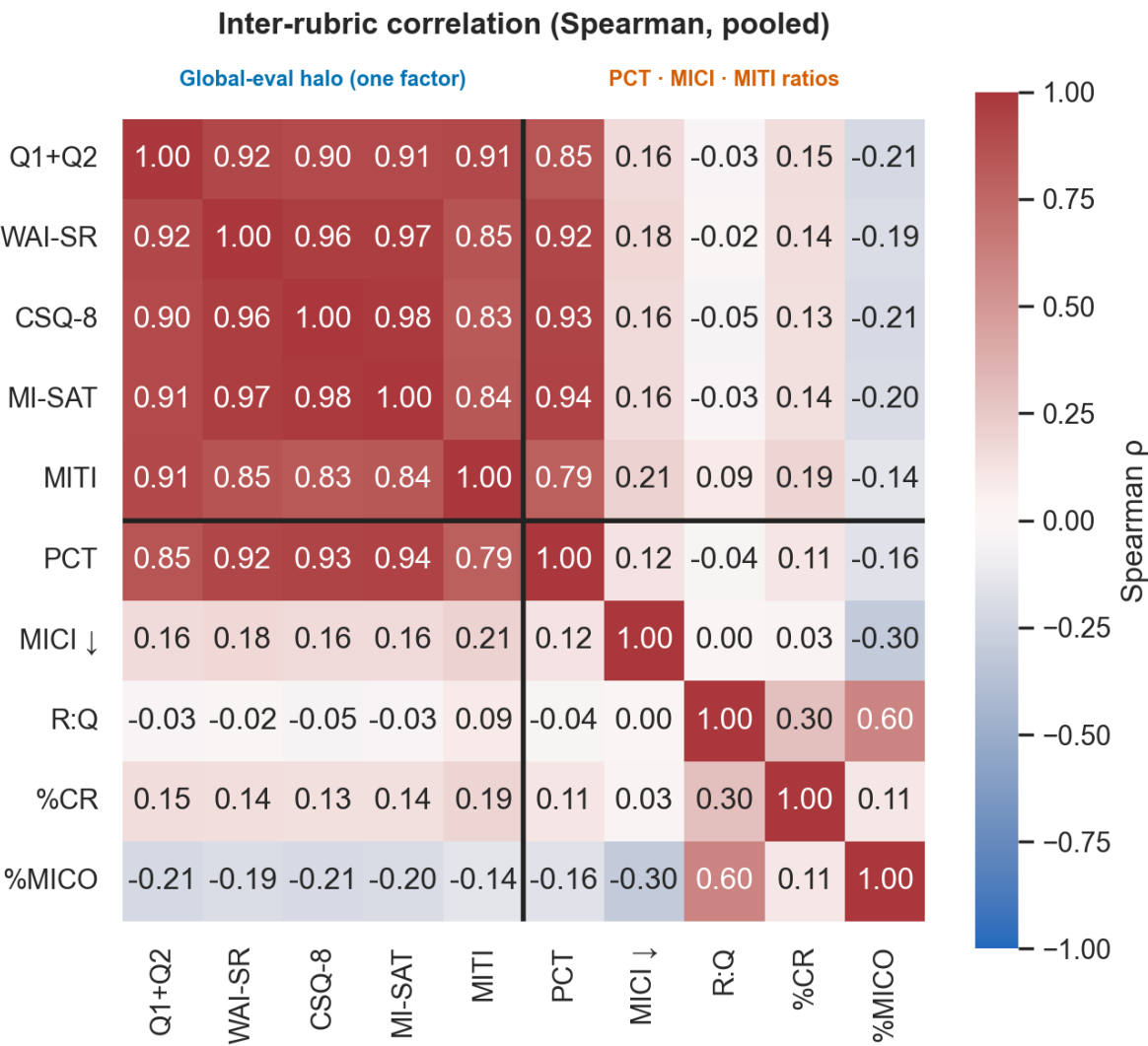
1.0 means the held-out judge sees the whole of the gain the training oracle credited.

EXP3 · RUBRICS

Exp3 — the instruments themselves

How the eight rubrics relate to each other, and what the item-level detail looks like.

How correlated are the eight rubrics?



AXES

Rubric × rubric.

CELL

Correlation between the two rubrics' per-conversation scores, pooled over all model states.

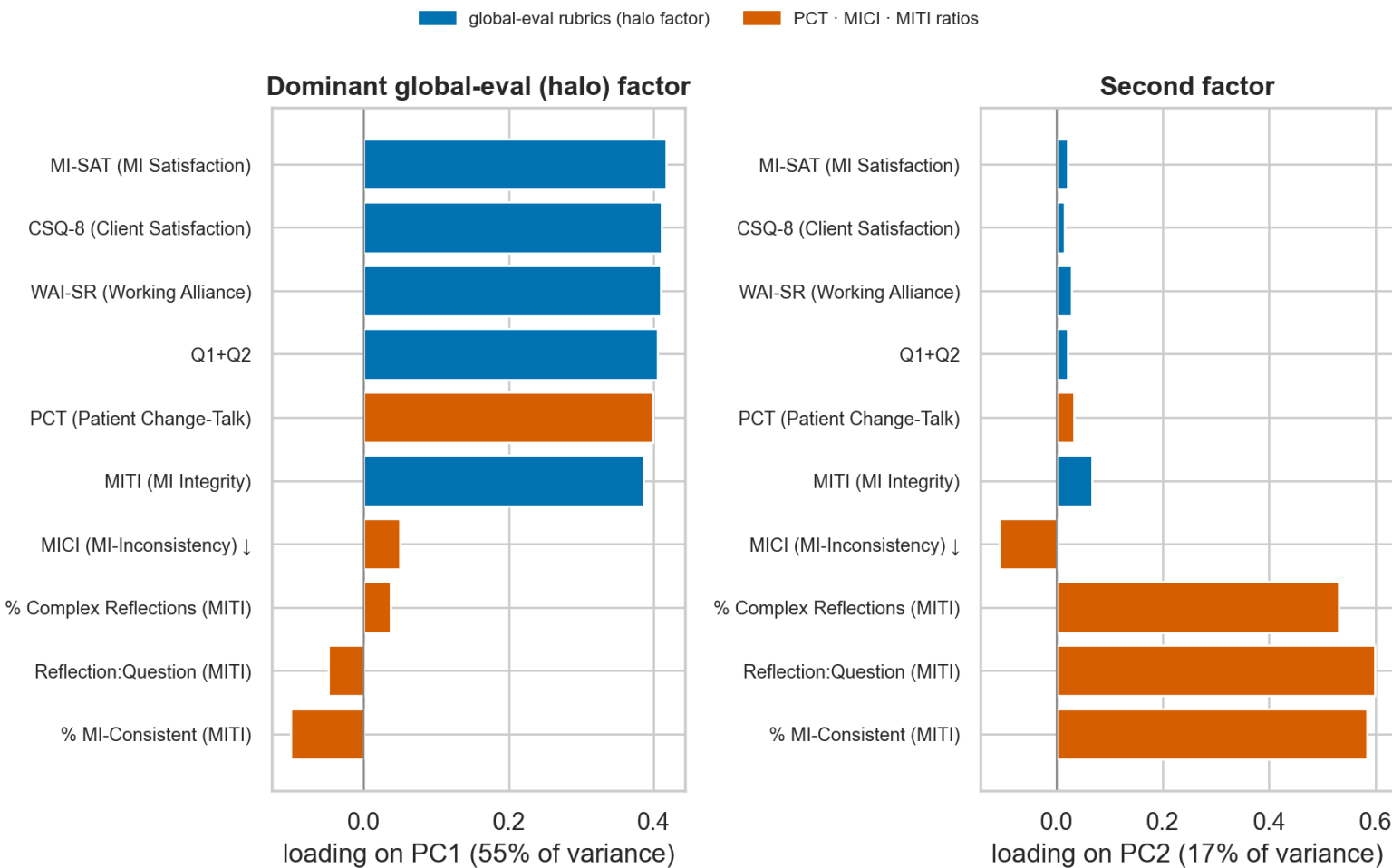
DATA

Spearman ρ on per-conversation scores, pooled over all model states. Grader: gpt-4o-mini.

NOTE

The matrix has ten rows and columns, not eight: the six questionnaires plus PCT, MICI and the MITI-derived ratios. MICI is lower-is-better — read its signs against that.

Factor structure across the rubrics



X AXIS
Loading. One panel per component: PC1 (55% of variance) left, PC2 (17%) right. The two panels have different x ranges.

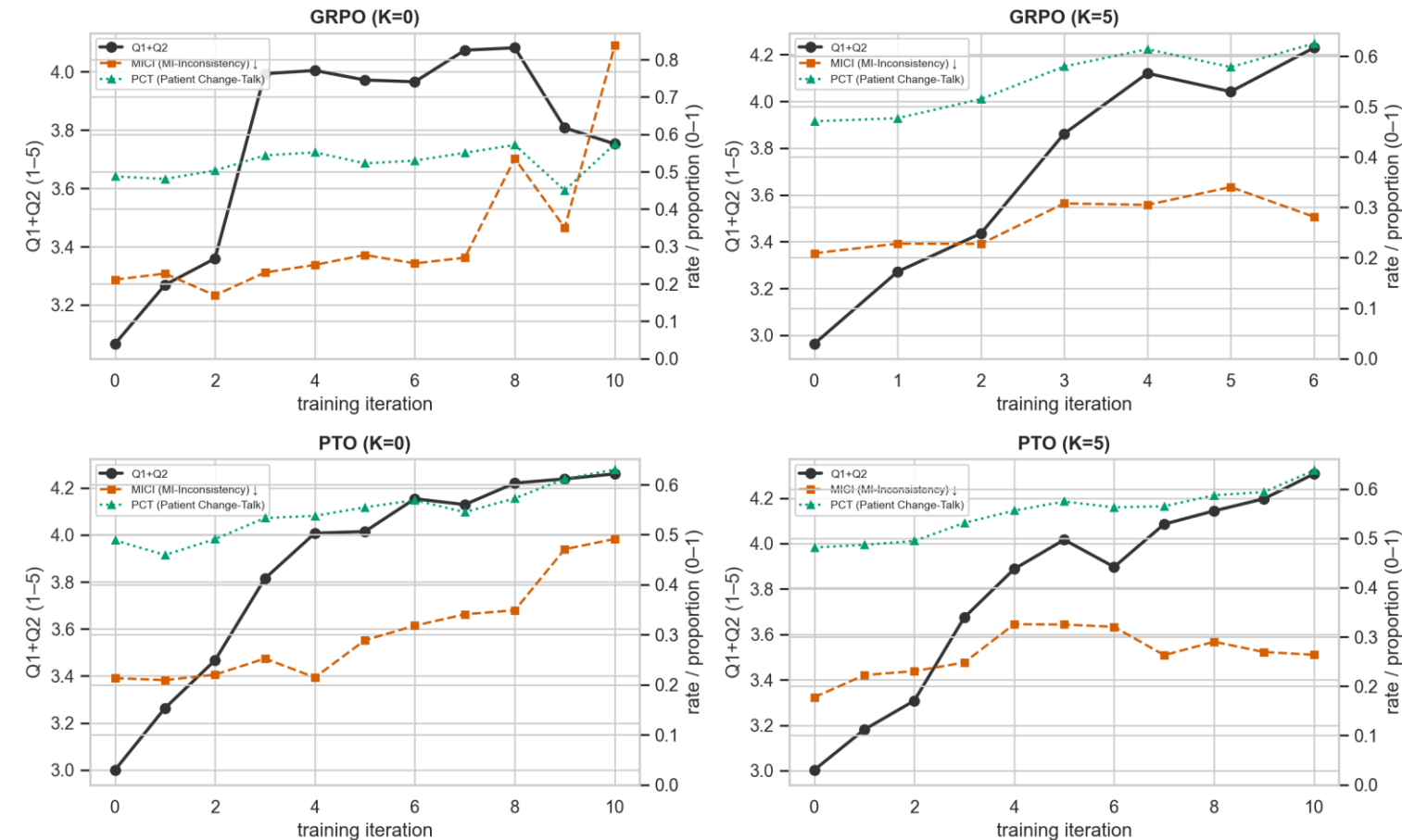
Y AXIS
The ten rubrics and MITI-derived ratios, ordered by PC1 loading.

SERIES
Bar colour marks the two blocks the figure groups by: the global-eval rubrics, and the PCT / MICI / MITI ratios.

DATA
Unrotated PCA on the per-conversation score matrix. Grader: gpt-4o-mini.

The training reward beside two behaviour rubrics

Reward-hacking: global-eval reward ↑ and MI-inconsistency ↑ together, patient change-talk ~flat



Left axis = the Q1+Q2 reward proxy (global evaluation; higher = better). Right axis: MI-Inconsistency (red, higher = worse) · Patient Change-Talk (green, higher = better — the actual MI goal).

PANELS

Four, one per arm. X = training iteration.

Y AXES

TWO per panel: left = Q1+Q2 on 1–5; right = rate / proportion on 0–1, shared by the other two series.

SERIES

Three per panel: Q1+Q2 (solid), MICI (dashed, lower is better) and PCT (dotted).

NOTE

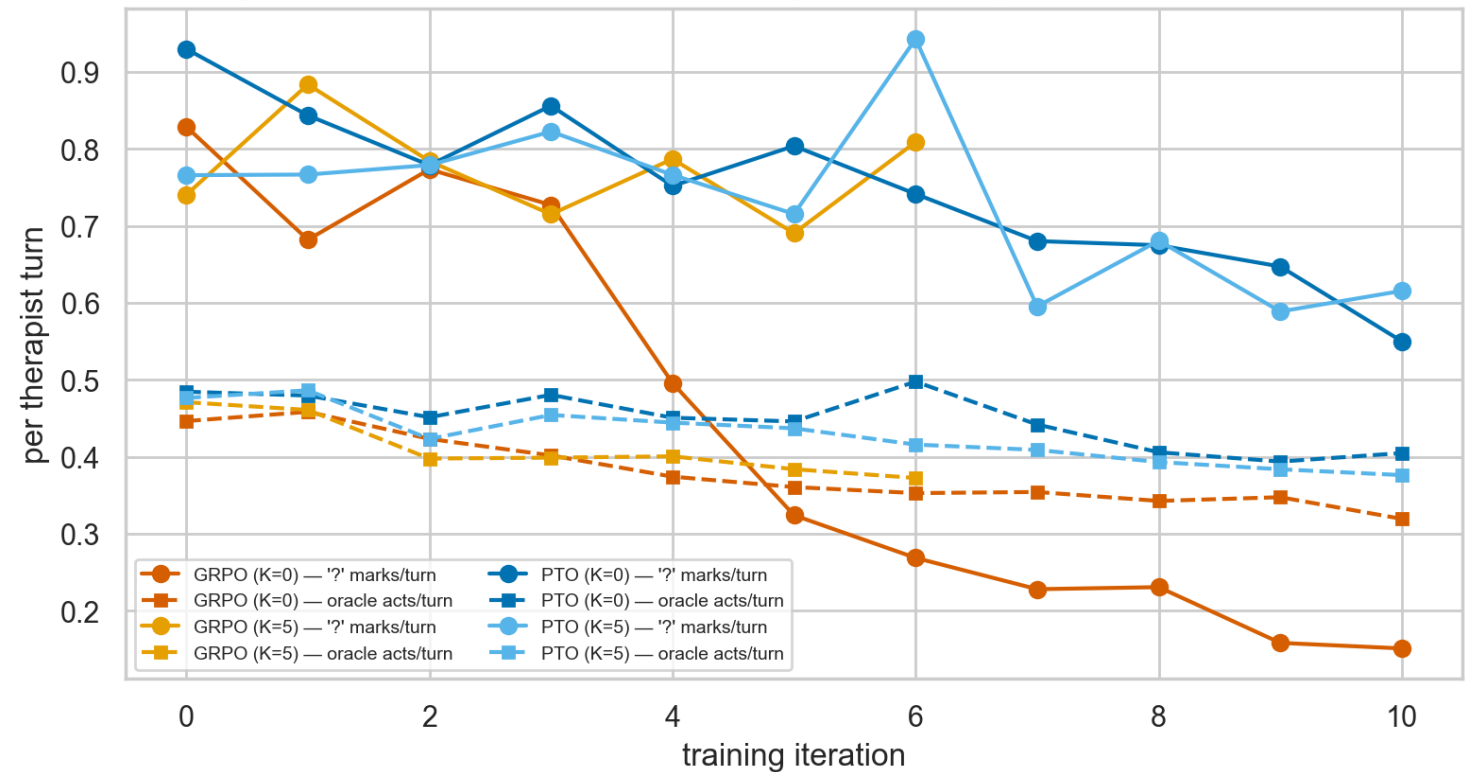
The figure's own supertitle states a conclusion. Read the lines, not the title.

DATA

Grader: gpt-4o-mini. The held-out twin of this figure exists at the same path under claude-haiku-4-5.

Two crosschecks: questions and over-praise

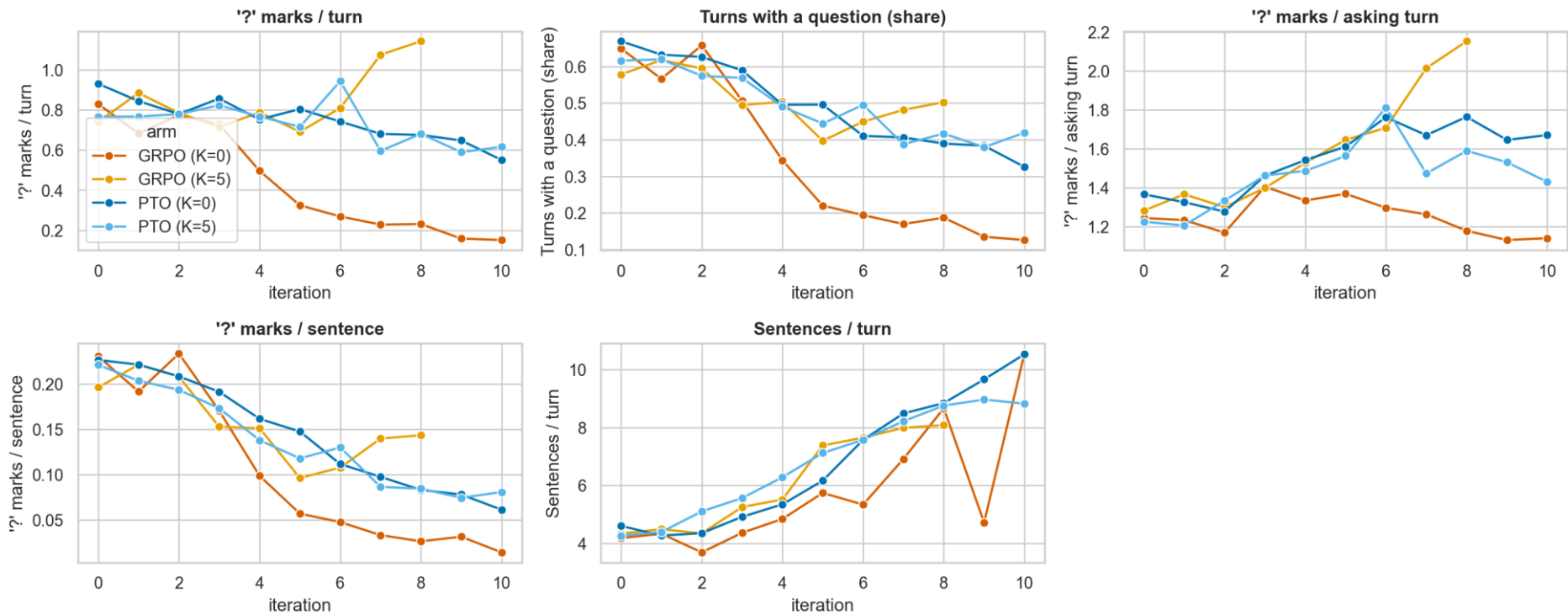
Questions two ways: literal '?' marks vs oracle-coded question acts (same denominator, different numerators)



X AXIS Training iteration.	Y AXIS Questions per therapist turn, two counts: literal '?' MARKS (solid) vs oracle-coded question ACTS (dashed).	SERIES One line per counting method, per arm.	LEVELS Marks run roughly 1.4–2.3× above acts in three of the four arms; in GRPO K=0 the two series cross. Same denominator (therapist turns), different numerators.	ALSO The over-praise twin of this crosscheck is overpraise_crosscheck.png, at the same path.
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What the question rate pools

Question metric decomposition — asking vs stacking vs speech volume (no oracle)

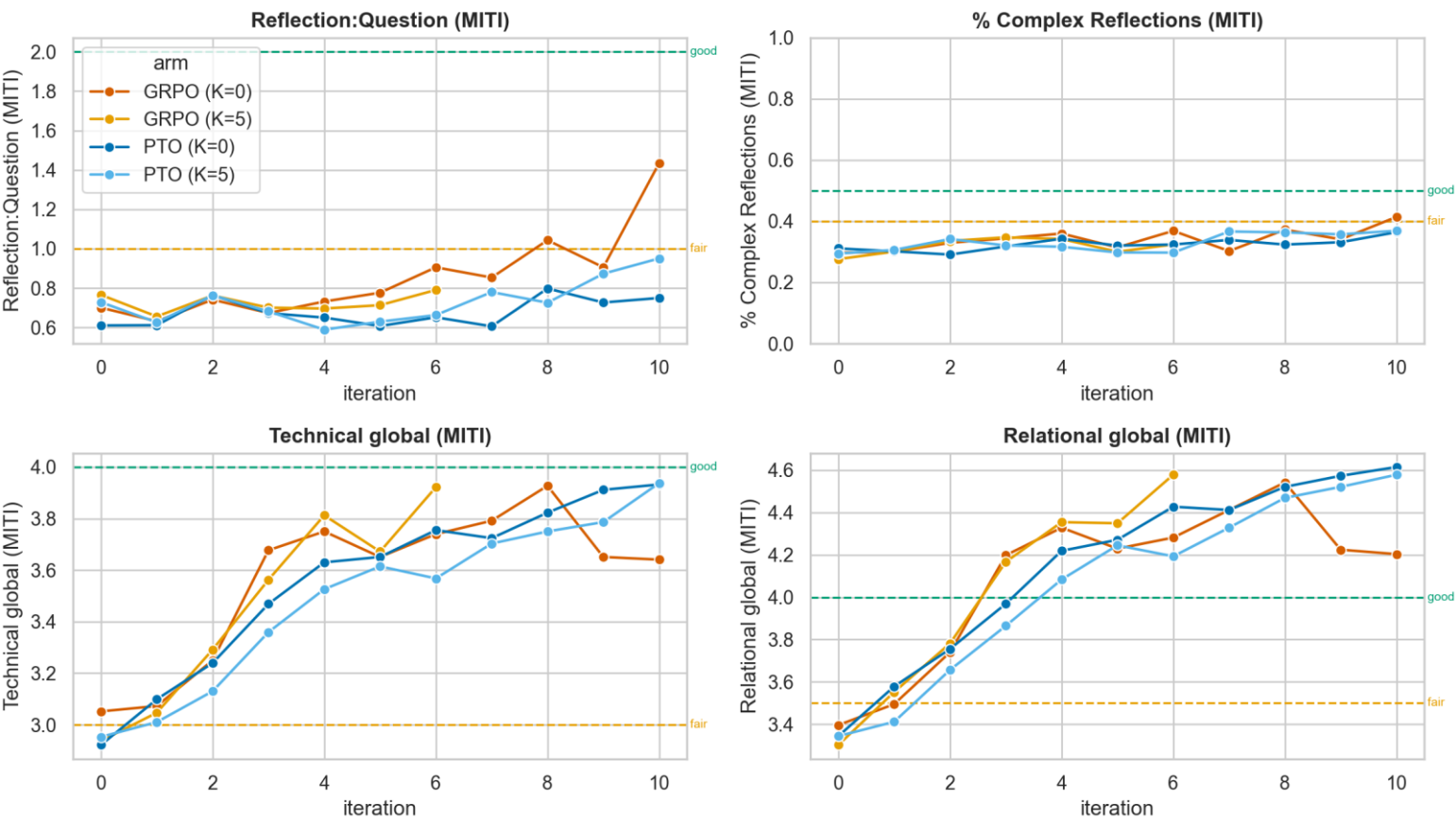


PANELS	WHY	THE EXCEPTION	WATCH	DATA
Headline '?' marks/turn, then its parts: share of turns asking, marks per asking turn, marks per sentence.	Headline = panels 2 × 3, and they move oppositely. All four arms ask in FEWER turns; three stack MORE.	GRPO K=0 collapses on both: asking 0.65 → 0.13, stacking flat 1.24 → 1.14. The one arm that stops asking.	GRPO K=5's headline RISES from iteration 5 (0.69 → 1.14) and BOTH parts move: asking share 0.40 → 0.50 (×1.26) and marks per asking turn 1.65 → 2.16 (×1.31); 1.26 × 1.31 = 1.65.	Deterministic, no grader. GRPO K=5 has transcripts to iteration 8, scores only to 5/6.

source: arms/validity/figures/gpt-4o-mini/question_decomposition.png

MITI against published proficiency thresholds

MITI 4.2.1 summary scores vs the manual's competency thresholds (dashed: fair / good)



Thresholds from the MITI 4.2.1 manual (Moyers, Manuel & Ernst 2014, rev. 2015) — expert opinion, not normatively validated, and defined for ~20-min human audio sessions (short text-chat sessions are out-of-domain).

X AXIS

Model state.

Y AXIS

MITI global and component scores.

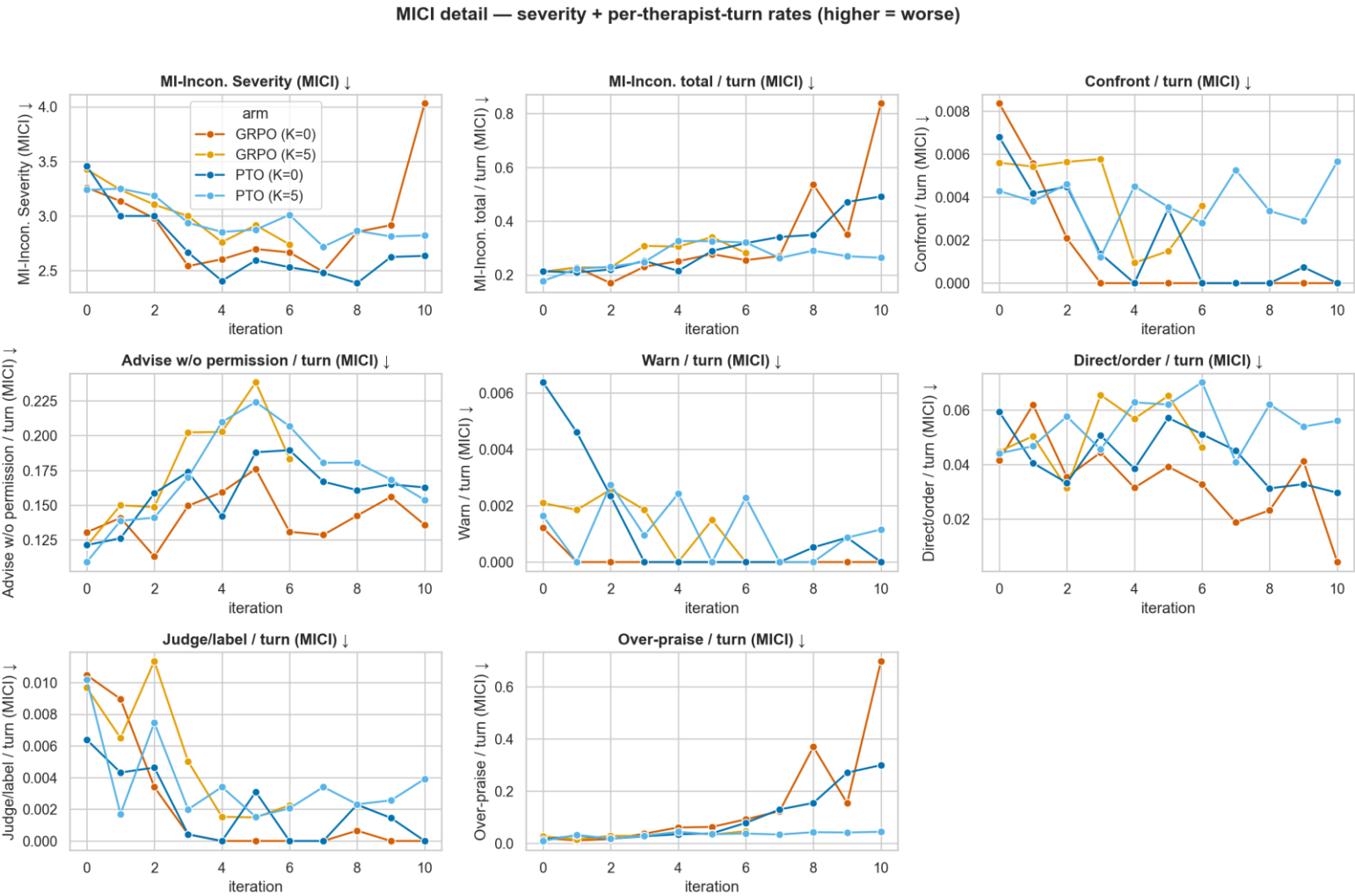
REFERENCE LINES

The published MITI 4.2 'fair' and 'good' proficiency thresholds.

DATA

Grader: gpt-4o-mini. MITI is the weakest instrument in this battery on single-judge dependability.

MI-inconsistency, item by item



X AXIS

Training iteration, per panel.

Y AXIS

Rate of that specific MI-inconsistent behaviour.

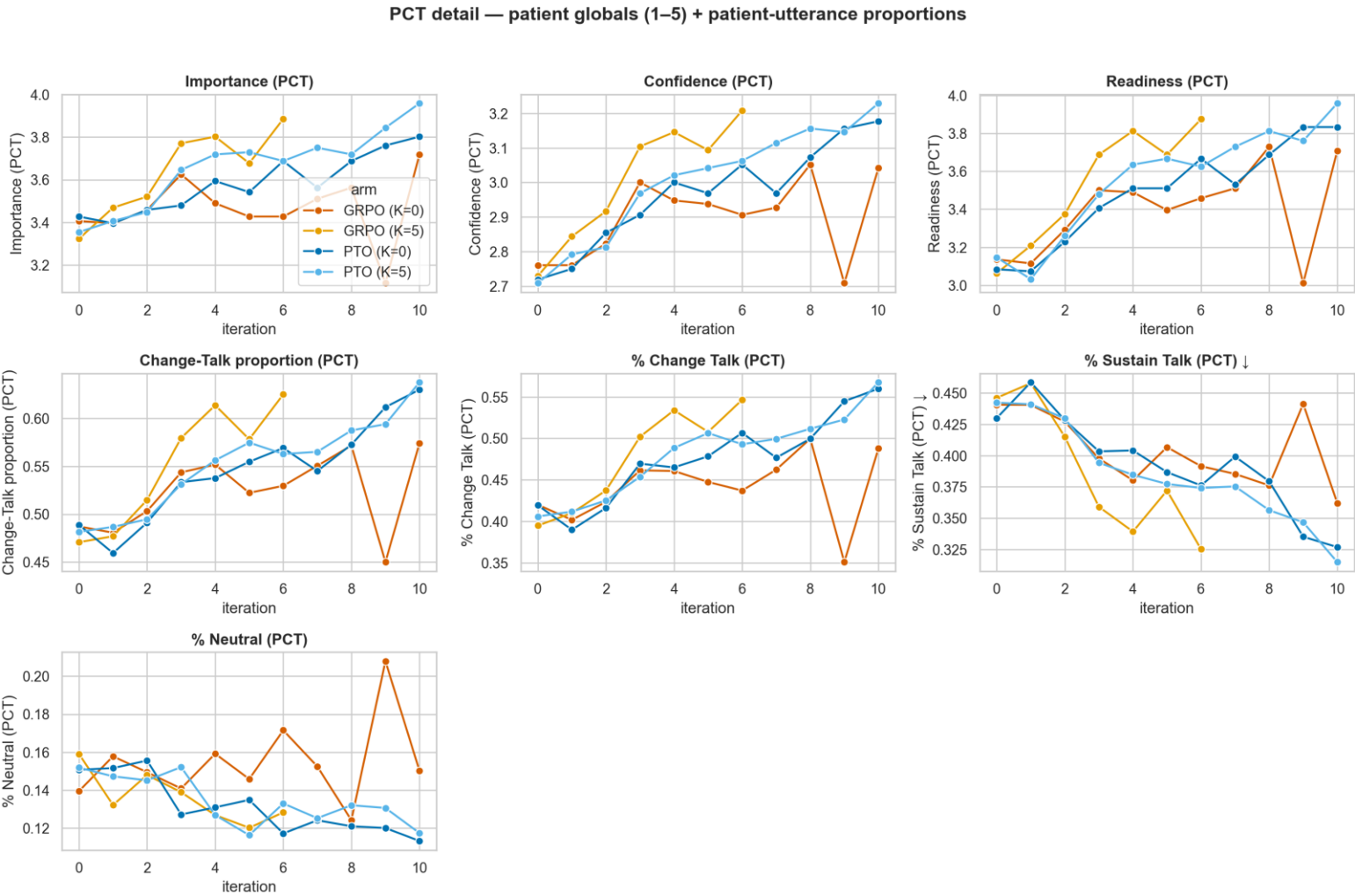
PANELS

One per coded behaviour: over-praise, confront, direct, judge, warn, advise without permission, plus the severity global.

SERIES

One line per arm. Higher = worse throughout.
Grader: gpt-4o-mini.

Patient change talk, item by item



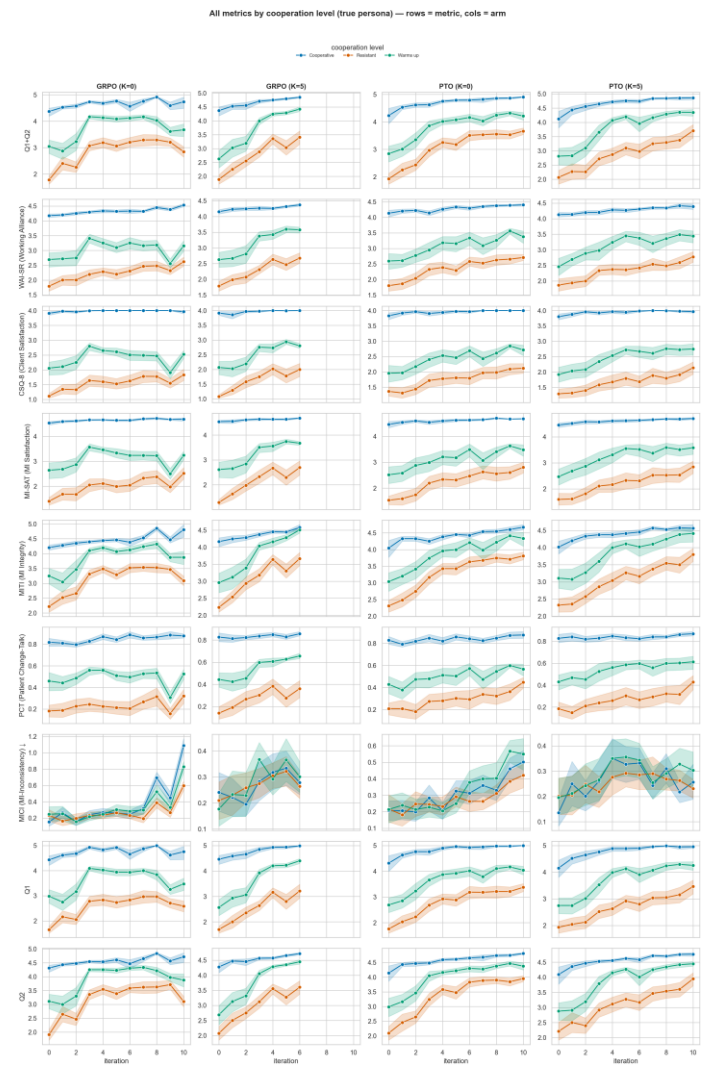
X AXIS
Training iteration, per panel.

Y AXIS
The panel's PCT component — change-talk proportion, sustain-talk proportion, neutral proportion, importance, confidence, readiness.

SERIES
One line per arm.

NOTE
PCT codes the PATIENT's language, not the therapist's. Grader: gpt-4o-mini.

Outcomes split by patient cooperation level



LAYOUT

Rows = rubric (nine), columns = arm (all four).
Every panel has its own y scale.

X AXIS

Training iteration.

SERIES

THREE lines per panel, one per patient cooperation stratum: Cooperative, Warms up, Resistant — 32 personas each, disjoint, unpaired.
Ribbons = 95% bootstrap CI.

DATA

Grader: gpt-4o-mini. The same grid split by presenting problem is at problem_all_metrics.png.

Rendered but not shown here

- **Per-judge twins.** Almost every figure under arms/* exists once per grader. This deck shows the gpt-4o-mini copy for the detail grids and both copies for the headline outcome figures. The held-out twin sits at the same path with claude-haiku-4-5 substituted.
- **Per-item figures.** arms/questionnaires/figures/<judge>/{mici,miti,pct}/ hold one figure per individual coded item; the detail grids in this deck aggregate them.
- **Per-arm training curves.** arms/training/figures/<judge>/tb_curves/<arm>.png and reward_distribution/<arm>.png hold the single-arm versions.
- **PTO preference drift.** arms/preference/figures/<judge>/PTO_LA{0,5}_pref_*.png — category, direction, word-level and learn/unlearn drift across iterations, per arm.
- **Single-column variants.** compute_trajectory_col.png and crossgen_col.png are the same data laid out for a paper column.
- **Six figures are denser than a slide can carry.** The two 36-row channel forests, the 40-state dumbbell, the judge-agreement scatter, the gain-retention grid and the 9×4 cooperation grid have row labels smaller than this text at slide size. Open the PNG at full size from the results tree when we reach them — the paths are on each slide.
- **Everything is regenerable.** Exp3: python tools/render_results.py from Exp3_PTO_GRPO/eda/. Exp1 and Exp2: python make_exp1_exp2_figs.py from meetings/build/.